

Topological Energy Condensate Theory (TECT): 32

Jusang Lee*
Independent Researcher, Incheon, Korea
 (Dated: March, 2026)

I. CORRECTED RESIDUAL QUARTIC CLOSURE: EXACT FOUR-CLASS UPGRADE

II. CURRENT STAGE

We freeze the operational backend and the corrected local branch as

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096). \quad (1)$$

The correct reduced object is the projected residual map

$$G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix}, \quad G_X(\delta) := \Re \langle \Phi_X, R[\Psi(\delta)] \rangle_\Omega, \quad X \in \{A, E, O\}, \quad (2)$$

not a naive reduced free-energy critical point.

The corrected branch diagnostics are

$$\|G(\delta^{\text{final}})\| \approx 0.03791852545979427, \quad \text{rel_residual}(\delta^{\text{final}}) \approx 0.036459908397927156, \quad (3)$$

$$\text{shell_fraction}(\delta^{\text{final}}) \approx 0.9156153162713265, \quad \text{orbit_balance}(\delta^{\text{final}}) \approx 0.8860846137625883. \quad (4)$$

III. CORRECTED PROJECTED QUARTIC DATA

The residual-based extractor gives the quartic class means

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_\perp = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813. \quad (5)$$

If one compresses these numbers into the old three-parameter shell kernel

$$U_{\alpha\beta}^{\text{simple}} = u \delta_{\alpha\beta} + v + \kappa (\hat{k}_\alpha \cdot \hat{k}_\beta)^2, \quad (6)$$

then one obtains

$$u = -12381.062523497216, \quad v = 9027.17290917827, \quad \kappa = -27129.133911956756, \quad (7)$$

but the corresponding exact defect is

$$\Delta_{\text{ker}} = 32976.132300215555. \quad (8)$$

Hence the (u, v, κ) truncation fails strongly even on the corrected branch.

* jusanglee@kakao.com

IV. WHY THE THREE-PARAMETER TRUNCATION FAILS MATHEMATICALLY

For a BCC shell, ordered shell pairs (α, β) fall into exactly four classes:

$$\alpha = \beta, \quad \hat{k}_\beta = -\hat{k}_\alpha, \quad \hat{k}_\alpha \cdot \hat{k}_\beta = 0, \quad |\hat{k}_\alpha \cdot \hat{k}_\beta| = \frac{1}{2}. \quad (9)$$

For the old ansatz

$$U_{\alpha\beta}^{\text{simple}} = u \delta_{\alpha\beta} + v + \kappa (\hat{k}_\alpha \cdot \hat{k}_\beta)^2, \quad (10)$$

the four class values are forced to satisfy

$$U_d = u + v + \kappa, \quad U_{\text{opp}} = v + \kappa, \quad U_\perp = v, \quad U_{\text{obl}} = v + \frac{\kappa}{4}. \quad (11)$$

Therefore the old truncation imposes the exact algebraic constraint

$$\boxed{4U_{\text{obl}} - 3U_\perp - U_{\text{opp}} = 0.} \quad (12)$$

Equivalently,

$$\boxed{\Delta_{\text{ker}} := 4U_{\text{obl}} - 3U_\perp - U_{\text{opp}}} \quad (13)$$

must vanish identically on the old three-parameter submanifold.

Using the corrected projected data,

$$4U_{\text{obl}} = 8979.5577247563252, \quad (14)$$

$$3U_\perp = 27081.51872753481, \quad (15)$$

$$4U_{\text{obl}} - 3U_\perp - U_{\text{opp}} = -32976.132300215555. \quad (16)$$

Thus

$$\boxed{|\Delta_{\text{ker}}| = 32976.132300215555 \gg 0,} \quad (17)$$

so the simple kernel is not approximately wrong but structurally impossible.

V. MINIMAL EXACT ENLARGEMENT: FOUR-CLASS SHELL KERNEL

Let

$$P_{\alpha\beta}^{(d)} := \mathbf{1}_{\{\alpha=\beta\}}, \quad P_{\alpha\beta}^{(\text{opp})} := \mathbf{1}_{\{\hat{k}_\beta = -\hat{k}_\alpha\}}, \quad (18)$$

$$P_{\alpha\beta}^{(\perp)} := \mathbf{1}_{\{\hat{k}_\alpha \cdot \hat{k}_\beta = 0\}}, \quad P_{\alpha\beta}^{(\text{obl})} := \mathbf{1}_{\{|\hat{k}_\alpha \cdot \hat{k}_\beta| = 1/2\}}. \quad (19)$$

Then

$$\boxed{P^{(d)} + P^{(\text{opp})} + P^{(\perp)} + P^{(\text{obl})} = \mathbf{1}} \quad (20)$$

on ordered shell-pair space.

The minimal exact enlarged quartic shell kernel is therefore

$$\boxed{U_{\alpha\beta}^{(4c)} = u_d P_{\alpha\beta}^{(d)} + u_{\text{opp}} P_{\alpha\beta}^{(\text{opp})} + u_\perp P_{\alpha\beta}^{(\perp)} + u_{\text{obl}} P_{\alpha\beta}^{(\text{obl})}.} \quad (21)$$

Equivalently,

$$U_{\alpha\beta}^{(4c)} = \begin{cases} u_d, & \alpha = \beta, \\ u_{\text{opp}}, & \hat{k}_\beta = -\hat{k}_\alpha, \\ u_\perp, & \hat{k}_\alpha \cdot \hat{k}_\beta = 0, \\ u_{\text{obl}}, & |\hat{k}_\alpha \cdot \hat{k}_\beta| = \frac{1}{2}. \end{cases} \quad (22)$$

This is the smallest exact shell kernel that resolves all four BCC pair classes independently.

VI. DIRECT COEFFICIENT IDENTIFICATION ON THE CORRECTED BRANCH

Because the four-class kernel resolves the classes exactly, its coefficients are not fitted indirectly but read off directly:

$$\boxed{u_d = U_d, \quad u_{\text{opp}} = U_{\text{opp}}, \quad u_{\perp} = U_{\perp}, \quad u_{\text{obl}} = U_{\text{obl}}.} \quad (23)$$

Hence on the corrected branch,

$$\boxed{u_d = 2493.1087739398513, \quad u_{\text{opp}} = 14874.171297437068, \quad u_{\perp} = 9027.17290917827, \quad u_{\text{obl}} = 2244.8894311890813.} \quad (24)$$

So the four-class upgrade is not a heuristic ansatz. It is the exact minimal interpolation of the already extracted corrected quartic data.

VII. REDUCED QUARTIC FORM ON THE ORBIT-AMPLITUDE SECTOR

Let the shell amplitudes be reduced to three orbit sectors

$$A, \quad E, \quad O,$$

with real amplitudes

$$a_A, \quad a_E, \quad a_O,$$

and define

$$r_A = a_A^2, \quad r_E = a_E^2, \quad r_O = a_O^2.$$

The quartic shell contribution is

$$F_4^{\text{red}} = \frac{1}{4} \sum_{\alpha, \beta \in S_*} U_{\alpha\beta}^{(4c)} |\psi_{\alpha}|^2 |\psi_{\beta}|^2. \quad (25)$$

After orbit reduction, it becomes

$$\boxed{F_4^{\text{red}}(r_A, r_E, r_O) = \frac{1}{4} \sum_{X, Y \in \{A, E, O\}} M_{XY} r_X r_Y,} \quad (26)$$

with

$$\boxed{M_{XY} = N_{XY}^{(d)} u_d + N_{XY}^{(\text{opp})} u_{\text{opp}} + N_{XY}^{(\perp)} u_{\perp} + N_{XY}^{(\text{obl})} u_{\text{obl}}.} \quad (27)$$

Here

$$N_{XY}^{(c)}$$

is the purely combinatorial number of ordered shell pairs with first index in orbit X , second index in orbit Y , and pair-class c .

Thus the enlarged quartic closure problem is reduced to a finite combinatorial computation:

$$\boxed{\{N_{XY}^{(d)}, N_{XY}^{(\text{opp})}, N_{XY}^{(\perp)}, N_{XY}^{(\text{obl})}\} \implies M_{XY}.} \quad (28)$$

VIII. RESIDUAL-LEVEL UPGRADE

Because the correct reduced object is the residual map, the quartic contribution must enter through

$$\boxed{G_X^{(4)}(a_A, a_E, a_O) = \frac{\partial F_4^{\text{red}}}{\partial a_X}, \quad X \in \{A, E, O\}.} \quad (29)$$

Using $r_X = a_X^2$, this gives

$$G_X^{(4)} = a_X \sum_{Y \in \{A, E, O\}} M_{XY} r_Y. \quad (30)$$

Therefore the corrected reduced residual system must be upgraded to

$$G_X(a_A, a_E, a_O) = G_X^{(2)} + a_X \sum_Y M_{XY} r_Y + \dots, \quad X \in \{A, E, O\}. \quad (31)$$

IX. MINIMAL THEOREM-LEVEL CONCLUSION

Theorem IX.1. *Let the BCC shell quartic data be reduced to the four class means*

$$(U_d, U_{\text{opp}}, U_{\perp}, U_{\text{obl}}).$$

Then:

1. *the simple kernel $u\delta_{\alpha\beta} + v + \kappa(\hat{k}_\alpha \cdot \hat{k}_\beta)^2$ is exact if and only if*

$$4U_{\text{obl}} - 3U_{\perp} - U_{\text{opp}} = 0;$$

2. *if this condition fails, the minimal exact enlargement is the four-class kernel*

$$U_{\alpha\beta}^{(4c)} = u_d P_{\alpha\beta}^{(d)} + u_{\text{opp}} P_{\alpha\beta}^{(\text{opp})} + u_{\perp} P_{\alpha\beta}^{(\perp)} + u_{\text{obl}} P_{\alpha\beta}^{(\text{obl})};$$

3. *the resulting orbit-reduced quartic form is determined by the finite matrix*

$$M_{XY} = N_{XY}^{(d)} u_d + N_{XY}^{(\text{opp})} u_{\text{opp}} + N_{XY}^{(\perp)} u_{\perp} + N_{XY}^{(\text{obl})} u_{\text{obl}},$$

and the quartic residual contribution is

$$G_X^{(4)} = a_X \sum_Y M_{XY} r_Y.$$

Proof. Part (1) follows from the explicit class values of the three-parameter kernel. Part (2) follows because the BCC shell has exactly four pair classes, so any exact pair-class-resolved kernel must have at least four independent coefficients. Part (3) is the orbit reduction of the shell quartic form followed by differentiation with respect to a_X . $\square$

X. SHARP NEXT TASK

The next nonredundant executable step is now uniquely fixed:

$$(u, v, \kappa) \longrightarrow (u_d, u_{\text{opp}}, u_{\perp}, u_{\text{obl}}) \longrightarrow M_{XY} \longrightarrow G_X^{(4)}. \quad (32)$$

If the residual misfit remains large after this step, then the next obstruction is no longer pair-class compression. It is genuine higher shell-tensor structure beyond four pair classes.

XI. FINAL SUMMARY

$$\boxed{\text{Current status: corrected residual quartic closure stage.}} \quad (33)$$

$$\boxed{\text{The working-branch carrier witness is no longer the bottleneck.}} \quad (34)$$

$$\boxed{(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096)} \quad (35)$$

are frozen.

The corrected projected quartic class data are

$$\boxed{U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813.} \quad (36)$$

The old three-parameter shell kernel

$$U_{\alpha\beta} = u\delta_{\alpha\beta} + v + \kappa(\hat{k}_{\alpha} \cdot \hat{k}_{\beta})^2$$

necessarily imposes

$$4U_{\text{obl}} - 3U_{\perp} - U_{\text{opp}} = 0.$$

But the corrected branch violates this strongly:

$$\boxed{|\Delta_{\text{ker}}| = |4U_{\text{obl}} - 3U_{\perp} - U_{\text{opp}}| = 32976.132300215555.} \quad (37)$$

Hence the simple (u, v, κ) truncation is exactly ruled out.

The minimal exact enlarged reduced quartic kernel is therefore

$$\boxed{U_{\alpha\beta}^{(4c)} = u_d P_{\alpha\beta}^{(d)} + u_{\text{opp}} P_{\alpha\beta}^{(\text{opp})} + u_{\perp} P_{\alpha\beta}^{(\perp)} + u_{\text{obl}} P_{\alpha\beta}^{(\text{obl})},} \quad (38)$$

with direct identification

$$\boxed{u_d = U_d, \quad u_{\text{opp}} = U_{\text{opp}}, \quad u_{\perp} = U_{\perp}, \quad u_{\text{obl}} = U_{\text{obl}}.} \quad (39)$$

After orbit reduction, the quartic form is

$$\boxed{F_4^{\text{red}} = \frac{1}{4} \sum_{X,Y} M_{XY} r_X r_Y, \quad M_{XY} = N_{XY}^{(d)} u_d + N_{XY}^{(\text{opp})} u_{\text{opp}} + N_{XY}^{(\perp)} u_{\perp} + N_{XY}^{(\text{obl})} u_{\text{obl}},} \quad (40)$$

and the residual contribution is

$$\boxed{G_X^{(4)} = a_X \sum_Y M_{XY} r_Y.} \quad (41)$$

Therefore the exact next step is

$$\boxed{(u, v, \kappa) \rightarrow (u_d, u_{\text{opp}}, u_{\perp}, u_{\text{obl}}) \rightarrow M_{XY} \rightarrow G_X^{(4)}.} \quad (42)$$

$$\boxed{\text{Goal C: complete the corrected residual branch by enlarged quartic shell-kernel closure.}} \quad (43)$$

$$\boxed{\text{Goal M: compute the orbit-count matrices } N_{XY}^{(c)} \text{ and assemble } M_{XY}.} \quad (44)$$

$$\boxed{\text{Goal S: insert } M_{XY} \text{ into } G_X^{(4)} \text{ and rerun the corrected residual system.}} \quad (45)$$

XII. EXACT COARSE-ORBIT COMBINATORICS OF THE CORRECTED FOUR-CLASS QUARTIC KERNEL

XIII. SHORT EXPLANATION

At the current corrected residual stage, the next finite task is to compute the orbit-count matrices

$$N_{XY}^{(d)}, \quad N_{XY}^{(\text{opp})}, \quad N_{XY}^{(\perp)}, \quad N_{XY}^{(\text{obl})}, \quad X, Y \in \{A, E, O\},$$

for the coarse orbit split of the commensurate BCC shell.

This part can now be closed exactly.

The key result is that, for the coarse orbit decomposition

$$|A| = 2, \quad |E| = 2, \quad |O| = 8,$$

the ordered-pair class counts are completely explicit. As a consequence, the four-class reduced quartic matrix is explicit as well.

A second important consequence is structural:

the four-class pair-class closure is automatically $A \leftrightarrow E$ symmetric at coarse-orbit level.

Therefore any A/E asymmetry in the corrected residual system must come from the quadratic sector

$$\sigma_A \neq \sigma_E,$$

or from a refinement beyond coarse pair-class data, e.g. sign-resolved suborbit quartics.

XIV. 1. COARSE ORBIT SPLIT OF THE SELECTED BCC SHELL

Choose the shell axis and tangent frame

$$\hat{n} = \frac{(1, 1, 0)}{\sqrt{2}}, \quad e_1 = \frac{(1, -1, 0)}{\sqrt{2}}, \quad e_2 = (0, 0, 1). \quad (46)$$

Then the 12 first-shell BCC directions split as follows:

$$A = \left\{ \frac{(1, 1, 0)}{\sqrt{2}}, \frac{(-1, -1, 0)}{\sqrt{2}} \right\}, \quad |A| = 2, \quad (47)$$

$$E = \left\{ \frac{(1, -1, 0)}{\sqrt{2}}, \frac{(-1, 1, 0)}{\sqrt{2}} \right\}, \quad |E| = 2, \quad (48)$$

$$O = \left\{ \frac{(\pm 1, 0, \pm 1)}{\sqrt{2}}, \frac{(0, \pm 1, \pm 1)}{\sqrt{2}} \right\}, \quad |O| = 8. \quad (49)$$

Thus

$n_A = 2, \quad n_E = 2, \quad n_O = 8.$

 \quad (50)

XV. 2. ORDERED-PAIR CLASS-COUNT MATRICES

We count ordered pairs (α, β) with first index in orbit X , second index in orbit Y , and class

$$d, \quad \text{opp}, \quad \perp, \quad \text{obl}.$$

A. Diagonal counts

Diagonal pairs occur only when $\alpha = \beta$. Hence

$$N^{(d)} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{pmatrix}. \quad (51)$$

B. Opposite counts

Each shell direction has exactly one opposite partner in the same coarse orbit. Therefore

$$N^{(\text{opp})} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{pmatrix}. \quad (52)$$

C. Perpendicular counts

The A -vectors are perpendicular only to the E -vectors, and the oblique orbit O contains 16 ordered perpendicular pairs internally. Thus

$$N^{(\perp)} = \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 16 \end{pmatrix}. \quad (53)$$

D. Oblique counts

The A -vectors are oblique to all 8 O -vectors, likewise for the E -vectors, and the O -orbit contains 32 ordered oblique pairs internally. Hence

$$N^{(\text{obl})} = \begin{pmatrix} 0 & 0 & 16 \\ 0 & 0 & 16 \\ 16 & 16 & 32 \end{pmatrix}. \quad (54)$$

E. Consistency check

Summing the four class matrices gives the full ordered-pair count matrix:

$$N^{(d)} + N^{(\text{opp})} + N^{(\perp)} + N^{(\text{obl})} = \begin{pmatrix} 4 & 4 & 16 \\ 4 & 4 & 16 \\ 16 & 16 & 64 \end{pmatrix} = \begin{pmatrix} n_A n_A & n_A n_E & n_A n_O \\ n_E n_A & n_E n_E & n_E n_O \\ n_O n_A & n_O n_E & n_O n_O \end{pmatrix}. \quad (55)$$

So the combinatorics is complete and exact.

XVI. 3. RAW FOUR-CLASS ORBIT MATRIX ON THE CORRECTED BRANCH

The corrected class means are

$$u_d = 2493.1087739398513, \quad u_{\text{opp}} = 14874.171297437068, \quad u_{\perp} = 9027.17290917827, \quad u_{\text{obl}} = 2244.8894311890813. \quad (56)$$

Under the raw ordered-pair convention,

$$M_{XY}^{\text{raw}} = N_{XY}^{(d)} u_d + N_{XY}^{(\text{opp})} u_{\text{opp}} + N_{XY}^{(\perp)} u_{\perp} + N_{XY}^{(\text{obl})} u_{\text{obl}}. \quad (57)$$

Therefore

$$M^{\text{raw}} = \begin{pmatrix} 34734.560142753835 & 36108.69163671308 & 35918.23089902530 \\ 36108.69163671308 & 34734.560142753835 & 35918.23089902530 \\ 35918.23089902530 & 35918.23089902530 & 355209.46891591826 \end{pmatrix}. \quad (58)$$

XVII. 4. NORMALIZED ORBIT-SUM CONVENTION RELEVANT FOR THE CURRENT RESIDUAL ANSATZ

The current corrected residual ansatz is written in terms of normalized orbit sums

$$\Phi_A, \quad \Phi_E, \quad \Phi_O,$$

so if the orbit amplitudes a_X multiply normalized orbit vectors, then the shell amplitude per patch is

$$\psi_\alpha = \frac{a_X}{\sqrt{n_X}} \quad (\alpha \in X).$$

In that convention, the quartic coefficient matrix entering

$$F_4^{\text{red}} = \frac{1}{4} \sum_{X,Y} \bar{M}_{XY} r_X r_Y, \quad r_X = a_X^2,$$

is

$$\bar{M}_{XY} = \frac{1}{n_X n_Y} M_{XY}^{\text{raw}} = \sum_c \frac{N_{XY}^{(c)}}{n_X n_Y} u_c. \quad (59)$$

Hence

$$\bar{M} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}. \quad (60)$$

This is the coefficient matrix compatible with the normalized-orbit reduced variables appearing in

$$\Psi(\delta) = \Psi_{\text{BCC}} + \delta_A \Phi_A + \delta_E \Phi_E + \delta_O \Phi_O.$$

XVIII. 5. RESIDUAL-LEVEL QUARTIC CONTRIBUTION

Under the normalized orbit convention,

$$F_4^{\text{red}}(r_A, r_E, r_O) = \frac{1}{4} \sum_{X,Y \in \{A,E,O\}} \bar{M}_{XY} r_X r_Y. \quad (61)$$

Therefore the quartic residual contribution is

$$G_X^{(4)}(a_A, a_E, a_O) = a_X \sum_{Y \in \{A,E,O\}} \bar{M}_{XY} r_Y, \quad r_Y = a_Y^2. \quad (62)$$

Explicitly,

$$G_A^{(4)} = a_A \left(8683.640035688459 r_A + 9027.17290917827 r_E + 2244.8894311890813 r_O \right), \quad (63)$$

$$G_E^{(4)} = a_E \left(9027.17290917827 r_A + 8683.640035688459 r_E + 2244.8894311890813 r_O \right), \quad (64)$$

$$G_O^{(4)} = a_O \left(2244.8894311890813 r_A + 2244.8894311890813 r_E + 5550.147951811223 r_O \right). \quad (65)$$

XIX. 6. STRUCTURAL CONSEQUENCE: COARSE FOUR-CLASS CLOSURE CANNOT BREAK A/E SYMMETRY

From the exact count matrices,

$$N_{AA}^{(c)} = N_{EE}^{(c)}, \quad N_{AO}^{(c)} = N_{EO}^{(c)}, \quad N_{AE}^{(c)} = N_{EA}^{(c)}, \quad \forall c \in \{d, \text{opp}, \perp, \text{obl}\}. \quad (66)$$

Therefore

$$\boxed{\bar{M}_{AA} = \bar{M}_{EE}, \quad \bar{M}_{AO} = \bar{M}_{EO}, \quad \bar{M}_{AE} = \bar{M}_{EA}.} \quad (67)$$

Hence the coarse four-class quartic sector is exactly invariant under

$$A \leftrightarrow E.$$

So the corrected branch asymmetry

$$\delta_A^{\text{final}} = +0.0096, \quad \delta_E^{\text{final}} = -0.0288$$

cannot be explained by the coarse four-class quartic matrix alone.

The only possibilities are:

1. the quadratic orbit means already break that symmetry,

$$\sigma_A \neq \sigma_E;$$

2. higher reduced terms beyond coarse pair-class closure are relevant;
3. the true residual-level reduction must be refined to the sign-resolved suborbit basis

$$(A_+, A_-, E_+, E_-, O_{++}, O_{+-}, O_{-+}, O_{--}).$$

XX. 7. EXACT STATUS AFTER THE COMBINATORIAL CLOSURE

The coarse-orbit pair-class combinatorics is now completely closed.

What remains open is not

$$N_{XY}^{(c)},$$

but the next insertion step:

$$\boxed{(\sigma_A, \sigma_E, \sigma_O) \quad + \quad \bar{M} \quad \longrightarrow \quad G_X^{(2)} + G_X^{(4)}} \quad (68)$$

inside the actual projected residual system.

If this still fails to account for the corrected branch geometry, then the next irreducible refinement is not another pair-class kernel, but the sign-resolved suborbit quartic system.

XXI. FINAL SUMMARY

The exact ordered-pair class-count matrices of the coarse BCC orbit split are

$$\boxed{N^{(d)} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{pmatrix}, \quad N^{(\text{opp})} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{pmatrix},} \quad (69)$$

$$\boxed{N^{(\perp)} = \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 16 \end{pmatrix}, \quad N^{(\text{obl})} = \begin{pmatrix} 0 & 0 & 16 \\ 0 & 0 & 16 \\ 16 & 16 & 32 \end{pmatrix}.} \quad (70)$$

For the corrected four-class quartic data, the normalized coarse-orbit quartic matrix is

$$\bar{M} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}. \quad (71)$$

Therefore the quartic contribution to the reduced residual system is now explicit:

$$G_X^{(4)} = a_X \sum_Y \bar{M}_{XY} r_Y.$$

But this closure is still exactly $A \leftrightarrow E$ symmetric. So the corrected A/E asymmetry cannot come from the coarse four-class quartic sector alone.

Hence the next nonredundant step is sharply fixed:

$$\boxed{\text{insert } (\sigma_A, \sigma_E, \sigma_O) \text{ and } \bar{M} \text{ into the residual equations,}} \quad (72)$$

and if that is still insufficient,

$$\boxed{\text{upgrade from coarse } (A, E, O) \text{ to the sign-resolved suborbit quartic basis.}} \quad (73)$$

XXII. FINAL SUMMARY

$$\boxed{\text{Current status: the coarse-orbit four-class combinatorics is now closed.}} \quad (74)$$

The frozen corrected residual stage keeps

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096),$$

and the corrected quartic class data

$$u_d = 2493.1087739398513, \quad u_{\text{opp}} = 14874.171297437068, \quad u_{\perp} = 9027.17290917827, \quad u_{\text{obl}} = 2244.8894311890813.$$

For the coarse orbit split

$$|A| = 2, \quad |E| = 2, \quad |O| = 8,$$

the exact ordered-pair count matrices are

$$N^{(d)} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{pmatrix}, \quad N^{(\text{opp})} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{pmatrix},$$

$$N^{(\perp)} = \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 16 \end{pmatrix}, \quad N^{(\text{obl})} = \begin{pmatrix} 0 & 0 & 16 \\ 0 & 0 & 16 \\ 16 & 16 & 32 \end{pmatrix}.$$

Under the normalized orbit-sum convention, the corrected quartic matrix is

$$\bar{M} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}.$$

Hence

$$G_X^{(4)} = a_X \sum_Y \bar{M}_{XY} r_Y$$

is now explicit.

However, the coarse four-class quartic sector is exactly symmetric under

$$A \leftrightarrow E,$$

so it cannot by itself explain the corrected asymmetry between

$$\delta_A^{\text{final}} = +0.0096 \quad \text{and} \quad \delta_E^{\text{final}} = -0.0288.$$

Therefore the next exact task is:

$$(\sigma_A, \sigma_E, \sigma_O) + \bar{M} \longrightarrow G_X^{(2)} + G_X^{(4)}$$

inside the actual projected residual system.

$$\boxed{\text{Goal C: complete the corrected residual branch by explicit residual insertion.}} \quad (75)$$

$$\boxed{\text{Goal M: test whether the coarse } (A, E, O) \text{ residual model already explains the corrected branch asymmetry.}} \quad (76)$$

$$\boxed{\text{Goal S: if not, upgrade immediately to the sign-resolved suborbit quartic basis.}} \quad (77)$$

XXIII. COARSE RESIDUAL INSERTION OF THE CORRECTED SEED AND THE NECESSITY OF A/E QUADRATIC SPLITTING

XXIV. SHORT EXPLANATION

At the current corrected residual stage, the correct reduced object is the projected residual map

$$G(\delta) = (G_A(\delta), G_E(\delta), G_O(\delta))^T,$$

with frozen backend

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62)$$

and frozen corrected seed

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096).$$

The four-class quartic kernel is already fixed exactly by

$$u_d = U_d, \quad u_{\text{opp}} = U_{\text{opp}}, \quad u_{\perp} = U_{\perp}, \quad u_{\text{obl}} = U_{\text{obl}}.$$

So the next nonredundant step is no longer combinatorics by itself, but insertion of the corrected seed into the class-resolved residual equations.

The main result is:

$$\boxed{\text{if the coarse quadratic sector is } A/E\text{-symmetric, then the corrected seed cannot satisfy the coarse residual stationarity equation}} \quad (78)$$

Hence a nonzero A/E quadratic splitting is necessary already at the coarse (A, E, O) level.

XXV. 1. COARSE NORMALIZED QUARTIC MATRIX

For the normalized orbit-sum convention, the class-resolved quartic matrix is

$$\bar{M} = \begin{pmatrix} \bar{M}_{AA} & \bar{M}_{AE} & \bar{M}_{AO} \\ \bar{M}_{EA} & \bar{M}_{EE} & \bar{M}_{EO} \\ \bar{M}_{OA} & \bar{M}_{OE} & \bar{M}_{OO} \end{pmatrix} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}. \quad (79)$$

Equivalently, in terms of the four class means,

$$\bar{M}_{AA} = \bar{M}_{EE} = \frac{u_d + u_{\text{opp}}}{2}, \quad \bar{M}_{AE} = \bar{M}_{EA} = u_{\perp}, \quad \bar{M}_{AO} = \bar{M}_{EO} = u_{\text{obl}}, \quad (80)$$

and

$$\bar{M}_{OO} = \frac{u_d + u_{\text{opp}} + 2u_{\perp} + 4u_{\text{obl}}}{8}. \quad (81)$$

Thus the coarse quartic sector is exactly invariant under

$$A \leftrightarrow E.$$

XXVI. 2. COARSE RESIDUAL ANSATZ

Write the coarse reduced residual system in the minimal diagonal quadratic form

$$G_X(a_A, a_E, a_O) = a_X \left(q_X + \sum_{Y \in \{A, E, O\}} \bar{M}_{XY} r_Y \right), \quad r_Y := a_Y^2, \quad X \in \{A, E, O\}. \quad (82)$$

Here q_A, q_E, q_O are the unknown effective coarse quadratic coefficients in the same normalization as $\bar{M}$.

a. Important honesty note. These q_X are *effective coefficients of the present coarse reduced ansatz*. They should not be automatically identified with the numerically extracted orbit means

$$\sigma_A, \sigma_E, \sigma_O$$

unless the normalization map between the extractor output and (82) is checked explicitly.

XXVII. 3. INSERT THE CORRECTED SEED

For the frozen corrected seed,

$$(a_A, a_E, a_O) = (0.0096, -0.0288, 0.0096), \quad (83)$$

so

$$r_A = r_O = \rho, \quad r_E = 9\rho, \quad \rho := (0.0096)^2 = 9.216 \times 10^{-5}. \quad (84)$$

Define the quartic brackets

$$Q_A := \sum_Y \bar{M}_{AY} r_Y, \quad Q_E := \sum_Y \bar{M}_{EY} r_Y, \quad Q_O := \sum_Y \bar{M}_{OY} r_Y. \quad (85)$$

Using the above $\bar{M}$, one obtains

$$Q_A = \bar{M}_{AA}\rho + \bar{M}_{AE}(9\rho) + \bar{M}_{AO}\rho = 8.4946715734562584, \quad (86)$$

$$Q_E = \bar{M}_{EA}\rho + \bar{M}_{EE}(9\rho) + \bar{M}_{EO}\rho = 8.2413916564896905, \quad (87)$$

$$Q_O = \bar{M}_{OA}\rho + \bar{M}_{OE}(9\rho) + \bar{M}_{OO}\rho = 2.5803917350227796. \quad (88)$$

Hence the quartic residual part at the corrected seed is

$$G^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} a_A Q_A \\ a_E Q_E \\ a_O Q_O \end{pmatrix} = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}. \quad (89)$$

Its norm is

$$\|G^{(4)}(\delta^{\text{final}})\| \approx 0.252190135277701. \quad (90)$$

So the quartic contribution by itself is much larger than the actual corrected residual norm

$$\|G(\delta^{\text{final}})\| \approx 0.03791852545979427.$$

Therefore substantial cancellation by the lower-order residual sector is necessary.

XXVIII. 4. EXACT OBSTRUCTION UNDER A/E -SYMMETRIC QUADRATIC SECTOR

Assume first the coarse quadratic sector is A/E -symmetric:

$$q_A = q_E =: q. \quad (91)$$

If the corrected seed were an exact stationary point of the coarse residual system, then since

$$a_A \neq 0, \quad a_E \neq 0,$$

we would need

$$q + Q_A = 0, \quad q + Q_E = 0. \quad (92)$$

Subtracting gives

$$Q_A - Q_E = 0. \quad (93)$$

But

$$Q_A - Q_E = (\bar{M}_{AA} - \bar{M}_{AE})(r_A - r_E). \quad (94)$$

Now

$$\bar{M}_{AA} - \bar{M}_{AE} = 8683.640035688459 - 9027.17290917827 = -343.5328734898113 \neq 0, \quad (95)$$

and

$$r_A - r_E = \rho - 9\rho = -8\rho = -0.00073728 \neq 0. \quad (96)$$

Hence

$$Q_A - Q_E = 0.25327991696656785 \neq 0. \quad (97)$$

Therefore the corrected seed cannot satisfy the coarse residual stationarity equations under $q_A = q_E$.

Theorem XXVIII.1 (Necessity of coarse A/E quadratic splitting). *For the coarse residual ansatz (82) with the class-resolved quartic matrix $\bar{M}$ above, the corrected seed*

$$(a_A, a_E, a_O) = (0.0096, -0.0288, 0.0096)$$

cannot be an exact stationary point if

$$q_A = q_E.$$

Equivalently, any exact coarse residual realization of the corrected seed requires

$$q_A \neq q_E.$$

Proof. The calculation above shows that exact stationarity would force

$$Q_A = Q_E,$$

but the corrected seed gives

$$Q_A - Q_E = 0.25327991696656785 \neq 0.$$

Hence $q_A = q_E$ is impossible. □

XXIX. 5. EXACT REQUIRED QUADRATIC COEFFICIENTS FOR THE CORRECTED SEED

If one insists that the corrected seed be an exact coarse stationary point, then the required quadratic coefficients are uniquely fixed by

$$q_A = -Q_A, \quad q_E = -Q_E, \quad q_O = -Q_O. \quad (98)$$

Therefore

$$\boxed{q_A^* = -8.4946715734562584, \quad q_E^* = -8.2413916564896905, \quad q_O^* = -2.5803917350227796.} \quad (99)$$

In particular, the required coarse A/E splitting is

$$\boxed{q_A^* - q_E^* = -(Q_A - Q_E) = -0.25327991696656785.} \quad (100)$$

Thus the corrected seed requires the quadratic sector to satisfy

$$\boxed{q_A < q_E.} \quad (101)$$

XXX. 6. COMPARISON WITH THE NUMERICALLY EXTRACTED ORBIT ASYMMETRY

The corrected projection already yields orbit means

$$\boxed{\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933.} \quad (102)$$

So the numerical extractor already displays the same directional asymmetry

$$\boxed{\sigma_A < \sigma_E.}$$

This does *not* by itself prove that

$$q_A^* - q_E^* = \sigma_A - \sigma_E,$$

because the normalization map between the extractor output and the coarse residual ansatz has not yet been closed.

However, it does show that the corrected branch is numerically aligned with the same A/E -splitting direction required by the coarse residual insertion.

XXXI. 7. SHARP CONSEQUENCE

The coarse class-resolved quartic closure is not the end of the story. What it proves is more precise:

1. the corrected seed is incompatible with an A/E -symmetric coarse quadratic sector;
2. the corrected seed can be supported at coarse level only if a nonzero quadratic splitting

$$q_A - q_E = -0.25327991696656785$$

(in the coarse normalization) is present;

3. therefore the next reduced task is not more pair counting, but explicit insertion of the quadratic residual layer and its normalization map.

So the next exact reduced system to write is

$$\boxed{G_X(a_A, a_E, a_O) = a_X \left(q_X + \sum_Y \bar{M}_{XY} r_Y \right) + G_X^{(\text{rem})},} \quad (103)$$

where $G_X^{(\text{rem})}$ collects any still-unresolved residual pieces beyond the minimal coarse quadratic-plus-quartic truncation.

XXXII. FINAL SUMMARY

At the corrected residual stage, the frozen seed

$$(a_A, a_E, a_O) = (0.0096, -0.0288, 0.0096)$$

gives

$$r_A = r_O = \rho, \quad r_E = 9\rho, \quad \rho = 9.216 \times 10^{-5}.$$

Inserting this into the exact coarse four-class quartic matrix yields

$$\boxed{Q_A = 8.4946715734562584, \quad Q_E = 8.2413916564896905, \quad Q_O = 2.5803917350227796,} \quad (104)$$

and hence

$$\boxed{G^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}.} \quad (105)$$

Therefore the quartic contribution alone is not small; it must be cancelled by the quadratic residual layer.

More sharply, if the coarse quadratic sector is A/E -symmetric, then the corrected seed cannot be stationary, because exact stationarity would require

$$Q_A = Q_E,$$

but in fact

$$\boxed{Q_A - Q_E = 0.25327991696656785 \neq 0.} \quad (106)$$

Hence a nonzero coarse A/E quadratic splitting is necessary:

$$\boxed{q_A \neq q_E.} \quad (107)$$

The exact coarse stationary realization of the corrected seed requires

$$\boxed{q_A^* = -8.4946715734562584, \quad q_E^* = -8.2413916564896905, \quad q_O^* = -2.5803917350227796,} \quad (108)$$

so in particular

$$\boxed{q_A^* - q_E^* = -0.25327991696656785 < 0.} \quad (109)$$

Thus the next nonredundant step is not another quartic combinatorial refinement. It is the explicit identification and normalization of the quadratic residual layer that produces the required A/E splitting.

XXXIII. FINAL SUMMARY

$$\boxed{\text{Current status: coarse four-class quartic closure is inserted, and it already forces a nontrivial } A/E \text{ quadratic splitting.}} \quad (110)$$

The frozen corrected seed is

$$(a_A, a_E, a_O) = (0.0096, -0.0288, 0.0096),$$

so

$$r_A = r_O = \rho, \quad r_E = 9\rho, \quad \rho = 9.216 \times 10^{-5}.$$

Using the exact coarse normalized four-class quartic matrix

$$\bar{M} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix},$$

one obtains

$$Q_A = 8.4946715734562584, \quad Q_E = 8.2413916564896905, \quad Q_O = 2.5803917350227796,$$

and therefore

$$G^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}.$$

Hence the quartic contribution is not small by itself and must be cancelled by the lower-order residual sector. Most importantly, if the coarse quadratic sector were A/E -symmetric, then exact stationarity would force

$$Q_A = Q_E,$$

but in fact

$$Q_A - Q_E = 0.25327991696656785 \neq 0.$$

Therefore the corrected seed cannot be supported by an A/E -symmetric coarse residual model. A nonzero quadratic splitting is necessary:

$$q_A \neq q_E.$$

The exact coarse stationary realization of the corrected seed requires

$$q_A^* = -8.4946715734562584, \quad q_E^* = -8.2413916564896905, \quad q_O^* = -2.5803917350227796,$$

so that

$$q_A^* - q_E^* = -0.25327991696656785 < 0.$$

Thus the next exact reduced task is no longer pair counting. It is:

$$\boxed{\text{identify the quadratic residual layer and its normalization map.}}$$

$$\boxed{\text{Goal C: complete the corrected residual branch by closing the quadratic-plus-quartic residual system.}} \quad (111)$$

$$\boxed{\text{Goal M: determine the actual coarse quadratic coefficients } (q_A, q_E, q_O) \text{ from the extractor output.}} \quad (112)$$

$$\boxed{\text{Goal S: write and test the explicit normalized map from extracted orbit means to the coarse residual coefficients.}} \quad (113)$$

XXXIV. QUADRATIC NORMALIZATION AUDIT BEYOND THE NAIVE IDENTIFICATION $q_X = \sigma_X$

XXXV. SHORT EXPLANATION

At the corrected residual stage, the frozen data are:

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096), \quad \|G(\delta^{\text{final}})\| \approx 0.03791852545979427. \quad (114)$$

The corrected projection extractor gives

$$\sigma_0 = -109.41971030446824, \quad \eta = -6.502583100163014, \quad (115)$$

$$\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933, \quad (116)$$

and the four exact quartic class means

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813. \quad (117)$$

The key question is now:

$$\text{how do the extracted orbit means } (\sigma_A, \sigma_E, \sigma_O) \text{ map to the actual coarse quadratic residual layer } (q_A, q_E, q_O)?$$

The main result of this section is negative but decisive:

$$q_X = \sigma_X \text{ fails strongly,} \quad (118)$$

and even the more general two-parameter (σ_0, η) -based linear shell map fails. Therefore the correct next object is the full local residual Jacobian

$$J_{XY} = \partial_{\delta_Y} G_X|_{\delta^{\text{final}}},$$

not another compression to (σ_0, η) or a single orbitwise scale factor.

XXXVI. 1. REQUIRED COARSE QUADRATIC COEFFICIENTS FROM THE CORRECTED SEED

From the already-constructed coarse four-class quartic insertion, the normalized quartic matrix is

$$\bar{M} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}. \quad (119)$$

For

$$(a_A, a_E, a_O) = (0.0096, -0.0288, 0.0096),$$

we have

$$r_A = r_O = \rho, \quad r_E = 9\rho, \quad \rho = (0.0096)^2 = 9.216 \times 10^{-5}. \quad (120)$$

Hence the quartic brackets are

$$Q_A = 8.4946715734562584, \quad Q_E = 8.2413916564896905, \quad Q_O = 2.5803917350227796. \quad (121)$$

Therefore, in the minimal coarse ansatz

$$G_X(a_A, a_E, a_O) = a_X \left(q_X + \sum_Y \bar{M}_{XY} r_Y \right), \quad (122)$$

the exact stationary realization of the corrected seed requires

$$q_A^* = -Q_A = -8.4946715734562584, \quad (123)$$

$$q_E^* = -Q_E = -8.2413916564896905, \quad (124)$$

$$q_O^* = -Q_O = -2.5803917350227796. \quad (125)$$

So the target quadratic layer that must be explained is numerically fixed.

XXXVII. 2. CANONICAL DIRECT IDENTIFICATION AND WHY IT FAILS

The residual-based reduced object is

$$G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix} = \begin{pmatrix} \Re\langle \Phi_A, R[\Psi(\delta)] \rangle_\Omega \\ \Re\langle \Phi_E, R[\Psi(\delta)] \rangle_\Omega \\ \Re\langle \Phi_O, R[\Psi(\delta)] \rangle_\Omega \end{pmatrix}. \quad (126)$$

Suppose, for the moment, that:

1. the extracted shell coefficients h_α are the linear coefficients of the same reduced coordinates $\delta_A, \delta_E, \delta_O$;
2. the orbit basis Φ_A, Φ_E, Φ_O is normalized compatibly with the extractor convention;
3. the quadratic reduced residual layer is diagonal in the coarse orbit basis.

Then the canonical linear projection gives

$$G_X^{(2)} = \sigma_X a_X, \quad (127)$$

so that the direct identification would be

$$q_X^{\text{can}} = \sigma_X. \quad (128)$$

Numerically this predicts

$$q_A^{\text{can}} = -115.81015796064617, \quad q_E^{\text{can}} = -105.83201242243113, \quad q_O^{\text{can}} = -108.719022860933. \quad (129)$$

But the corrected seed requires instead

$$q_A^* = -8.4946715734562584, \quad q_E^* = -8.2413916564896905, \quad q_O^* = -2.5803917350227796. \quad (130)$$

Hence the ratios are

$$\frac{q_A^*}{\sigma_A} = 0.07334996966624345, \quad \frac{q_E^*}{\sigma_E} = 0.07787238915569299, \quad \frac{q_O^*}{\sigma_O} = 0.023734500799583762. \quad (131)$$

These are not equal. Therefore even a single common scale factor fails.

Theorem XXXVII.1 (Failure of the direct normalization map). *The corrected seed data are incompatible with the direct identification*

$$q_X = \sigma_X$$

and also incompatible with any single common rescaling

$$q_X = \zeta \sigma_X$$

of the extracted orbit means.

Proof. Direct comparison shows

$$q_X^* \neq \sigma_X$$

by more than an order of magnitude in each orbit. Moreover the three ratios

$$q_A^*/\sigma_A, \quad q_E^*/\sigma_E, \quad q_O^*/\sigma_O$$

are unequal, so no common ζ can satisfy

$$q_X^* = \zeta \sigma_X$$

simultaneously for $X = A, E, O$. □

XXXVIII. 3. TWO-PARAMETER (σ_0, η) MAP AND ITS FAILURE

The extractor formulas give

$$\sigma_A = \sigma_0 + \eta, \quad \sigma_E = \sigma_0 - \frac{\eta}{2}, \quad \sigma_O = \sigma_0 - \frac{\eta}{8}. \quad (132)$$

Therefore the most natural two-parameter linear shell model is

$$q_A = c_0\sigma_0 + c_2\eta, \quad q_E = c_0\sigma_0 - \frac{1}{2}c_2\eta, \quad q_O = c_0\sigma_0 - \frac{1}{8}c_2\eta. \quad (133)$$

Fitting q_A^* and q_E^* determines

$$c_0 = 0.07609066293733878, \quad c_2 = 0.025967108051795404. \quad (134)$$

This predicts

$$q_O^{\text{pred}} = c_0\sigma_0 - \frac{1}{8}c_2\eta = -8.304711635731334. \quad (135)$$

But the corrected seed requires

$$q_O^* = -2.5803917350227796. \quad (136)$$

So the defect is

$$q_O^{\text{pred}} - q_O^* = -5.724319900708554. \quad (137)$$

This is far too large to be attributed to the small corrected residual norm

$$\|G(\delta^{\text{final}})\| \approx 0.03791852545979427.$$

Theorem XXXVIII.1 (Failure of the (σ_0, η) -only quadratic map). *The required corrected-seed quadratic coefficients (q_A^*, q_E^*, q_O^*) cannot be generated by any linear map depending only on the two shell coefficients (σ_0, η) in the canonical orbit pattern*

$$(\sigma_0 + \eta, \sigma_0 - \eta/2, \sigma_0 - \eta/8).$$

Proof. The two-parameter ansatz (133) is uniquely fixed by q_A^* and q_E^* , but its prediction for q_O differs from the required corrected value by

$$-5.724319900708554,$$

which is not compatible with a merely small correction. Therefore no such two-parameter map can reproduce the corrected quadratic layer. $\square$

XXXIX. 4. HONEST INTERPRETATION

The files already warn that the extracted coefficients

$$(\sigma_0, \eta, U_d, U_{\text{opp}}, U_{\perp}, U_{\text{obl}})$$

must be interpreted honestly as

$$\text{effective reduced coefficients of the current solver family,}$$

not yet theorem-level coefficients of a fully closed microscopic functional.

The present audit now sharpens that warning.

What fails is not only the old quartic compression

$$(u, v, \kappa),$$

but also the naive assumption that the corrected quadratic residual layer is directly encoded by the extracted orbit means.

Therefore at least one of the following must be true:

1. the variables $(\delta_A, \delta_E, \delta_O)$ are not the canonically normalized orbit amplitudes used by the linear extractor;
2. the local quadratic residual layer is not diagonal in the coarse orbit basis;
3. the $\{1, P_2\}$ shell truncation is insufficient for the corrected branch, and the remainder

$$h_\alpha^{\text{rem}}$$

is quantitatively important;

4. the residual reduction around the corrected branch involves a nontrivial Jacobian between extraction coordinates and correction coordinates.

XL. 5. MINIMAL HONEST COARSE QUADRATIC LAYER

The failure of both $q_X = \sigma_X$ and the (σ_0, η) -map forces the following conclusion. The minimal honest linearized reduced residual layer must be

$$\boxed{G_X^{(2)}(\delta) = \sum_{Y \in \{A, E, O\}} L_{XY} \delta_Y, \quad L \in M_3(\mathbb{R}),} \quad (138)$$

rather than

$$G_X^{(2)} = q_X \delta_X$$

with q_X obtained from a direct shell-average identification.

At the corrected seed,

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096),$$

the local balance must satisfy

$$\boxed{L \delta^{\text{final}} + \begin{pmatrix} a_A Q_A \\ a_E Q_E \\ a_O Q_O \end{pmatrix} = G(\delta^{\text{final}}) + R_{\geq 5},} \quad (139)$$

where

$$\left\| \begin{pmatrix} a_A Q_A \\ a_E Q_E \\ a_O Q_O \end{pmatrix} \right\| \approx 0.252190135277701$$

and

$$\|G(\delta^{\text{final}})\| \approx 0.03791852545979427.$$

Thus the linear layer must cancel most of the quartic vector locally. This is an intrinsically matrix-valued statement, not a scalar orbit-mean statement.

XLI. 6. EXACT NEXT EXECUTABLE OBJECT: THE RESIDUAL JACOBIAN

Since the correct reduced object is the residual map itself, the next exact finite object is

$$\boxed{J_{XY} := \left. \frac{\partial G_X}{\partial \delta_Y} \right|_{\delta = \delta^{\text{final}}}, \quad X, Y \in \{A, E, O\}.} \quad (140)$$

This is the actual quadratic normalization map.

A centered finite-difference realization is

$$J_{XY} \approx \frac{G_X(\delta^{\text{final}} + \varepsilon e_Y) - G_X(\delta^{\text{final}} - \varepsilon e_Y)}{2\varepsilon}, \quad (141)$$

where e_Y is the unit vector in the Y -direction and ε is chosen inside the geometry-preserving local window. Once J is known, the corrected local reduced model becomes

$$G(\delta) = G(\delta^{\text{final}}) + J(\delta - \delta^{\text{final}}) + G^{(4)}(\delta) - G^{(4)}(\delta^{\text{final}}) + \dots \quad (142)$$

This is the first reduced model that is guaranteed to live on the correct residual object.

XLII. FINAL SUMMARY

The corrected branch data force the following sharp conclusion.
The direct identification

$$q_X = \sigma_X$$

fails strongly.

Even the more flexible two-parameter shell-linear map built from

$$(\sigma_0, \eta)$$

fails: after fitting q_A^* and q_E^* , it predicts

$$q_O^{\text{pred}} = -8.304711635731334,$$

whereas the corrected seed requires

$$q_O^* = -2.5803917350227796.$$

Therefore the quadratic residual layer is not encoded by the extracted shell means through any naive direct map.
The minimal honest conclusion is:

the corrected coarse quadratic layer is matrix-valued and must be extracted as the local Jacobian of the residual map itself.

(143)

So the next exact reduced object is

$$J_{XY} = \partial_{\delta_Y} G_X|_{\delta^{\text{final}}},$$

not another compression of $(\sigma_A, \sigma_E, \sigma_O)$.

XLIII. FINAL SUMMARY

Current status: the quartic four-class closure is in place, and the next obstruction is now the quadratic normalization map.
--

(144)

The corrected branch is fixed by

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096), \quad \|G(\delta^{\text{final}})\| \approx 0.03791852545979427.$$

The corrected extractor gives

$$\sigma_0 = -109.41971030446824, \quad \eta = -6.502583100163014,$$

$$\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933,$$

and the exact four-class quartic data

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813.$$

The corrected seed requires the coarse quadratic coefficients

$$q_A^* = -8.4946715734562584, \quad q_E^* = -8.2413916564896905, \quad q_O^* = -2.5803917350227796.$$

However, the direct map

$$q_X = \sigma_X$$

fails strongly, and even a general two-parameter shell-linear map based only on

$$(\sigma_0, \eta)$$

fails. So the quadratic residual layer is not captured by a naive shell-average identification.

Therefore the next honest reduced object is the local residual Jacobian

$$J_{XY} = \left. \frac{\partial G_X}{\partial \delta_Y} \right|_{\delta=\delta^{\text{final}}}.$$

This Jacobian is the actual normalization map between the corrected local coordinates

$$(\delta_A, \delta_E, \delta_O)$$

and the true quadratic residual layer.

$$\boxed{\text{Goal C: close the corrected residual branch by extracting the local quadratic Jacobian.}} \quad (145)$$

$$\boxed{\text{Goal M: replace the failed identifications } q_X = \sigma_X \text{ and } q_X = q_X(\sigma_0, \eta) \text{ by } J_{XY}.} \quad (146)$$

$$\boxed{\text{Goal S: evaluate } G(\delta^{\text{final}} \pm \epsilon e_Y) \text{ and build the centered-difference Jacobian.}} \quad (147)$$

XLIV. LOCAL RESIDUAL JACOBIAN AUDIT AT THE CORRECTED SEED

XLV. SHORT EXPLANATION

The corrected reduced object is the projected residual map

$$G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix}, \quad \delta = (\delta_A, \delta_E, \delta_O), \quad (148)$$

with frozen corrected seed

$$\boxed{\delta^{\text{final}} = (0.0096, -0.0288, 0.0096).} \quad (149)$$

The next exact local object is therefore not another shell-average coefficient, but the Jacobian

$$\boxed{J_{XY} := \left. \frac{\partial G_X}{\partial \delta_Y} \right|_{\delta=\delta^{\text{final}}}, \quad X, Y \in \{A, E, O\}.} \quad (150)$$

What can already be closed exactly is the quartic contribution $J^{(4)}$. What remains to be extracted numerically is the full residual Jacobian J itself.

XLVI. 1. EXACT QUARTIC REDUCED RESIDUAL AND ITS JACOBIAN

From the corrected four-class coarse closure, the normalized quartic residual model is

$$G_X^{(4)}(a_A, a_E, a_O) = a_X \sum_{Y \in \{A, E, O\}} \bar{M}_{XY} r_Y, \quad r_Y := a_Y^2, \quad (151)$$

with

$$\bar{M} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}. \quad (152)$$

Define

$$Q_X(a) := \sum_Y \bar{M}_{XY} a_Y^2. \quad (153)$$

Then

$$G_X^{(4)}(a) = a_X Q_X(a). \quad (154)$$

Therefore the exact quartic Jacobian is

$$J_{XZ}^{(4)}(a) := \frac{\partial G_X^{(4)}}{\partial a_Z} = \frac{\partial}{\partial a_Z} \left(a_X \sum_Y \bar{M}_{XY} a_Y^2 \right) \quad (155)$$

$$= \delta_{XZ} \sum_Y \bar{M}_{XY} a_Y^2 + 2a_X \bar{M}_{XZ} a_Z. \quad (156)$$

Equivalently,

$$J_{XZ}^{(4)}(a) = \delta_{XZ} Q_X(a) + 2a_X \bar{M}_{XZ} a_Z. \quad (157)$$

XLVII. 2. EXACT EVALUATION AT THE CORRECTED SEED

Insert

$$a_A = 0.0096, \quad a_E = -0.0288, \quad a_O = 0.0096.$$

Then

$$r_A = r_O = \rho, \quad r_E = 9\rho, \quad \rho = (0.0096)^2 = 9.216 \times 10^{-5}. \quad (158)$$

The quartic brackets become

$$Q_A = 8.4946715734562584, \quad Q_E = 8.2413916564896905, \quad Q_O = 2.5803917350227796. \quad (159)$$

Hence the quartic residual vector at the corrected seed is

$$G^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}, \quad (160)$$

with norm

$$\|G^{(4)}(\delta^{\text{final}})\| \approx 0.252190135277701. \quad (161)$$

Now use (157). One obtains

$$J^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} 10.095240102467715 & -4.991665534361308 & 0.41377801570768584 \\ -4.991665534361308 & 22.64650843997611 & -1.2413340471230576 \\ 0.41377801570768584 & -1.2413340471230576 & 3.6033950146440785 \end{pmatrix}. \quad (162)$$

This matrix is fully explicit and already theorem-level closed at the reduced coarse four-class level.

XLVIII. 3. IMPORTANT STRUCTURAL CONSEQUENCE

Although the coarse quartic matrix $\bar{M}$ is exactly $A \leftrightarrow E$ symmetric, its local Jacobian at the corrected asymmetric seed is not.

Indeed,

$$\boxed{J_{AA}^{(4)} \neq J_{EE}^{(4)}, \quad J_{AO}^{(4)} \neq J_{EO}^{(4)},} \quad (163)$$

because the evaluation point itself satisfies

$$a_A \neq a_E.$$

So the correct local statement is sharper than the earlier coarse-matrix statement:

$$\boxed{\text{the quartic coefficient matrix is } A/E\text{-symmetric, but the local quartic linearization at the corrected seed is already } A/E\text{-asymmetric}} \quad (164)$$

This means that the true local reduced audit must be formulated directly at Jacobian level.

XLIX. 4. FULL LOCAL RESIDUAL TAYLOR MODEL

Write the full corrected residual map near the corrected seed as

$$\boxed{G(\delta^{\text{final}} + \Delta) = G_f + J_f \Delta + \frac{1}{2} \mathcal{H}_f[\Delta, \Delta] + O(\|\Delta\|^3),} \quad (165)$$

where

$$G_f := G(\delta^{\text{final}}), \quad J_f := DG(\delta^{\text{final}}).$$

If we split

$$G = G^{(2)} + G^{(4)} + G^{(\geq 5)},$$

then

$$\boxed{J_f = J_f^{(2)} + J_f^{(4)} + J_f^{(\geq 5)}. } \quad (166)$$

The exact part already known is

$$J_f^{(4)} = J^{(4)}(\delta^{\text{final}}).$$

The actual unknown is therefore the remainder

$$J_f^{(2)} + J_f^{(\geq 5)}.$$

L. 5. WHY THE JACOBIAN, NOT THE SHELL MEANS, IS THE CORRECT NEXT OBJECT

The corrected extractor produces

$$(\sigma_0, \eta), \quad (\sigma_A, \sigma_E, \sigma_O),$$

but these were already identified as effective reduced coefficients of the current solver family, not yet a closed microscopic functional coefficient sheet.

The present local audit now sharpens that honesty condition.

The quantities

$$\sigma_A, \sigma_E, \sigma_O$$

do not determine the actual linearized corrected residual map by themselves. What determines the local reduced dynamics is

$$J_f = \left. \frac{\partial G}{\partial \delta} \right|_{\delta=\delta^{\text{final}}}. \quad (167)$$

Therefore the correct reduced local Newton/Gauss–Newton object is J_f , not a diagonal or orbitwise scalar approximation built from shell means.

LI. 6. COORDINATE-BASIS CENTERED-DIFFERENCE EXTRACTION

The first exact executable realization is the centered-difference stencil in the coordinate basis

$$e_A = (1, 0, 0), \quad e_E = (0, 1, 0), \quad e_O = (0, 0, 1).$$

For a sufficiently small admissible $\varepsilon > 0$,

$$J_{.Y} \approx \frac{G(\delta^{\text{final}} + \varepsilon e_Y) - G(\delta^{\text{final}} - \varepsilon e_Y)}{2\varepsilon}, \quad Y \in \{A, E, O\}. \quad (168)$$

This requires six residual evaluations around the frozen corrected seed.

LII. 7. GEOMETRY-ADAPTED JACOBIAN AUDIT

Because the admissible local valley is extremely narrow and follows the sign pattern

$$(+, -, +),$$

the more honest basis is the orthonormal adapted basis

$$t = \frac{1}{\sqrt{11}}(1, -3, 1), \quad n_1 = \frac{1}{\sqrt{2}}(1, 0, -1), \quad n_2 = \frac{1}{\sqrt{22}}(3, 2, 3). \quad (169)$$

Here t is the corrected-branch tangent direction, while n_1, n_2 span the local transverse plane. Then define the adapted Jacobian columns by

$$\tilde{J}_{.k} \approx \frac{G(\delta^{\text{final}} + \varepsilon b_k) - G(\delta^{\text{final}} - \varepsilon b_k)}{2\varepsilon}, \quad b_1 = t, \quad b_2 = n_1, \quad b_3 = n_2. \quad (170)$$

If

$$B = (t \mid n_1 \mid n_2)$$

is the orthogonal basis matrix, then

$$J \approx \tilde{J} B^T. \quad (171)$$

a. Why this is better. The coordinate-basis stencil (168) is mathematically correct, but the adapted stencil (170) resolves directly:

1. the longitudinal corrected-branch response Jt ,
2. the two geometry-breaking transverse responses Jn_1 and Jn_2 .

This is the correct local certificate if the admissible corrected valley is truly narrow.

LIII. 8. ADMISSIBILITY PROTOCOL FOR THE FINITE-DIFFERENCE RADIUS

The local stencil points must stay inside the geometry-admissible domain. Operationally, each test point must satisfy the same corrected reduced constraints used in the residual-refinement program:

$$\text{shell_fraction} \geq s_\star, \quad \text{orbit_balance} \geq b_\star,$$

with operational thresholds

$$s_\star = 0.90, \quad b_\star = 0.88.$$

Therefore ε must be chosen by backtracking:

$$\boxed{\varepsilon \downarrow \varepsilon/2 \quad \text{until all six stencil points are admissible.}} \quad (172)$$

This is not a numerical convenience. It is required because the corrected reduced object is only meaningful inside the geometry-preserving region.

LIV. 9. FIRST LOCAL CONSISTENCY CHECK

Once the full J_f is extracted, the first local consistency test is

$$\boxed{J_f - J_f^{(4)}}. \quad (173)$$

If this difference is small, then the corrected branch is already largely controlled by the coarse four-class quartic closure.

If it is not small, then the unresolved local structure is necessarily coming from

$$J_f^{(2)} + J_f^{(\geq 5)},$$

i.e. from the quadratic normalization layer and/or higher reduced residual terms.

This gives the first honest yes/no test for whether one must immediately upgrade to a sign-resolved suborbit residual model.

LV. FINAL SUMMARY

The correct next reduced object is the full local residual Jacobian

$$J_f = \left. \frac{\partial G}{\partial \delta} \right|_{\delta=\delta^{\text{final}}}.$$

What is already closed exactly is its quartic piece:

$$\boxed{J^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} 10.095240102467715 & -4.991665534361308 & 0.41377801570768584 \\ -4.991665534361308 & 22.64650843997611 & -1.2413340471230576 \\ 0.41377801570768584 & -1.2413340471230576 & 3.6033950146440785 \end{pmatrix}}. \quad (174)$$

Thus the next executable audit is finite and explicit:

1. choose an admissible ε ,
2. evaluate G at six centered stencil points,
3. reconstruct J_f ,
4. compare J_f with the exact quartic piece $J_f^{(4)}$.

That comparison, not another shell-average compression, is the correct local closure test.

LVI. FINAL SUMMARY

Current status: the next exact reduced object is the local residual Jacobian at the frozen corrected seed.

 (175)

The corrected seed and reduced object are fixed as

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096), \quad G(\delta) = (G_A(\delta), G_E(\delta), G_O(\delta))^T.$$

The corrected extractor outputs

$$(\sigma_0, \eta) = (-109.41971030446824, -6.502583100163014),$$

$$(\sigma_A, \sigma_E, \sigma_O) = (-115.81015796064617, -105.83201242243113, -108.719022860933),$$

but these are only effective reduced coefficients of the current solver family and do not yet determine the true local linearization.

The exact quartic reduced residual is already known:

$$G_X^{(4)}(a) = a_X \sum_Y \bar{M}_{XY} a_Y^2,$$

with

$$\bar{M} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}.$$

Therefore its exact Jacobian contribution is

$$J_{XZ}^{(4)} = \delta_{XZ} Q_X + 2a_X \bar{M}_{XZ} a_Z.$$

At the corrected seed this gives

$$J^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} 10.095240102467715 & -4.991665534361308 & 0.41377801570768584 \\ -4.991665534361308 & 22.64650843997611 & -1.2413340471230576 \\ 0.41377801570768584 & -1.2413340471230576 & 3.6033950146440785 \end{pmatrix}.$$

Hence the remaining unknown is not the quartic local linearization itself, but the difference

$$J_f - J_f^{(4)}, \quad J_f = \left. \frac{\partial G}{\partial \delta} \right|_{\delta = \delta^{\text{final}}}.$$

The correct next executable audit is therefore:

$$J_{\cdot Y} \approx \frac{G(\delta^{\text{final}} + \varepsilon e_Y) - G(\delta^{\text{final}} - \varepsilon e_Y)}{2\varepsilon}, \quad Y \in \{A, E, O\},$$

or, more honestly, the adapted-basis version along

$$t = \frac{1}{\sqrt{11}}(1, -3, 1), \quad n_1 = \frac{1}{\sqrt{2}}(1, 0, -1), \quad n_2 = \frac{1}{\sqrt{22}}(3, 2, 3).$$

Goal C: close the corrected local reduced model at Jacobian level.

 (176)

Goal M: extract the full residual Jacobian J_f and compare it to the exact quartic piece $J_f^{(4)}$.

 (177)

Goal S: decide from $J_f - J_f^{(4)}$ whether coarse four-class closure is locally sufficient or whether sign-resolved suborbit refinement

 (178)

LVII. JACOBIAN-LEVEL LOCAL CERTIFICATE AT THE CORRECTED SEED

LVIII. SHORT EXPLANATION

The corrected reduced object is the projected residual map

$$G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix}, \quad \delta = (\delta_A, \delta_E, \delta_O), \quad (179)$$

with frozen corrected seed

$$\delta_f := \delta^{\text{final}} = (0.0096, -0.0288, 0.0096). \quad (180)$$

The next exact local object is therefore

$$J_f := DG(\delta_f) = \left. \frac{\partial G}{\partial \delta} \right|_{\delta=\delta_f}. \quad (181)$$

The quartic part of this Jacobian is already exactly computable from the class-resolved coarse four-class closure, and provides the natural certificate baseline for the full finite audit.

LIX. 1. EXACT QUARTIC REDUCED RESIDUAL

At the coarse normalized-orbit level,

$$G_X^{(4)}(a) = a_X \sum_{Y \in \{A, E, O\}} \bar{M}_{XY} a_Y^2, \quad a = (a_A, a_E, a_O), \quad (182)$$

with

$$\bar{M} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}. \quad (183)$$

Define

$$Q_X(a) := \sum_Y \bar{M}_{XY} a_Y^2. \quad (184)$$

Then

$$G^{(4)}(a) = \begin{pmatrix} a_A Q_A(a) \\ a_E Q_E(a) \\ a_O Q_O(a) \end{pmatrix}. \quad (185)$$

At the corrected seed

$$a_A = 0.0096, \quad a_E = -0.0288, \quad a_O = 0.0096,$$

so

$$r_A = r_O = \rho, \quad r_E = 9\rho, \quad \rho = (0.0096)^2 = 9.216 \times 10^{-5}. \quad (186)$$

Hence

$$Q_A = 8.4946715734562584, \quad Q_E = 8.2413916564896905, \quad Q_O = 2.5803917350227796. \quad (187)$$

Therefore

$$G^{(4)}(\delta_f) = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}, \quad (188)$$

with

$$\|G^{(4)}(\delta_f)\| \approx 0.252190135277701. \quad (189)$$

LX. 2. EXACT QUARTIC JACOBIAN

Differentiate

$$G_X^{(4)}(a) = a_X \sum_Y \bar{M}_{XY} a_Y^2.$$

Then

$$J_{XZ}^{(4)}(a) := \frac{\partial G_X^{(4)}}{\partial a_Z} \quad (190)$$

$$= \delta_{XZ} \sum_Y \bar{M}_{XY} a_Y^2 + 2a_X \bar{M}_{XZ} a_Z. \quad (191)$$

So

$$J_{XZ}^{(4)}(a) = \delta_{XZ} Q_X(a) + 2a_X \bar{M}_{XZ} a_Z. \quad (192)$$

At the corrected seed this becomes

$$J_f^{(4)} := J^{(4)}(\delta_f) = \begin{pmatrix} 10.095240102467715 & -4.991665534361308 & 0.41377801570768584 \\ -4.991665534361308 & 22.64650843997611 & -1.2413340471230576 \\ 0.41377801570768584 & -1.2413340471230576 & 3.6033950146440785 \end{pmatrix}. \quad (193)$$

This matrix is symmetric. Its eigenvalues are

$$\lambda_1^{(4)} \approx 3.5212959, \quad \lambda_2^{(4)} \approx 8.35219818, \quad \lambda_3^{(4)} \approx 24.47164948. \quad (194)$$

Hence

$$J_f^{(4)} > 0. \quad (195)$$

So the quartic local linearization is already coercive.

LXI. 3. EULER IDENTITY AND THE EXACT QUARTIC NEWTON STEP

Because $G^{(4)}$ is homogeneous of degree 3, Euler's theorem gives

$$J^{(4)}(a) a = 3 G^{(4)}(a). \quad (196)$$

Therefore, at the corrected seed,

$$J_f^{(4)} \delta_f = 3 G^{(4)}(\delta_f). \quad (197)$$

If one forms the quartic-only Newton correction

$$\Delta_f^{(4)} := -(J_f^{(4)})^{-1} G^{(4)}(\delta_f), \quad (198)$$

then (196) implies immediately

$$\boxed{\Delta_f^{(4)} = -\frac{1}{3} \delta_f.} \quad (199)$$

Hence

$$\boxed{\Delta_f^{(4)} = (-0.0032, 0.0096, -0.0032).} \quad (200)$$

This is an exact local certificate of the quartic model, not a numerical accident.

a. Meaning. If the full extracted Jacobian J_f and full residual vector $G_f := G(\delta_f)$ were governed primarily by the coarse quartic closure, then the full local Newton correction

$$\Delta_f := -J_f^{-1} G_f$$

should lie close to

$$-\delta_f/3.$$

Thus the deviation

$$\boxed{\mathfrak{D}_{\text{Newt}} := \left\| \Delta_f + \frac{1}{3} \delta_f \right\|} \quad (201)$$

is a sharp quantitative measure of the non-quartic local structure.

LXII. 4. FULL LOCAL TAYLOR MODEL

Write the full corrected residual map as

$$\boxed{G(\delta_f + \Delta) = G_f + J_f \Delta + \frac{1}{2} \mathcal{H}_f[\Delta, \Delta] + O(\|\Delta\|^3),} \quad (202)$$

where

$$G_f := G(\delta_f), \quad J_f := DG(\delta_f).$$

Split

$$G = G^{(2)} + G^{(4)} + G^{(\geq 5)}. \quad (203)$$

Then

$$\boxed{J_f = J_f^{(2)} + J_f^{(4)} + J_f^{(\geq 5)}.} \quad (204)$$

Since $J_f^{(4)}$ is already known exactly, the real unknown part is

$$\boxed{J_f^{\text{rem}} := J_f - J_f^{(4)} = J_f^{(2)} + J_f^{(\geq 5)}.} \quad (205)$$

So the next finite audit is to extract J_f and then inspect J_f^{rem} .

LXIII. 5. COORDINATE-BASIS CENTERED-DIFFERENCE EXTRACTION

Let

$$e_A = (1, 0, 0), \quad e_E = (0, 1, 0), \quad e_O = (0, 0, 1).$$

For an admissible step $\varepsilon > 0$, define the centered-difference estimate

$$J_{\cdot Y}^{[\varepsilon]} = \frac{G(\delta_f + \varepsilon e_Y) - G(\delta_f - \varepsilon e_Y)}{2\varepsilon}, \quad Y \in \{A, E, O\}. \quad (206)$$

This uses six residual evaluations:

$$\delta_f \pm \varepsilon e_A, \quad \delta_f \pm \varepsilon e_E, \quad \delta_f \pm \varepsilon e_O.$$

The truncation error is

$$J^{[\varepsilon]} = J_f + O(\varepsilon^2). \quad (207)$$

A Richardson-improved estimate is

$$J^{\text{Rich}} = \frac{4J^{[\varepsilon/2]} - J^{[\varepsilon]}}{3} = J_f + O(\varepsilon^4), \quad (208)$$

which requires twelve evaluations rather than six.

LXIV. 6. ADAPTED-BASIS EXTRACTION

Because the corrected local valley is narrow and organized around the sign pattern

$$(+, -, +),$$

it is more honest to use the orthonormal adapted basis

$$t = \frac{1}{\sqrt{11}}(1, -3, 1), \quad n_1 = \frac{1}{\sqrt{2}}(1, 0, -1), \quad n_2 = \frac{1}{\sqrt{22}}(3, 2, 3). \quad (209)$$

Here t is the corrected-branch tangent direction, while n_1, n_2 span the transverse plane. Define

$$B = (t \mid n_1 \mid n_2) \in O(3).$$

Then the adapted finite-difference columns are

$$\tilde{J}_{\cdot k}^{[\varepsilon]} = \frac{G(\delta_f + \varepsilon b_k) - G(\delta_f - \varepsilon b_k)}{2\varepsilon}, \quad b_1 = t, \quad b_2 = n_1, \quad b_3 = n_2. \quad (210)$$

Since B is orthogonal, the coordinate-basis Jacobian is reconstructed by

$$J_f \approx \tilde{J} B^T. \quad (211)$$

a. Why this basis matters. The adapted stencil separates:

1. the along-branch linear response $J_f t$,
2. the first transverse symmetry-breaking response $J_f n_1$,
3. the second transverse symmetry-breaking response $J_f n_2$.

This is the correct local certificate if the admissible corrected valley is truly narrow.

LXV. 7. EXACT QUARTIC JACOBIAN IN THE ADAPTED BASIS

Transform the exact quartic Jacobian:

$$\boxed{\tilde{J}_f^{(4)} := B^T J_f^{(4)} B.} \quad (212)$$

One obtains

$$\boxed{\tilde{J}_f^{(4)} = \begin{pmatrix} 23.24934224 & 3.78278610 & -3.12859268 \\ 3.78278610 & 6.43553954 & 1.80527992 \\ -3.12859268 & 1.80527992 & 6.66026178 \end{pmatrix}.} \quad (213)$$

Thus the quartic local stiffness is dominated along the corrected-branch tangent t , while the two transverse directions remain softer but still positive.

This gives a physically meaningful comparison template for the extracted full Jacobian:

$$\tilde{J}_f - \tilde{J}_f^{(4)}.$$

LXVI. 8. ADMISSIBILITY PROTOCOL FOR THE STEP SIZE

The finite-difference stencil must remain inside the geometry-preserving corrected local region. A practical admissibility rule is:

$$\boxed{\text{accept } \delta \text{ only if } \text{shell_fraction}(\delta) \geq s_*, \quad \text{orbit_balance}(\delta) \geq b_*,} \quad (214)$$

with conservative thresholds chosen slightly below the corrected-seed values. For example, one may start with

$$s_* = 0.90, \quad b_* = 0.88.$$

Then perform backtracking:

$$\boxed{\varepsilon \mapsto \varepsilon/2 \quad \text{until all stencil points are admissible.}} \quad (215)$$

This is required because the residual-based reduced object is only meaningful within the geometry-preserving local window.

LXVII. 9. LOCAL CERTIFICATE QUANTITIES

Once J_f is extracted, the minimal local certificate should report the following finite quantities:

a. (i) *Jacobian remainder*

$$\boxed{\mathfrak{D}_J := \frac{\|J_f - J_f^{(4)}\|_F}{\|J_f^{(4)}\|_F}.} \quad (216)$$

b. (ii) *Newton deviation*

$$\boxed{\mathfrak{D}_{\text{Newt}} := \frac{\| -J_f^{-1} G_f + \delta_f/3 \|}{\|\delta_f\|}.} \quad (217)$$

c. (iii) *Tangent/transverse mismatch in adapted basis* Let

$$\tilde{J}_f = B^T J_f B.$$

Then define

$$\mathfrak{D}_t := \frac{|\tilde{J}_{11} - \tilde{J}_{11}^{(4)}|}{|\tilde{J}_{11}^{(4)}|}, \quad \mathfrak{D}_\perp := \frac{\left\| \tilde{J}_{(2,3) \times (2,3)} - \tilde{J}_{(2,3) \times (2,3)}^{(4)} \right\|_F}{\left\| \tilde{J}_{(2,3) \times (2,3)}^{(4)} \right\|_F}. \quad (218)$$

These are the correct certificate-level diagnostics of whether the coarse four-class quartic closure is locally sufficient.

LXVIII. 10. HONEST PASS/FAIL LOGIC

The next local decision should be made as follows.

a. *Pass (coarse four-class locally sufficient).* If

$$\mathfrak{D}_J \ll 1, \quad \mathfrak{D}_{\text{Newt}} \ll 1,$$

and the adapted tangent/transverse mismatches are also small, then the corrected local branch is already largely controlled by the coarse four-class quartic closure.

b. *Fail (immediate refinement required).* If either

$$\mathfrak{D}_J = O(1) \quad \text{or} \quad \mathfrak{D}_{\text{Newt}} = O(1),$$

then the unresolved local structure is not a small perturbation. In that case one must move immediately to one of:

1. a nontrivial quadratic Jacobian normalization layer,
2. higher reduced residual terms $G^{(\geq 5)}$,
3. sign-resolved suborbit quartic refinement.

LXIX. FINAL SUMMARY

The exact quartic reduced residual Jacobian at the corrected seed is already known:

$$J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665534361308 & 0.41377801570768584 \\ -4.991665534361308 & 22.64650843997611 & -1.2413340471230576 \\ 0.41377801570768584 & -1.2413340471230576 & 3.6033950146440785 \end{pmatrix}. \quad (219)$$

Its eigenvalues are all positive, and its exact quartic-only Newton correction is

$$\Delta_f^{(4)} = -\frac{1}{3}\delta_f = (-0.0032, 0.0096, -0.0032). \quad (220)$$

Therefore the remaining unknown local structure is completely captured by the finite audit object

$$J_f - J_f^{(4)}, \quad (221)$$

where

$$J_f = \left. \frac{\partial G}{\partial \delta} \right|_{\delta=\delta_f}.$$

So the next executable task is finite and explicit:

$$\text{extract } J_f \text{ by six- or twelve-point centered differences and compare it directly to } J_f^{(4)}.$$

LXX. FINAL SUMMARY

Current status: the correct next local object is the full residual Jacobian $J_f = DG(\delta_f)$.

(222)

The corrected seed is frozen at

$$\delta_f = (0.0096, -0.0288, 0.0096),$$

and the reduced object is the projected residual map

$$G(\delta) = (G_A(\delta), G_E(\delta), G_O(\delta))^T.$$

The exact quartic local linearization is already known:

$$J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665534361308 & 0.41377801570768584 \\ -4.991665534361308 & 22.64650843997611 & -1.2413340471230576 \\ 0.41377801570768584 & -1.2413340471230576 & 3.6033950146440785 \end{pmatrix}.$$

It is positive definite, with eigenvalues

$$3.5212959, \quad 8.35219818, \quad 24.47164948.$$

Because the quartic residual is homogeneous of degree 3,

$$J_f^{(4)} \delta_f = 3G^{(4)}(\delta_f),$$

so the exact quartic-only Newton correction is

$\Delta_f^{(4)} = -(J_f^{(4)})^{-1}G^{(4)}(\delta_f) = -\delta_f/3.$

Hence the finite local certificate should compare the extracted full Jacobian J_f to $J_f^{(4)}$ using

$$\mathfrak{D}_J = \frac{\|J_f - J_f^{(4)}\|_F}{\|J_f^{(4)}\|_F}, \quad \mathfrak{D}_{\text{Newt}} = \frac{\| -J_f^{-1}G_f + \delta_f/3 \|}{\|\delta_f\|}.$$

If these are small, the coarse four-class quartic closure is locally sufficient. If they are order one, then the unresolved local structure is not a small perturbation and one must immediately refine the reduced model.

Goal C: extract the full local Jacobian J_f at the corrected seed.

(223)

Goal M: compare J_f against the exact quartic baseline $J_f^{(4)}$.

(224)

Goal S: decide from $J_f - J_f^{(4)}$ whether coarse four-class closure is locally sufficient or whether a higher reduced refinement is un

(225)

LXXI. POST-JACOBIAN DECISION PROTOCOL: FROM J_f TO THE NEXT EXACT REFINEMENT

LXXII. SHORT EXPLANATION

The next executable object is the local residual Jacobian

$J_f := DG(\delta_f), \quad \delta_f = (0.0096, -0.0288, 0.0096).$

(226)

But extracting J_f is not yet the end of the local audit. The true next decision is controlled by the remainder

$$\boxed{J_f^{\text{rem}} := J_f - J_f^{(4)},} \quad (227)$$

where the exact quartic baseline is already known:

$$\boxed{J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665534361308 & 0.41377801570768584 \\ -4.991665534361308 & 22.64650843997611 & -1.2413340471230576 \\ 0.41377801570768584 & -1.2413340471230576 & 3.6033950146440785 \end{pmatrix}.} \quad (228)$$

So the next exact question is:

$$\boxed{\text{Is } J_f^{\text{rem}} \text{ small, diagonal-dominant, full-matrix but coarse, or genuinely beyond the coarse } (A, E, O) \text{ basis?}} \quad (229)$$

LXXIII. 1. EXACT QUARTIC BASELINE RECAP

The quartic baseline is positive definite with eigenvalues

$$\boxed{\lambda_1^{(4)} \approx 3.5212959, \quad \lambda_2^{(4)} \approx 8.35219818, \quad \lambda_3^{(4)} \approx 24.47164948.} \quad (230)$$

Its Frobenius norm is

$$\boxed{\|J_f^{(4)}\|_F \approx 26.09636693908683.} \quad (231)$$

Because the quartic residual is cubic-homogeneous, the exact quartic-only Newton step is

$$\boxed{\Delta_f^{(4)} = -(J_f^{(4)})^{-1} G^{(4)}(\delta_f) = -\frac{1}{3} \delta_f = (-0.0032, 0.0096, -0.0032).} \quad (232)$$

Therefore the full local deviation from quartic control is measured by how much the extracted full Jacobian J_f and the actual Newton step differ from this exact quartic prediction.

LXXIV. 2. FINITE EXTRACTION OF THE FULL JACOBIAN

Let b_1, b_2, b_3 be either the coordinate basis

$$e_A, e_E, e_O$$

or the adapted basis

$$t = \frac{1}{\sqrt{11}}(1, -3, 1), \quad n_1 = \frac{1}{\sqrt{2}}(1, 0, -1), \quad n_2 = \frac{1}{\sqrt{22}}(3, 2, 3).$$

For an admissible step $\varepsilon > 0$, define the centered-difference estimate

$$\boxed{\tilde{J}_{\cdot k}^{[\varepsilon]} = \frac{G(\delta_f + \varepsilon b_k) - G(\delta_f - \varepsilon b_k)}{2\varepsilon}, \quad k = 1, 2, 3.} \quad (233)$$

If the basis is the adapted one and

$$B = (b_1 | b_2 | b_3) \in O(3),$$

then

$$\boxed{J_f \approx \tilde{J}^{[\varepsilon]} B^T.} \quad (234)$$

A Richardson-upgraded estimate is

$$\boxed{\tilde{J}^{\text{Rich}} = \frac{4\tilde{J}^{[\varepsilon/2]} - \tilde{J}^{[\varepsilon]}}{3},} \quad (235)$$

with

$$\tilde{J}^{\text{Rich}} = \tilde{J}_f + O(\varepsilon^4).$$

LXXV. 3. FIRST DECOMPOSITION OF THE REMAINDER

Once J_f is extracted, define the remainder

$$\boxed{J_f^{\text{rem}} := J_f - J_f^{(4)}}. \quad (236)$$

Now split it into symmetric and antisymmetric parts:

$$\boxed{S_f := \frac{1}{2} (J_f^{\text{rem}} + (J_f^{\text{rem}})^T), \quad A_f := \frac{1}{2} (J_f^{\text{rem}} - (J_f^{\text{rem}})^T)}. \quad (237)$$

This is the first sharp local test.

a. Interpretation.

- S_f is the true additional local linear stiffness / instability layer.
- A_f measures local non-gradient behavior of the residual map in the chosen reduced coordinates.

If A_f is numerically negligible, then the local reduced residual map is approximately gradient-like. If A_f is not negligible, then a naive local effective free-energy interpretation is still invalid even at Jacobian level.

Define the antisymmetry diagnostic

$$\boxed{\mathfrak{D}_{\text{skew}} := \frac{\|A_f\|_F}{\|J_f\|_F}}. \quad (238)$$

LXXVI. 4. BEST DIAGONAL COARSE QUADRATIC FIT

The minimal coarse quadratic correction model is diagonal:

$$\boxed{J_f \approx J_f^{(4)} + D_f, \quad D_f = \text{diag}(d_A, d_E, d_O)}. \quad (239)$$

The Frobenius-best diagonal fit is simply

$$\boxed{D_f^* := \text{diag}((J_f^{\text{rem}})_{AA}, (J_f^{\text{rem}})_{EE}, (J_f^{\text{rem}})_{OO})}. \quad (240)$$

The corresponding diagonal-fit defect is

$$\boxed{\mathfrak{D}_{\text{diag}} := \frac{\|J_f^{\text{rem}} - D_f^*\|_F}{\|J_f^{\text{rem}}\|_F}}. \quad (241)$$

a. Meaning.

- If $\mathfrak{D}_{\text{diag}} \ll 1$, then the next correction beyond the quartic baseline is essentially an orbitwise quadratic layer.
- If $\mathfrak{D}_{\text{diag}} = O(1)$, then a diagonal quadratic correction is not enough.

LXXVII. 5. BEST SYMMETRIC COARSE QUADRATIC FIT

If diagonal fitting fails, the next minimal model is a full symmetric 3×3 coarse matrix:

$$\boxed{J_f \approx J_f^{(4)} + K_f, \quad K_f = K_f^T}. \quad (242)$$

The Frobenius-best symmetric fit is

$$\boxed{K_f^* := S_f}. \quad (243)$$

Then the remaining non-symmetric defect is exactly A_f , and the symmetric-coarse-fit defect is

$$\mathfrak{D}_{\text{sym}} := \frac{\|A_f\|_F}{\|J_f^{\text{rem}}\|_F}. \quad (244)$$

If $\mathfrak{D}_{\text{sym}} \ll 1$ but $\mathfrak{D}_{\text{diag}} = O(1)$, then the corrected local branch is still describable at the coarse (A, E, O) level, but only through a full symmetric quadratic matrix, not an orbitwise scalar correction.

LXXVIII. 6. ADAPTED-BASIS BRANCH DECOMPOSITION

Now transform to the adapted basis

$$B = (t \mid n_1 \mid n_2).$$

Define

$$\tilde{J}_f := B^T J_f B, \quad \tilde{J}_f^{(4)} := B^T J_f^{(4)} B, \quad \tilde{J}_f^{\text{rem}} := \tilde{J}_f - \tilde{J}_f^{(4)}. \quad (245)$$

Then write

$$\tilde{J}_f^{\text{rem}} = \begin{pmatrix} r_{tt} & r_{t\perp_1} & r_{t\perp_2} \\ r_{\perp_1 t} & r_{\perp_1 \perp_1} & r_{\perp_1 \perp_2} \\ r_{\perp_2 t} & r_{\perp_2 \perp_1} & r_{\perp_2 \perp_2} \end{pmatrix}.$$

This gives the physically sharp local decomposition:

- r_{tt} : additional along-branch stiffness,
- $r_{t\perp_i}$: tangent/transverse mixing,
- $r_{\perp_i \perp_j}$: transverse residual structure.

Define

$$\mathfrak{D}_{t\leftrightarrow\perp} := \frac{\sqrt{|r_{t\perp_1}|^2 + |r_{t\perp_2}|^2 + |r_{\perp_1 t}|^2 + |r_{\perp_2 t}|^2}}{\|\tilde{J}_f^{\text{rem}}\|_F}, \quad (246)$$

and

$$\mathfrak{D}_{\perp\text{-}blk} := \frac{\left\| \begin{pmatrix} r_{\perp_1 \perp_1} & r_{\perp_1 \perp_2} \\ r_{\perp_2 \perp_1} & r_{\perp_2 \perp_2} \end{pmatrix} \right\|_F}{\|\tilde{J}_f^{\text{rem}}\|_F}. \quad (247)$$

a. Meaning.

- If $\mathfrak{D}_{t\leftrightarrow\perp} \ll 1$, the corrected valley decouples cleanly at linear order.
- If $\mathfrak{D}_{t\leftrightarrow\perp} = O(1)$, the coarse corrected branch is not a clean one-dimensional valley even locally.

LXXIX. 7. NEWTON-LEVEL DECISION

Let

$$G_f := G(\delta_f), \quad \Delta_f := -J_f^{-1} G_f.$$

The quartic baseline predicts

$$\Delta_f^{(4)} = -\delta_f/3.$$

Define the Newton mismatch

$$\mathfrak{D}_{\text{Newt}} := \frac{\|\Delta_f + \delta_f/3\|}{\|\delta_f\|}. \quad (248)$$

Then:

- If $\mathfrak{D}_{\text{Newt}} \ll 1$, the local corrected branch is quartic-dominated.
- If $\mathfrak{D}_{\text{Newt}} = O(1)$, then the non-quartic local structure is not perturbative.

LXXX. 8. EXACT LOCAL DECISION TREE

The next-step decision can now be made without ambiguity.

a. Case I: quartic-local sufficiency If

$$\mathfrak{D}_J = \frac{\|J_f - J_f^{(4)}\|_F}{\|J_f^{(4)}\|_F} \ll 1$$

and

$$\mathfrak{D}_{\text{Newt}} \ll 1,$$

then the coarse four-class quartic closure is locally sufficient. No further reduced refinement is needed yet.

b. Case II: diagonal quadratic correction sufficient If $\mathfrak{D}_J$ is not small but

$$\mathfrak{D}_{\text{diag}} \ll 1, \quad \mathfrak{D}_{\text{skew}} \ll 1,$$

then the next correct reduced model is

$$J_f \approx J_f^{(4)} + D_f^*.$$

That is, one adds an orbitwise quadratic normalization layer and stays in the coarse (A, E, O) basis.

c. Case III: full symmetric coarse correction required If

$$\mathfrak{D}_{\text{diag}} = O(1), \quad \mathfrak{D}_{\text{sym}} \ll 1,$$

then the correct next reduced model is

$$J_f \approx J_f^{(4)} + K_f^*, \quad K_f^* = S_f,$$

namely a full symmetric coarse 3×3 quadratic matrix.

d. Case IV: residual map remains genuinely non-gradient If

$$\mathfrak{D}_{\text{skew}} = O(1),$$

then even the corrected local reduced map is not approximately gradient-like. In that case one must keep the projected residual system itself as primary and abandon any local reduced free-energy interpretation at this level.

e. Case V: coarse orbit basis itself fails If

$$\mathfrak{D}_{t \leftrightarrow \perp} = O(1) \quad \text{or} \quad \mathfrak{D}_{\perp\text{-}blk} = O(1)$$

in a way inconsistent with the coarse (A, E, O) truncation, then the next unavoidable refinement is the sign-resolved suborbit basis

$$(A_+, A_-, E_+, E_-, O_{++}, O_{+-}, O_{-+}, O_{--}).$$

LXXXI. 9. MINIMAL EXECUTABLE PSEUDOCODE LOGIC

The full audit logic is:

$$\boxed{\delta_f \rightarrow \{G(\delta_f \pm \varepsilon b_k)\}_{k=1}^3 \rightarrow J_f \rightarrow J_f^{\text{rem}} = J_f - J_f^{(4)} \rightarrow (S_f, A_f, D_f^*, \tilde{J}_f^{\text{rem}}, \Delta_f) \rightarrow \text{decision tree.}} \quad (249)$$

Thus the next local frontier is no longer conceptual. It is a finite matrix audit with a unique pass/fail/refine logic.

LXXXII. FINAL SUMMARY

Once the full local Jacobian J_f is extracted, the only honest remaining question is how to classify

$$J_f^{\text{rem}} := J_f - J_f^{(4)}.$$

This classification is exact and finite:

- compare J_f to the quartic baseline $J_f^{(4)}$,
- test diagonal versus full symmetric versus non-gradient remainder structure,
- inspect tangent/transverse mixing in the adapted basis,
- compare the full Newton step to the exact quartic prediction $-\delta_f/3$.

This completely determines the next mathematically correct move: stay with coarse four-class closure, add a diagonal quadratic layer, add a full symmetric coarse quadratic layer, or jump to a sign-resolved suborbit refinement.

LXXXIII. FINAL SUMMARY

$$\boxed{\text{Current status: after extracting } J_f, \text{ the true local unknown is the remainder } J_f^{\text{rem}} = J_f - J_f^{(4)}.} \quad (250)$$

The corrected branch is frozen at

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta_f = (0.0096, -0.0288, 0.0096),$$

and the correct reduced object is the projected residual map

$$G(\delta) = (G_A(\delta), G_E(\delta), G_O(\delta))^T.$$

The exact quartic local baseline is already known:

$$J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665534361308 & 0.41377801570768584 \\ -4.991665534361308 & 22.64650843997611 & -1.2413340471230576 \\ 0.41377801570768584 & -1.2413340471230576 & 3.6033950146440785 \end{pmatrix},$$

with exact quartic Newton correction

$$\Delta_f^{(4)} = -\delta_f/3.$$

Therefore the next local audit is finite and unambiguous:

$$\boxed{J_f \rightarrow J_f^{\text{rem}} \rightarrow (S_f, A_f, D_f^*, \tilde{J}_f^{\text{rem}}, \Delta_f) \rightarrow \text{decision.}}$$

The decision logic is:

1. If $J_f - J_f^{(4)}$ is small and the Newton mismatch is small, the coarse four-class quartic closure is locally sufficient.
2. If the remainder is not small but is approximately diagonal and symmetric, add an orbitwise quadratic normalization layer.
3. If the remainder is symmetric but not diagonal, add a full symmetric coarse 3×3 quadratic layer.
4. If the remainder has a large antisymmetric part, keep the projected residual map as primary and do not reintroduce a local free-energy interpretation.
5. If tangent/transverse mixing is large in the adapted basis, the coarse (A, E, O) basis itself fails and one must move to the sign-resolved suborbit basis.

$$\boxed{\text{Goal C: extract } J_f \text{ and classify } J_f - J_f^{(4)}. \quad (251)}$$

$$\boxed{\text{Goal M: determine whether the next correction is diagonal, full symmetric coarse, or genuinely non-gradient.} \quad (252)}$$

$$\boxed{\text{Goal S: decide whether the coarse } (A, E, O) \text{ basis still survives locally or whether sign-resolved suborbit refinement is unavoidable.} \quad (253)}$$

LXXXIV. QUANTITATIVE REMAINDER THEOREM AROUND THE CORRECTED SEED

LXXXV. SETUP

The corrected residual stage is frozen at

$$\boxed{(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta_f = (0.0096, -0.0288, 0.0096). \quad (254)}$$

The correct reduced object is the projected residual map

$$\boxed{G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix}, \quad (255)}$$

and the corrected diagnostics give

$$\boxed{\|G(\delta_f)\| \approx 0.03791852545979427. \quad (256)}$$

The corrected extractor outputs

$$\boxed{\sigma_0 = -109.41971030446824, \quad \eta = -6.502583100163014, \quad (257)}$$

$$\boxed{\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933, \quad (258)}$$

and

$$\boxed{U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813. \quad (259)}$$

These quantities must be interpreted honestly as residual-based effective reduced coefficients of the current solver family, not yet theorem-level coefficients of a fully closed microscopic energy functional.

LXXXVI. 1. EXACT QUARTIC BASELINE OBJECTS

From the coarse four-class closure, the quartic residual vector at the corrected seed is

$$G_f^{(4)} = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}, \quad \|G_f^{(4)}\| \approx 0.2521901352777009. \quad (260)$$

The exact quartic Jacobian at the corrected seed is

$$J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665534361308 & 0.41377801570768584 \\ -4.991665534361308 & 22.64650843997611 & -1.2413340471230576 \\ 0.41377801570768584 & -1.2413340471230576 & 3.6033950146440785 \end{pmatrix}. \quad (261)$$

Its smallest eigenvalue is

$$\lambda_* := \lambda_{\min}(J_f^{(4)}) \approx 3.5212959. \quad (262)$$

Also,

$$\|\delta_f\| = \sqrt{(0.0096)^2 + (-0.0288)^2 + (0.0096)^2} \approx 0.03183959798741184. \quad (263)$$

The quartic-only Newton correction is exactly

$$\Delta_f^{(4)} = -(J_f^{(4)})^{-1} G_f^{(4)} = -\frac{1}{3} \delta_f = (-0.0032, 0.0096, -0.0032), \quad (264)$$

so

$$\|\Delta_f^{(4)}\| = \frac{1}{3} \|\delta_f\| \approx 0.01061319932913728. \quad (265)$$

LXXXVII. 2. EXACT DECOMPOSITION OF THE UNKNOWN LOCAL STRUCTURE

Let the full local Jacobian and full residual vector be

$$J_f := DG(\delta_f), \quad G_f := G(\delta_f). \quad (266)$$

Define the remainders

$$R_f := J_f - J_f^{(4)}, \quad E_f := G_f - G_f^{(4)}. \quad (267)$$

Then

$$J_f = J_f^{(4)} + R_f, \quad G_f = G_f^{(4)} + E_f. \quad (268)$$

This is the exact local split.

LXXXVIII. 3. STRONG NONQUARTIC FORCING ALREADY VISIBLE AT VECTOR LEVEL

We know only the norm of the full corrected residual vector:

$$\|G_f\| \approx 0.03791852545979427.$$

Therefore the triangle inequality yields

$$\left| \|G_f^{(4)}\| - \|G_f\| \right| \leq \|E_f\| \leq \|G_f^{(4)}\| + \|G_f\|. \quad (269)$$

Substituting the known norms gives

$$\boxed{0.21427160981790663 \leq \|E_f\| \leq 0.2901086607374952.} \quad (270)$$

Dividing by $\|G_f^{(4)}\| \approx 0.2521901352777009$, one gets

$$\boxed{0.8496431059127669 \leq \frac{\|E_f\|}{\|G_f^{(4)}\|} \leq 1.1503568940872333.} \quad (271)$$

Theorem LXXXVIII.1 (Strong nonquartic forcing at the corrected seed). *The nonquartic residual vector $E_f = G_f - G_f^{(4)}$ is necessarily of order one relative to the quartic baseline:*

$$\frac{\|E_f\|}{\|G_f^{(4)}\|} \geq 0.8496431059127669.$$

Hence quartic-only local sufficiency is already excluded at the residual-vector level.

Proof. Immediate from the reverse triangle inequality. $\square$

a. Interpretation. This does *not* yet prove that the Jacobian remainder R_f is large. But it proves that the full corrected residual vector is obtained only after a substantial cancellation against the quartic baseline. Therefore the next local question is not whether nonquartic structure exists, but what matrix type it has.

LXXXIX. 4. REMAINDER THEOREM FOR INVERTIBILITY

Assume

$$\boxed{\|R_f\|_2 < \lambda_*}. \quad (272)$$

Then J_f is invertible, because

$$J_f = J_f^{(4)} (I + (J_f^{(4)})^{-1} R_f),$$

and

$$\|(J_f^{(4)})^{-1} R_f\|_2 \leq \|(J_f^{(4)})^{-1}\|_2 \|R_f\|_2 = \frac{\|R_f\|_2}{\lambda_*} < 1.$$

Therefore the Neumann series converges, and

$$\boxed{\|J_f^{-1}\|_2 \leq \frac{1}{\lambda_* - \|R_f\|_2}.} \quad (273)$$

Theorem LXXXIX.1 (Quantitative invertibility criterion). *If*

$$\|R_f\|_2 < \lambda_* \approx 3.5212959,$$

then the full local Jacobian J_f is invertible and satisfies

$$\|J_f^{-1}\|_2 \leq \frac{1}{3.5212959 - \|R_f\|_2}.$$

XC. 5. EXACT NEWTON-STEP REMAINDER IDENTITY

Let

$$\boxed{\Delta_f := -J_f^{-1}G_f} \quad (274)$$

be the full local Newton correction.

Using

$$J_f = J_f^{(4)} + R_f, \quad G_f = G_f^{(4)} + E_f, \quad J_f^{(4)}\Delta_f^{(4)} = -G_f^{(4)},$$

we get

$$\Delta_f = -(J_f^{(4)} + R_f)^{-1}(G_f^{(4)} + E_f) \quad (275)$$

$$= \Delta_f^{(4)} - J_f^{-1}(R_f\Delta_f^{(4)} + E_f). \quad (276)$$

Hence the exact difference formula is

$$\boxed{\Delta_f - \Delta_f^{(4)} = -J_f^{-1}(R_f\Delta_f^{(4)} + E_f).} \quad (277)$$

Taking norms and using the invertibility bound above gives

$$\boxed{\|\Delta_f - \Delta_f^{(4)}\| \leq \frac{\|R_f\|_2 \|\Delta_f^{(4)}\| + \|E_f\|}{\lambda_* - \|R_f\|_2}.} \quad (278)$$

Since

$$\|\Delta_f^{(4)}\| = \|\delta_f\|/3,$$

this becomes

$$\boxed{\|\Delta_f + \delta_f/3\| \leq \frac{\|E_f\| + (\|R_f\|_2/3)\|\delta_f\|}{\lambda_* - \|R_f\|_2}.} \quad (279)$$

Dividing by $\|\delta_f\|$, the normalized Newton mismatch obeys

$$\boxed{\mathfrak{D}_{\text{Newt}} := \frac{\|\Delta_f + \delta_f/3\|}{\|\delta_f\|} \leq \frac{\|E_f\|/\|\delta_f\| + \|R_f\|_2/3}{\lambda_* - \|R_f\|_2}.} \quad (280)$$

XCI. 6. CONSEQUENCE OF THE KNOWN NORM DATA

Because $\|E_f\| \geq 0.21427160981790663$ and

$$\|\delta_f\| \approx 0.03183959798741184,$$

the forcing term in (280) satisfies

$$\boxed{\frac{\|E_f\|}{\|\delta_f\|} \geq 6.729308620179073.} \quad (281)$$

So even before J_f is extracted, the full Newton step can only be close to the quartic predictor $-\delta_f/3$ if the inverse J_f^{-1} and the vector $R_f\Delta_f^{(4)} + E_f$ produce a substantial directional cancellation.

In particular:

$$\boxed{\text{the strong form of quartic-local sufficiency is already ruled out.}} \quad (282)$$

What remains open is the weaker question:

$$\boxed{\text{is } R_f \text{ small and structured, or is the local nonquartic sector intrinsically matrix-large?}}$$

XCII. 7. QUANTITATIVE DECISION PARAMETERS AFTER EXTRACTING J_f

Once J_f is extracted, define

$$\mathfrak{D}_J^{(2)} := \frac{\|R_f\|_2}{\lambda_*}, \quad \mathfrak{D}_J^{(F)} := \frac{\|R_f\|_F}{\|J_f^{(4)}\|_F}. \quad (283)$$

The mathematically sharp thresholds are:

$$\mathfrak{D}_J^{(2)} < 1 \implies J_f \text{ is certainly invertible.} \quad (284)$$

A useful stronger perturbative regime is

$$\mathfrak{D}_J^{(2)} \leq 0.2 \iff \|R_f\|_2 \leq 0.70425918, \quad (285)$$

and a moderate one is

$$\mathfrak{D}_J^{(2)} \leq 0.5 \iff \|R_f\|_2 \leq 1.76064795. \quad (286)$$

If $\mathfrak{D}_J^{(2)}$ is already $O(1)$, then the quartic baseline is not the dominant local Jacobian.

XCIII. 8. WHAT THE NEXT EXTRACTION WILL ACTUALLY DECIDE

The next finite audit,

$$J_f \implies R_f = J_f - J_f^{(4)},$$

will decide between the following logically distinct possibilities:

1. R_f is small: the quartic baseline controls the local Jacobian, but not the residual vector.
2. R_f is not small but approximately diagonal: the next correct reduced model is a diagonal coarse quadratic normalization layer.
3. R_f is symmetric but not diagonal: the next correct reduced model is a full symmetric coarse 3×3 quadratic layer.
4. R_f has a large antisymmetric part: the local reduced residual map is genuinely non-gradient even at Jacobian level.
5. R_f destroys the tangent/transverse split in the adapted basis: the coarse (A, E, O) reduction itself fails locally and sign-resolved suborbit refinement is unavoidable.

XCIV. FINAL SUMMARY

The corrected residual vector already proves a strong fact:

$$\|G_f - G_f^{(4)}\| \geq 0.21427160981790663, \quad (287)$$

hence

$$\frac{\|G_f - G_f^{(4)}\|}{\|G_f^{(4)}\|} \geq 0.8496431059127669. \quad (288)$$

So quartic-only local sufficiency is already excluded at vector level.
The only remaining local unknown is the matrix remainder

$$R_f = J_f - J_f^{(4)}.$$

If

$$\|R_f\|_2 < 3.5212959,$$

then J_f is automatically invertible. Moreover the Newton-step deviation satisfies

$$\|\Delta_f + \delta_f/3\| \leq \frac{\|E_f\| + (\|R_f\|_2/3)\|\delta_f\|}{3.5212959 - \|R_f\|_2}.$$

Therefore the next exact task is uniquely fixed:

$$\text{extract } J_f, \text{ compute } R_f = J_f - J_f^{(4)}, \text{ and classify } R_f.$$

XCV. FINAL SUMMARY

$$\boxed{\text{Current status: quartic-local sufficiency is already excluded in the strong vector-level sense.}} \quad (289)$$

The corrected seed and residual map are fixed as

$$\delta_f = (0.0096, -0.0288, 0.0096), \quad G(\delta) = (G_A(\delta), G_E(\delta), G_O(\delta))^T,$$

with corrected residual norm

$$\|G(\delta_f)\| \approx 0.03791852545979427.$$

The corrected extractor outputs

$$\sigma_0 = -109.41971030446824, \quad \eta = -6.502583100163014,$$

$$\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933,$$

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813,$$

and these must be interpreted honestly as residual-based effective reduced coefficients of the current solver family, not yet as theorem-level microscopic coefficients of a fully closed energy functional. :contentReference[oaicite:2]index=2
:contentReference[oaicite:3]index=3

The exact quartic local baseline gives

$$\|G_f^{(4)}\| \approx 0.2521901352777009, \quad \lambda_{\min}(J_f^{(4)}) \approx 3.5212959.$$

Therefore the nonquartic residual vector

$$E_f := G_f - G_f^{(4)}$$

already satisfies

$$0.21427160981790663 \leq \|E_f\| \leq 0.2901086607374952,$$

so in relative size

$$0.8496431059127669 \leq \frac{\|E_f\|}{\|G_f^{(4)}\|} \leq 1.1503568940872333.$$

Hence quartic-only local sufficiency is already ruled out at vector level.
The only remaining local unknown is the Jacobian remainder

$$R_f := J_f - J_f^{(4)}.$$

If

$$\|R_f\|_2 < 3.5212959,$$

then J_f is automatically invertible, and the full Newton-step deviation from the exact quartic predictor $-\delta_f/3$ satisfies

$$\|\Delta_f + \delta_f/3\| \leq \frac{\|E_f\| + (\|R_f\|_2/3)\|\delta_f\|}{3.5212959 - \|R_f\|_2}.$$

So the next exact task is now sharply fixed:

extract J_f , compute $R_f = J_f - J_f^{(4)}$, and classify R_f as small / diagonal / symmetric / skew / coarse-basis-failing.

XCVI. QUANTITATIVE NO-GO FOR STRONG QUARTIC-LOCAL SUFFICIENCY

XCVII. SETUP

The corrected residual stage is frozen at

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta_f = (0.0096, -0.0288, 0.0096). \quad (290)$$

The correct reduced object is the projected residual map

$$G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix}, \quad (291)$$

with corrected residual norm

$$\|G(\delta_f)\| \approx 0.03791852545979427. \quad (292)$$

The corrected extractor outputs

$$\sigma_0 = -109.41971030446824, \quad \eta = -6.502583100163014, \quad (293)$$

$$\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933, \quad (294)$$

and the quartic class means

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813. \quad (295)$$

These must be interpreted honestly as residual-based effective reduced coefficients of the current solver family, not yet theorem-level coefficients of a fully closed microscopic energy functional.

XCVIII. 1. EXACT QUARTIC BASELINE OBJECTS

The coarse four-class quartic closure gives

$$G_f^{(4)} = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}, \quad \|G_f^{(4)}\| \approx 0.2521901352777009. \quad (296)$$

The exact quartic Jacobian baseline is

$$J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665534361308 & 0.41377801570768584 \\ -4.991665534361308 & 22.64650843997611 & -1.2413340471230576 \\ 0.41377801570768584 & -1.2413340471230576 & 3.6033950146440785 \end{pmatrix}. \quad (297)$$

Its smallest eigenvalue is

$$\lambda_* := \lambda_{\min}(J_f^{(4)}) \approx 3.5212959, \quad (298)$$

its operator norm is

$$\|J_f^{(4)}\|_2 \approx 24.4716494789482, \quad (299)$$

and

$$\|\delta_f\| = \sqrt{(0.0096)^2 + (-0.0288)^2 + (0.0096)^2} \approx 0.03183959798741184. \quad (300)$$

Because the quartic residual is cubic-homogeneous,

$$J_f^{(4)}\delta_f = 3G_f^{(4)},$$

so the exact quartic-only Newton correction is

$$\Delta_f^{(4)} = -(J_f^{(4)})^{-1}G_f^{(4)} = -\frac{1}{3}\delta_f = (-0.0032, 0.0096, -0.0032). \quad (301)$$

Hence

$$\|\Delta_f^{(4)}\| = \frac{1}{3}\|\delta_f\| \approx 0.01061319932913728. \quad (302)$$

XCIX. 2. EXACT DECOMPOSITION OF THE UNKNOWN LOCAL STRUCTURE

Define the full Jacobian and full corrected residual vector:

$$J_f := DG(\delta_f), \quad G_f := G(\delta_f). \quad (303)$$

Now define the remainders

$$R_f := J_f - J_f^{(4)}, \quad E_f := G_f - G_f^{(4)}. \quad (304)$$

Then

$$J_f = J_f^{(4)} + R_f, \quad G_f = G_f^{(4)} + E_f. \quad (305)$$

The local uncertainty is therefore completely encoded by the pair

$$(R_f, E_f).$$

C. 3. STRONG NONQUARTIC FORCING ALREADY VISIBLE AT VECTOR LEVEL

Because $\|G_f\|$ and $\|G_f^{(4)}\|$ are known, the reverse triangle inequality gives

$$\left| \|G_f^{(4)}\| - \|G_f\| \right| \leq \|E_f\| \leq \|G_f^{(4)}\| + \|G_f\|. \quad (306)$$

Substituting the numerical values:

$$0.21427160981790663 \leq \|E_f\| \leq 0.2901086607374952. \quad (307)$$

Dividing by $\|G_f^{(4)}\| \approx 0.2521901352777009$, we obtain

$$0.8496431059127669 \leq \frac{\|E_f\|}{\|G_f^{(4)}\|} \leq 1.1503568940872333. \quad (308)$$

Theorem C.1 (Vector-level no-go for strong quartic sufficiency). *The nonquartic residual vector E_f is necessarily order-one relative to the quartic baseline:*

$$\frac{\|E_f\|}{\|G_f^{(4)}\|} \geq 0.8496431059127669.$$

Therefore the strong form of quartic-local sufficiency is already excluded at vector level.

Proof. Immediate from the reverse triangle inequality. $\square$

a. Meaning. Even if the Jacobian remainder R_f turns out to be small, the vector remainder E_f is not. So quartic control can at best survive in the weak Jacobian-dominance sense, not in the strong full-residual sense.

CI. 4. QUANTITATIVE INVERTIBILITY CRITERION FOR THE FULL JACOBIAN

If

$$\|R_f\|_2 < \lambda_* \approx 3.5212959, \quad (309)$$

then the full Jacobian

$$J_f = J_f^{(4)} + R_f$$

is invertible, because

$$J_f = J_f^{(4)} \left(I + (J_f^{(4)})^{-1} R_f \right),$$

and

$$\|(J_f^{(4)})^{-1} R_f\|_2 \leq \frac{\|R_f\|_2}{\lambda_*} < 1.$$

Moreover,

$$\|J_f^{-1}\|_2 \leq \frac{1}{\lambda_* - \|R_f\|_2}. \quad (310)$$

So the hard numerical threshold is

$$\|R_f\|_2 < 3.5212959 \implies J_f \text{ certainly invertible.} \quad (311)$$

CII. 5. EXACT NEWTON-STEP REMAINDER IDENTITY

Define the full Newton correction

$$\boxed{\Delta_f := -J_f^{-1}G_f.} \quad (312)$$

Using

$$J_f = J_f^{(4)} + R_f, \quad G_f = G_f^{(4)} + E_f, \quad J_f^{(4)}\Delta_f^{(4)} = -G_f^{(4)},$$

one gets the exact identity

$$\boxed{\Delta_f - \Delta_f^{(4)} = -J_f^{-1}(R_f\Delta_f^{(4)} + E_f).} \quad (313)$$

So the deviation from the quartic Newton predictor is driven jointly by the matrix remainder R_f and vector remainder E_f .

CIII. 6. UPPER BOUND ON THE NEWTON MISMATCH

If $\|R_f\|_2 < \lambda_*$, then from (313),

$$\|\Delta_f - \Delta_f^{(4)}\| \leq \|J_f^{-1}\|_2 (\|R_f\|_2 \|\Delta_f^{(4)}\| + \|E_f\|), \quad (314)$$

hence

$$\boxed{\|\Delta_f + \delta_f/3\| \leq \frac{\|E_f\| + (\|R_f\|_2/3)\|\delta_f\|}{\lambda_* - \|R_f\|_2}.} \quad (315)$$

This gives the standard perturbative control inequality.

CIV. 7. LOWER BOUND ON THE NEWTON MISMATCH

A more useful statement for the present corrected branch is the lower bound.

From (313),

$$R_f\Delta_f^{(4)} + E_f = -J_f(\Delta_f - \Delta_f^{(4)}),$$

so

$$\|R_f\Delta_f^{(4)} + E_f\| \leq \|J_f\|_2 \|\Delta_f - \Delta_f^{(4)}\|.$$

Therefore

$$\|\Delta_f - \Delta_f^{(4)}\| \geq \frac{\|E_f\| - \|R_f\|_2 \|\Delta_f^{(4)}\|}{\|J_f\|_2}. \quad (316)$$

Using

$$\|J_f\|_2 \leq \|J_f^{(4)}\|_2 + \|R_f\|_2,$$

we obtain

$$\boxed{\|\Delta_f + \delta_f/3\| \geq \frac{\|E_f\| - (\|R_f\|_2/3)\|\delta_f\|}{\|J_f^{(4)}\|_2 + \|R_f\|_2},} \quad (317)$$

whenever the numerator is positive.

Dividing by $\|\delta_f\|$, this yields the normalized lower bound

$$\boxed{\mathfrak{D}_{\text{Newt}} := \frac{\|\Delta_f + \delta_f/3\|}{\|\delta_f\|} \geq \frac{\|E_f\|/\|\delta_f\| - \|R_f\|_2/3}{\|J_f^{(4)}\|_2 + \|R_f\|_2}.} \quad (318)$$

CV. 8. CONCRETE NUMERICAL CONSEQUENCES

Because

$$\|E_f\| \geq 0.21427160981790663, \quad \|\delta_f\| \approx 0.03183959798741184,$$

we have

$$\boxed{\frac{\|E_f\|}{\|\delta_f\|} \geq 6.729308620179073.} \quad (319)$$

Therefore (318) becomes

$$\boxed{\mathfrak{D}_{\text{Newt}} \geq \frac{6.729308620179073 - \|R_f\|_2/3}{24.4716494789482 + \|R_f\|_2}.} \quad (320)$$

Now evaluate this in the strong and moderate Jacobian-dominance regimes.

a. Strong quartic Jacobian dominance. If

$$\boxed{\|R_f\|_2 \leq 0.2 \lambda_* = 0.70425918,}$$

then

$$\boxed{\mathfrak{D}_{\text{Newt}} \geq 0.2579834549109606.} \quad (321)$$

b. Very strong quartic Jacobian dominance. If

$$\boxed{\|R_f\|_2 \leq 0.1 \lambda_* = 0.35212959,}$$

then

$$\boxed{\mathfrak{D}_{\text{Newt}} \geq 0.2663713855973499.} \quad (322)$$

c. Moderate quartic Jacobian dominance. If

$$\boxed{\|R_f\|_2 \leq 0.5 \lambda_* = 1.76064795,}$$

then

$$\boxed{\mathfrak{D}_{\text{Newt}} \geq 0.23417080882818148.} \quad (323)$$

Theorem CV.1 (No-go for strong Newton-level quartic sufficiency). *Even if the Jacobian remainder R_f lies in the strong quartic-dominance regime*

$$\|R_f\|_2 \leq 0.2 \lambda_*,$$

the normalized Newton mismatch still obeys

$$\mathfrak{D}_{\text{Newt}} \geq 0.2579834549109606.$$

Therefore the corrected branch cannot be strongly quartic-local even in the optimistic small- R_f regime.

Proof. Substitute $\|R_f\|_2 \leq 0.70425918$ into (318). □

CVI. 9. HONEST INTERPRETATION

The quartic baseline can still dominate the *Jacobian* while failing to dominate the *residual vector*. That is now the only viable weak notion of quartic-local sufficiency.

So after extracting J_f , the correct question is no longer

“Is the corrected branch almost purely quartic?”

because the answer is already no.

The correct question is instead

$$\boxed{\text{What matrix type does } R_f = J_f - J_f^{(4)} \text{ have?}} \quad (324)$$

Namely:

- small,
- approximately diagonal,
- symmetric but full,
- substantially antisymmetric,
- or coarse-basis-failing in the adapted tangent/transverse split.

CVII. FINAL SUMMARY

The corrected residual vector already forces a large nonquartic contribution:

$$\boxed{0.21427160981790663 \leq \|E_f\| \leq 0.2901086607374952,} \quad (325)$$

hence

$$\boxed{\frac{\|E_f\|}{\|G_f^{(4)}\|} \geq 0.8496431059127669.} \quad (326)$$

So strong quartic-local sufficiency is already excluded at vector level.

If the Jacobian remainder satisfies

$$\|R_f\|_2 < 3.5212959,$$

then J_f is certainly invertible. But even in the strong Jacobian-dominance regime

$$\|R_f\|_2 \leq 0.70425918,$$

the normalized Newton mismatch still satisfies

$$\boxed{\mathfrak{D}_{\text{Newt}} \geq 0.2579834549109606.}$$

Therefore the next exact task is sharply fixed:

$$\boxed{\text{extract } J_f, \text{ compute } R_f = J_f - J_f^{(4)}, \text{ and classify the matrix type of } R_f.}$$

CVIII. FINAL SUMMARY

$$\boxed{\text{Current status: strong quartic-local sufficiency is already ruled out before } J_f \text{ is extracted.}} \quad (327)$$

The corrected branch is frozen at

$$\delta_f = (0.0096, -0.0288, 0.0096), \quad \|G(\delta_f)\| \approx 0.03791852545979427,$$

while the exact quartic baseline satisfies

$$\|G_f^{(4)}\| \approx 0.2521901352777009.$$

Hence the nonquartic vector remainder

$$E_f := G_f - G_f^{(4)}$$

already obeys

$$\boxed{0.21427160981790663 \leq \|E_f\| \leq 0.2901086607374952,}$$

so in relative size

$$\boxed{\frac{\|E_f\|}{\|G_f^{(4)}\|} \geq 0.8496431059127669.}$$

Therefore quartic-local sufficiency fails in the strong vector-level sense.

The only remaining local unknown is the Jacobian remainder

$$R_f := J_f - J_f^{(4)}.$$

If

$$\|R_f\|_2 < 3.5212959,$$

then J_f is automatically invertible. But even in the strong Jacobian-dominance regime

$$\|R_f\|_2 \leq 0.70425918,$$

the normalized Newton mismatch must still satisfy

$$\boxed{\mathfrak{D}_{\text{Newt}} \geq 0.2579834549109606.}$$

So after extracting J_f , the correct next question is not whether the corrected branch is almost purely quartic. That is already false. The correct next question is:

$$\boxed{\text{what matrix type does } R_f = J_f - J_f^{(4)} \text{ have?}}$$

$$\boxed{\text{Goal C: extract } J_f \text{ and compute } R_f = J_f - J_f^{(4)}.} \quad (328)$$

$$\boxed{\text{Goal M: classify } R_f \text{ as small / diagonal / symmetric / skew / coarse-basis-failing.}} \quad (329)$$

$$\boxed{\text{Goal S: determine whether the next reduced refinement is a quadratic matrix layer or a sign-resolved suborbit upgrade.}} \quad (330)$$

CIX. EXACT CLASSIFICATION THEOREM FOR THE JACOBIAN REMAINDER

CX. SHORT EXPLANATION

At the corrected residual stage, the next finite object is

$$\boxed{J_f := DG(\delta_f), \quad \delta_f = (0.0096, -0.0288, 0.0096).} \quad (331)$$

The exact quartic baseline is already known:

$$\boxed{J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665534361308 & 0.41377801570768584 \\ -4.991665534361308 & 22.64650843997611 & -1.2413340471230576 \\ 0.41377801570768584 & -1.2413340471230576 & 3.6033950146440785 \end{pmatrix},} \quad (332)$$

with

$$\boxed{\lambda_* := \lambda_{\min}(J_f^{(4)}) \approx 3.5212959, \quad \|J_f^{(4)}\|_F \approx 26.09636693908683.} \quad (333)$$

The only remaining local matrix uncertainty is therefore

$$\boxed{R_f := J_f - J_f^{(4)}.} \quad (334)$$

The purpose of this section is to classify R_f exactly by orthogonal projection, so that one line of norm computations determines the next correct refinement.

CXI. 1. STRONG QUARTIC-LOCAL SUFFICIENCY IS ALREADY EXCLUDED

Let

$$G_f := G(\delta_f), \quad G_f^{(4)} := G^{(4)}(\delta_f).$$

The corrected branch data give

$$\boxed{\|G_f\| \approx 0.03791852545979427, \quad \|G_f^{(4)}\| \approx 0.2521901352777009.} \quad (335)$$

Hence for

$$E_f := G_f - G_f^{(4)},$$

the reverse triangle inequality gives

$$\boxed{0.21427160981790663 \leq \|E_f\| \leq 0.2901086607374952,} \quad (336)$$

so in relative size

$$\boxed{\frac{\|E_f\|}{\|G_f^{(4)}\|} \geq 0.8496431059127669.} \quad (337)$$

Therefore the strong form of quartic-local sufficiency is already ruled out at vector level. The only remaining question is the *matrix type* of R_f .

CXII. 2. FIRST SCALAR SIZE PARAMETER

Define the quartic-relative Jacobian size

$$\alpha_f := \frac{\|R_f\|_2}{\lambda_*}. \quad (338)$$

This is the unique hard perturbative parameter.

Theorem CXII.1 (Invertibility threshold). *If*

$$\alpha_f < 1,$$

then J_f is automatically invertible. Moreover,

$$\|J_f^{-1}\|_2 \leq \frac{1}{\lambda_*(1 - \alpha_f)}. \quad (339)$$

Proof. Write

$$J_f = J_f^{(4)} \left(I + (J_f^{(4)})^{-1} R_f \right).$$

Since

$$\|(J_f^{(4)})^{-1} R_f\|_2 \leq \frac{\|R_f\|_2}{\lambda_*} = \alpha_f < 1,$$

the Neumann series converges. □

a. Operational values. The exact numerical thresholds are

$$\alpha_f \leq 0.1 \iff \|R_f\|_2 \leq 0.35212959, \quad (340)$$

$$\alpha_f \leq 0.2 \iff \|R_f\|_2 \leq 0.70425918, \quad (341)$$

$$\alpha_f \leq 0.5 \iff \|R_f\|_2 \leq 1.76064795. \quad (342)$$

CXIII. 3. EXACT ORTHOGONAL DECOMPOSITION IN THE COARSE BASIS

Write

$$R_f = D_f + C_f + A_f, \quad (343)$$

where

$$D_f := \text{diag}(R_f), \quad (344)$$

$$C_f := \frac{1}{2}(R_f + R_f^T) - D_f, \quad (345)$$

$$A_f := \frac{1}{2}(R_f - R_f^T). \quad (346)$$

Then:

- D_f is diagonal,
- C_f is symmetric and off-diagonal,
- A_f is antisymmetric.

These three subspaces are Frobenius-orthogonal. Therefore:

$$\|R_f\|_F^2 = \|D_f\|_F^2 + \|C_f\|_F^2 + \|A_f\|_F^2. \quad (347)$$

Theorem CXIII.1 (Best-approximation theorem). D_f is the unique Frobenius-best diagonal approximation to R_f , and

$$D_f + C_f = \frac{1}{2}(R_f + R_f^T)$$

is the unique Frobenius-best symmetric approximation to R_f .

Proof. Both statements are orthogonal projection theorems onto closed linear subspaces of $M_3(\mathbb{R})$ under the Frobenius inner product. $\square$

CXIV. 4. EXACT RELATIVE DEFECTS

Define the diagonal-fit defect

$$\varepsilon_{\text{diag}} := \frac{\|R_f - D_f\|_F}{\|R_f\|_F} = \frac{\|C_f + A_f\|_F}{\|R_f\|_F} = \sqrt{\frac{\|C_f\|_F^2 + \|A_f\|_F^2}{\|R_f\|_F^2}}. \quad (348)$$

Define the symmetric-fit defect

$$\varepsilon_{\text{sym}} := \frac{\|R_f - (D_f + C_f)\|_F}{\|R_f\|_F} = \frac{\|A_f\|_F}{\|R_f\|_F}. \quad (349)$$

Define the skew fraction

$$\varepsilon_{\text{skew}} := \frac{\|A_f\|_F}{\|R_f\|_F} = \varepsilon_{\text{sym}}. \quad (350)$$

Define the pure off-diagonal symmetric fraction

$$\varepsilon_{\text{off}} := \frac{\|C_f\|_F}{\|R_f\|_F}. \quad (351)$$

By (347),

$$1 = \frac{\|D_f\|_F^2}{\|R_f\|_F^2} + \varepsilon_{\text{off}}^2 + \varepsilon_{\text{skew}}^2. \quad (352)$$

So the remainder classification is reduced to two independent relative fractions:

$$\varepsilon_{\text{off}}, \quad \varepsilon_{\text{skew}}.$$

CXV. 5. EXACT DECISION THEOREM IN THE COARSE BASIS

Let $\tau \in (0, 1)$ be the desired structural tolerance.

Theorem CXV.1 (Coarse-basis decision theorem). *For the extracted Jacobian remainder R_f :*

1. If

$$\varepsilon_{\text{diag}} \leq \tau,$$

then the best diagonal coarse correction D_f reproduces R_f with relative Frobenius error at most τ . In this case an orbitwise quadratic normalization layer is sufficient to relative accuracy τ .

2. If

$$\varepsilon_{\text{diag}} > \tau \quad \text{but} \quad \varepsilon_{\text{sym}} \leq \tau,$$

then no diagonal correction is sufficient at relative accuracy τ , but a full symmetric coarse 3×3 quadratic matrix is sufficient.

3. If

$$\varepsilon_{\text{sym}} > \tau,$$

then even the best symmetric coarse correction fails at relative accuracy τ . In this case the local reduced residual map is genuinely non-gradient at order τ , and one must keep the projected residual system itself as primary.

Proof. Immediate from the best-approximation theorem and the exact defect formulas. $\square$

CXVI. 6. ADAPTED-BASIS DECOMPOSITION

Now move to the corrected local orthonormal basis

$$\boxed{t = \frac{1}{\sqrt{11}}(1, -3, 1), \quad n_1 = \frac{1}{\sqrt{2}}(1, 0, -1), \quad n_2 = \frac{1}{\sqrt{22}}(3, 2, 3),} \quad (353)$$

and let

$$B = (t \mid n_1 \mid n_2) \in O(3).$$

Define

$$\boxed{\tilde{R}_f := B^T R_f B.} \quad (354)$$

Write

$$\tilde{R}_f = \begin{pmatrix} r_{tt} & u^T \\ v & P \end{pmatrix}, \quad u, v \in \mathbb{R}^2, \quad P \in M_2(\mathbb{R}). \quad (355)$$

Because orthogonal changes of basis preserve the Frobenius norm,

$$\boxed{\|R_f\|_F^2 = |r_{tt}|^2 + \|u\|^2 + \|v\|^2 + \|P\|_F^2.} \quad (356)$$

Define the tangent-transverse mixing fraction

$$\boxed{\varepsilon_{\text{mix}} := \frac{\sqrt{\|u\|^2 + \|v\|^2}}{\|R_f\|_F},} \quad (357)$$

and the transverse-block fraction

$$\boxed{\varepsilon_{\perp} := \frac{\|P\|_F}{\|R_f\|_F}.} \quad (358)$$

Then

$$\boxed{1 = \frac{|r_{tt}|^2}{\|R_f\|_F^2} + \varepsilon_{\text{mix}}^2 + \varepsilon_{\perp}^2.} \quad (359)$$

CXVII. 7. EXACT SUBORBIT TRIGGER THEOREM

Theorem CXVII.1 (Adapted-basis suborbit trigger). *Fix a structural tolerance $\tau \in (0, 1)$.*

1. *If*

$$\varepsilon_{\text{mix}} \leq \tau,$$

then the corrected tangent direction t is linearly decoupled from the two transverse directions to relative accuracy τ .

2. *If*

$$\varepsilon_{\text{mix}} > \tau,$$

then the corrected branch is not cleanly one-dimensional at Jacobian level. In this case the coarse (A, E, O) branch picture is already failing locally, and sign-resolved suborbit refinement becomes the natural next upgrade.

3. *Even if $\varepsilon_{\text{mix}} \leq \tau$, a large $\varepsilon_{\perp}$ means that the transverse sector has substantial internal residual structure that cannot be ignored in the next reduced model.*

CXVIII. 8. OPERATIONAL ONE-LINE CLASSIFICATION

Choose a single tolerance τ_{cert} , for example

$$\boxed{\tau_{\text{cert}} = 0.2.} \tag{360}$$

Then the local decision is immediate from one line of numbers:

$$\alpha_f, \quad \varepsilon_{\text{diag}}, \quad \varepsilon_{\text{sym}}, \quad \varepsilon_{\text{mix}}.$$

The exact operational rule is:

$\alpha_f \geq 1$	$\implies$ quartic baseline no longer controls invertibility;
$\alpha_f < 1, \varepsilon_{\text{diag}} \leq \tau_{\text{cert}}$	$\implies$ diagonal coarse quadratic layer;
$\alpha_f < 1, \varepsilon_{\text{diag}} > \tau_{\text{cert}}, \varepsilon_{\text{sym}} \leq \tau_{\text{cert}}, \varepsilon_{\text{mix}} \leq \tau_{\text{cert}}$	$\implies$ full symmetric coarse 3×3 layer;
$\alpha_f < 1, \varepsilon_{\text{sym}} > \tau_{\text{cert}}$	$\implies$ locally non-gradient residual map;
$\alpha_f < 1, \varepsilon_{\text{mix}} > \tau_{\text{cert}}$	$\implies$ coarse branch basis fails locally; move to sign-resolved suborbits.

(361)

CXIX. 9. HONEST INTERPRETATION

Because the vector-level theorem already gives

$$\frac{\|E_f\|}{\|G_f^{(4)}\|} \geq 0.8496431059127669,$$

the branch

“quartic-only local sufficiency”

is no longer logically available.

So after extracting J_f , the only honest remaining branches are:

1. quartic Jacobian dominance plus diagonal coarse correction;
2. quartic Jacobian dominance plus full symmetric coarse correction;
3. locally non-gradient residual map;
4. coarse basis failure and suborbit refinement.

CXX. FINAL SUMMARY

The next local frontier is completely finite.
Once J_f is extracted, define

$$R_f := J_f - J_f^{(4)},$$

then compute

$$\alpha_f = \frac{\|R_f\|_2}{\lambda_*}, \quad \varepsilon_{\text{diag}}, \quad \varepsilon_{\text{sym}}, \quad \varepsilon_{\text{mix}}.$$

These four numbers determine the next mathematically correct move.
Thus the next exact task is:

extract J_f , form R_f , compute $(\alpha_f, \varepsilon_{\text{diag}}, \varepsilon_{\text{sym}}, \varepsilon_{\text{mix}})$, and branch accordingly.

(362)

CXXI. FINAL SUMMARY

Current status: after J_f is extracted, the next branch is determined by an exact norm classification theorem for $R_f = J_f - J_f^{(4)}$

(363)

The corrected residual stage is frozen at

$$\delta_f = (0.0096, -0.0288, 0.0096), \quad \|G(\delta_f)\| \approx 0.03791852545979427.$$

The exact quartic baseline is

$$J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665534361308 & 0.41377801570768584 \\ -4.991665534361308 & 22.64650843997611 & -1.2413340471230576 \\ 0.41377801570768584 & -1.2413340471230576 & 3.6033950146440785 \end{pmatrix},$$

with

$$\lambda_* = \lambda_{\min}(J_f^{(4)}) \approx 3.5212959.$$

Strong quartic-local sufficiency is already excluded at vector level because

$$\frac{\|G_f - G_f^{(4)}\|}{\|G_f^{(4)}\|} \geq 0.8496431059127669.$$

Therefore the only remaining local unknown is the matrix type of

$$R_f := J_f - J_f^{(4)}.$$

The exact one-line classification parameters are

$$\alpha_f = \frac{\|R_f\|_2}{\lambda_*}, \quad \varepsilon_{\text{diag}} = \frac{\|R_f - \text{diag}(R_f)\|_F}{\|R_f\|_F},$$

$$\varepsilon_{\text{sym}} = \frac{\left\| R_f - \frac{1}{2}(R_f + R_f^T) \right\|_F}{\|R_f\|_F}, \quad \varepsilon_{\text{mix}} = \frac{\sqrt{\|u\|^2 + \|v\|^2}}{\|R_f\|_F} \quad \text{in the adapted basis } (t, n_1, n_2).$$

With a chosen certificate tolerance τ_{cert} , for example $\tau_{\text{cert}} = 0.2$, the exact decision is:

$$\begin{aligned}
&\alpha_f < 1, \varepsilon_{\text{diag}} \leq \tau_{\text{cert}} \implies \text{diagonal coarse quadratic layer;} \\
&\alpha_f < 1, \varepsilon_{\text{diag}} > \tau_{\text{cert}}, \varepsilon_{\text{sym}} \leq \tau_{\text{cert}}, \varepsilon_{\text{mix}} \leq \tau_{\text{cert}} \implies \text{full symmetric coarse } 3 \times 3 \text{ layer;} \\
&\alpha_f < 1, \varepsilon_{\text{sym}} > \tau_{\text{cert}} \implies \text{locally non-gradient residual map;} \\
&\alpha_f < 1, \varepsilon_{\text{mix}} > \tau_{\text{cert}} \implies \text{coarse basis failure and sign-resolved suborbit refinement.}
\end{aligned}$$

$$\boxed{\text{Goal C: extract } J_f \text{ and form } R_f = J_f - J_f^{(4)}}. \quad (364)$$

$$\boxed{\text{Goal M: compute } (\alpha_f, \varepsilon_{\text{diag}}, \varepsilon_{\text{sym}}, \varepsilon_{\text{mix}})}. \quad (365)$$

$$\boxed{\text{Goal S: choose uniquely between diagonal coarse correction, full symmetric coarse correction, non-gradient residual dynamics, a}} \quad (366)$$

CXXII. ENTRY-LEVEL ONE-LINE CLASSIFICATION THEOREM FOR THE JACOBIAN REMAINDER

CXXIII. SHORT EXPLANATION

Once the full local residual Jacobian

$$J_f = DG(\delta_f)$$

is extracted, the only remaining matrix uncertainty is

$$\boxed{R_f := J_f - J_f^{(4)}}.$$

The correct next branch is determined entirely by the matrix type of R_f . This section rewrites the previously abstract diagnostics

$$\alpha_f, \quad \varepsilon_{\text{diag}}, \quad \varepsilon_{\text{sym}}, \quad \varepsilon_{\text{mix}}$$

directly in terms of the entries of R_f , so that the next step can be decided by one line of norm calculations.

CXXIV. 1. EXACT BASELINE DATA

The corrected branch is frozen at

$$\delta_f = (0.0096, -0.0288, 0.0096), \quad \|G(\delta_f)\| \approx 0.03791852545979427,$$

and the quartic baseline Jacobian is

$$\boxed{J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665534361308 & 0.41377801570768584 \\ -4.991665534361308 & 22.64650843997611 & -1.2413340471230576 \\ 0.41377801570768584 & -1.2413340471230576 & 3.6033950146440785 \end{pmatrix}}. \quad (367)$$

Its smallest eigenvalue is

$$\boxed{\lambda_* := \lambda_{\min}(J_f^{(4)}) \approx 3.5212959}. \quad (368)$$

So for the extracted full Jacobian J_f , the quartic-relative size parameter is

$$\alpha_f = \|R_f\|_2 / \lambda_*.$$

CXXV. 2. DIRECT ENTRY FORMULAS IN THE COARSE (A, E, O) BASIS

Write

$$R_f = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{pmatrix}. \quad (369)$$

Define the total Frobenius energy

$$S_f := \|R_f\|_F^2 = r_{11}^2 + r_{22}^2 + r_{33}^2 + r_{12}^2 + r_{13}^2 + r_{21}^2 + r_{23}^2 + r_{31}^2 + r_{32}^2. \quad (370)$$

Then the diagonal fit defect is simply

$$\varepsilon_{\text{diag}}^2 = \frac{r_{12}^2 + r_{13}^2 + r_{21}^2 + r_{23}^2 + r_{31}^2 + r_{32}^2}{S_f}. \quad (371)$$

So $\varepsilon_{\text{diag}}$ is exactly the square root of the off-diagonal energy fraction.

Next, the best symmetric approximation is

$$\frac{1}{2}(R_f + R_f^T),$$

hence the symmetric-fit defect is

$$\varepsilon_{\text{sym}}^2 = \frac{(r_{12} - r_{21})^2 + (r_{13} - r_{31})^2 + (r_{23} - r_{32})^2}{2S_f}. \quad (372)$$

This is exactly the skew-energy fraction.

Likewise, the pure symmetric off-diagonal fraction is

$$\varepsilon_{\text{off}}^2 = \frac{(r_{12} + r_{21})^2 + (r_{13} + r_{31})^2 + (r_{23} + r_{32})^2}{2S_f}. \quad (373)$$

The diagonal energy fraction is

$$\varepsilon_{\text{D}}^2 = \frac{r_{11}^2 + r_{22}^2 + r_{33}^2}{S_f}. \quad (374)$$

Hence the exact identity is

$$1 = \varepsilon_{\text{D}}^2 + \varepsilon_{\text{off}}^2 + \varepsilon_{\text{sym}}^2. \quad (375)$$

CXXVI. 3. DIRECT ENTRY FORMULAS IN THE ADAPTED BASIS

Now transform to the corrected local orthonormal basis

$$t = \frac{1}{\sqrt{11}}(1, -3, 1), \quad n_1 = \frac{1}{\sqrt{2}}(1, 0, -1), \quad n_2 = \frac{1}{\sqrt{22}}(3, 2, 3),$$

and let

$$B = (t | n_1 | n_2) \in O(3).$$

Define

$$\tilde{R}_f := B^T R_f B = \begin{pmatrix} \tilde{r}_{11} & \tilde{r}_{12} & \tilde{r}_{13} \\ \tilde{r}_{21} & \tilde{r}_{22} & \tilde{r}_{23} \\ \tilde{r}_{31} & \tilde{r}_{32} & \tilde{r}_{33} \end{pmatrix}. \quad (376)$$

Because orthogonal transformations preserve the Frobenius norm,

$$\|\tilde{R}_f\|_F^2 = \|R_f\|_F^2 = S_f.$$

The tangent-transverse mixing fraction is then

$$\varepsilon_{\text{mix}}^2 = \frac{\tilde{r}_{12}^2 + \tilde{r}_{13}^2 + \tilde{r}_{21}^2 + \tilde{r}_{31}^2}{S_f}. \quad (377)$$

The pure transverse-block fraction is

$$\varepsilon_{\perp}^2 = \frac{\tilde{r}_{22}^2 + \tilde{r}_{23}^2 + \tilde{r}_{32}^2 + \tilde{r}_{33}^2}{S_f}. \quad (378)$$

The pure tangent fraction is

$$\varepsilon_t^2 = \frac{\tilde{r}_{11}^2}{S_f}. \quad (379)$$

Hence

$$1 = \varepsilon_t^2 + \varepsilon_{\text{mix}}^2 + \varepsilon_{\perp}^2. \quad (380)$$

CXXVII. 4. EXACT THRESHOLD MEANING FOR $\tau_{\text{cert}} = 0.2$

Fix the certificate tolerance

$$\tau_{\text{cert}} = 0.2. \quad (381)$$

Then the conditions become immediately interpretable:

a. Diagonal coarse correction sufficient

$$\varepsilon_{\text{diag}} \leq 0.2$$

is equivalent to

$$r_{12}^2 + r_{13}^2 + r_{21}^2 + r_{23}^2 + r_{31}^2 + r_{32}^2 \leq 0.04 S_f. \quad (382)$$

So at least 96% of the remainder energy lies on the diagonal.

b. Full symmetric coarse correction sufficient

$$\varepsilon_{\text{sym}} \leq 0.2$$

is equivalent to

$$(r_{12} - r_{21})^2 + (r_{13} - r_{31})^2 + (r_{23} - r_{32})^2 \leq 0.08 S_f. \quad (383)$$

So at least 96% of the remainder energy is symmetric.

c. Coarse branch picture survives in the adapted basis

$$\varepsilon_{\text{mix}} \leq 0.2$$

is equivalent to

$$\tilde{r}_{12}^2 + \tilde{r}_{13}^2 + \tilde{r}_{21}^2 + \tilde{r}_{31}^2 \leq 0.04 S_f. \quad (384)$$

So at least 96% of the remainder energy does not mix the branch tangent with the transverse plane.

CXXVIII. 5. ONE-LINE CLASSIFICATION THEOREM

Theorem CXXVIII.1 (One-line classification theorem). *Let*

$$R_f = J_f - J_f^{(4)}$$

be the extracted Jacobian remainder at the corrected seed, and define

$$\alpha_f = \frac{\|R_f\|_2}{\lambda_*}, \quad \varepsilon_{\text{diag}}, \quad \varepsilon_{\text{sym}}, \quad \varepsilon_{\text{mix}}$$

by (371), (372), and (377). Fix $\tau_{\text{cert}} = 0.2$. Then the next correct local refinement is determined exactly as follows:

1. *If*

$$\alpha_f \geq 1,$$

then the quartic baseline no longer controls even invertibility, so a quartic-dominance interpretation must be abandoned immediately.

2. *If*

$$\alpha_f < 1, \quad \varepsilon_{\text{sym}} > 0.2,$$

then the local reduced residual map is genuinely non-gradient at 20% accuracy, so a symmetric reduced free-energy correction is not valid.

3. *If*

$$\alpha_f < 1, \quad \varepsilon_{\text{sym}} \leq 0.2, \quad \varepsilon_{\text{mix}} > 0.2,$$

then the coarse (A, E, O) branch basis fails locally, and sign-resolved suborbit refinement is the correct next move.

4. *If*

$$\alpha_f < 1, \quad \varepsilon_{\text{sym}} \leq 0.2, \quad \varepsilon_{\text{mix}} \leq 0.2, \quad \varepsilon_{\text{diag}} \leq 0.2,$$

then an orbitwise diagonal quadratic normalization layer is sufficient.

5. *If*

$$\alpha_f < 1, \quad \varepsilon_{\text{sym}} \leq 0.2, \quad \varepsilon_{\text{mix}} \leq 0.2, \quad \varepsilon_{\text{diag}} > 0.2,$$

then the correct next model is a full symmetric coarse 3×3 quadratic layer.

Proof. The statement follows from the exact orthogonal decompositions

$$1 = \varepsilon_{\text{D}}^2 + \varepsilon_{\text{off}}^2 + \varepsilon_{\text{sym}}^2$$

in the coarse basis and

$$1 = \varepsilon_t^2 + \varepsilon_{\text{mix}}^2 + \varepsilon_{\perp}^2$$

in the adapted basis, together with the perturbative invertibility threshold $\alpha_f < 1$. □

CXXIX. 6. PRACTICAL EVALUATION ORDER

The next extraction should therefore be evaluated in the following exact order:

$$\boxed{J_f \implies R_f \implies \alpha_f \implies \varepsilon_{\text{sym}} \implies \varepsilon_{\text{mix}} \implies \varepsilon_{\text{diag}}.} \tag{385}$$

The reason for this order is logical:

1. α_f decides whether quartic invertibility control survives at all.
2. ε_{sym} decides whether a gradient-like coarse correction is even meaningful.
3. ε_{mix} decides whether the coarse branch basis survives locally.
4. $\varepsilon_{\text{diag}}$ only matters after the previous two tests pass.

CXXX. 7. HONEST INTERPRETATION

Because the vector-level theorem already gave

$$\frac{\|G_f - G_f^{(4)}\|}{\|G_f^{(4)}\|} \geq 0.8496431059127669,$$

the only viable weak notion of quartic-local sufficiency is:

quartic baseline may still dominate the Jacobian type, even though it does not dominate the corrected residual vector.

So the next local frontier is no longer “is the corrected branch almost quartic?” That is already false. The remaining local frontier is exactly:

what matrix type does R_f have?

CXXXI. FINAL SUMMARY

Once J_f is extracted, the next step is mathematically automatic.
Compute

$$R_f = J_f - J_f^{(4)},$$

then evaluate

$$\alpha_f, \quad \varepsilon_{\text{sym}}, \quad \varepsilon_{\text{mix}}, \quad \varepsilon_{\text{diag}}.$$

With $\tau_{\text{cert}} = 0.2$, this gives an exact one-line branching rule:

$$\alpha_f \geq 1 \implies \text{quartic baseline loses control;}$$

$$\alpha_f < 1, \varepsilon_{\text{sym}} > 0.2 \implies \text{non-gradient local residual map;}$$

$$\alpha_f < 1, \varepsilon_{\text{sym}} \leq 0.2, \varepsilon_{\text{mix}} > 0.2 \implies \text{suborbit refinement;}$$

$$\alpha_f < 1, \varepsilon_{\text{sym}} \leq 0.2, \varepsilon_{\text{mix}} \leq 0.2, \varepsilon_{\text{diag}} \leq 0.2 \implies \text{diagonal coarse quadratic layer;}$$

$$\alpha_f < 1, \varepsilon_{\text{sym}} \leq 0.2, \varepsilon_{\text{mix}} \leq 0.2, \varepsilon_{\text{diag}} > 0.2 \implies \text{full symmetric coarse } 3 \times 3 \text{ layer.}$$

CXXXII. FINAL SUMMARY

Current status: the next branch after extracting J_f is now fixed by an exact one-line norm classification of $R_f = J_f - J_f^{(4)}$.

(386)

The corrected stage is frozen at

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta_f = (0.0096, -0.0288, 0.0096),$$

and the correct reduced object is the projected residual map

$$G(\delta) = (G_A(\delta), G_E(\delta), G_O(\delta))^T.$$

The corrected projection coefficient sheet must still be interpreted honestly as a residual-based effective reduced coefficient set of the current solver family, not yet as theorem-level microscopic coefficients of a fully closed energy functional. :contentReference[oaicite:1]index=1

Because strong quartic-local sufficiency is already excluded at vector level, the only remaining local unknown is the matrix type of

$$R_f := J_f - J_f^{(4)}.$$

Writing

$$R_f = (r_{ij})_{i,j=1}^3, \quad S_f = \|R_f\|_F^2,$$

the exact direct formulas are

$$\alpha_f = \frac{\|R_f\|_2}{\lambda_*},$$

$$\varepsilon_{\text{diag}}^2 = \frac{r_{12}^2 + r_{13}^2 + r_{21}^2 + r_{23}^2 + r_{31}^2 + r_{32}^2}{S_f},$$

$$\varepsilon_{\text{sym}}^2 = \frac{(r_{12} - r_{21})^2 + (r_{13} - r_{31})^2 + (r_{23} - r_{32})^2}{2S_f},$$

and in the adapted basis $\tilde{R}_f = B^T R_f B$,

$$\varepsilon_{\text{mix}}^2 = \frac{\tilde{r}_{12}^2 + \tilde{r}_{13}^2 + \tilde{r}_{21}^2 + \tilde{r}_{31}^2}{S_f}.$$

With the certificate tolerance $\tau_{\text{cert}} = 0.2$, the next branch is determined exactly by:

$$\alpha_f \geq 1 \implies \text{quartic baseline loses invertibility control};$$

$$\alpha_f < 1, \varepsilon_{\text{sym}} > 0.2 \implies \text{non-gradient local residual map};$$

$$\alpha_f < 1, \varepsilon_{\text{sym}} \leq 0.2, \varepsilon_{\text{mix}} > 0.2 \implies \text{sign-resolved suborbit refinement};$$

$$\alpha_f < 1, \varepsilon_{\text{sym}} \leq 0.2, \varepsilon_{\text{mix}} \leq 0.2, \varepsilon_{\text{diag}} \leq 0.2 \implies \text{diagonal coarse quadratic layer};$$

$$\alpha_f < 1, \varepsilon_{\text{sym}} \leq 0.2, \varepsilon_{\text{mix}} \leq 0.2, \varepsilon_{\text{diag}} > 0.2 \implies \text{full symmetric coarse } 3 \times 3 \text{ quadratic layer}.$$

Goal C: extract J_f and form $R_f = J_f - J_f^{(4)}$.

(387)

Goal M: compute $(\alpha_f, \varepsilon_{\text{sym}}, \varepsilon_{\text{mix}}, \varepsilon_{\text{diag}})$.

(388)

Goal S: branch uniquely to diagonal coarse correction, full symmetric coarse correction, non-gradient residual dynamics, or sub

(389)

CXXXIII. ENTRY-ONLY NO-EIGENSOLVER CERTIFICATE FOR THE JACOBIAN REMAINDER**CXXXIV. SHORT EXPLANATION**

At the corrected residual stage, the frozen local object is

$$\delta_f = (0.0096, -0.0288, 0.0096), \quad G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix}. \quad (390)$$

The exact quartic baseline Jacobian is already known:

$$J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665534361308 & 0.41377801570768584 \\ -4.991665534361308 & 22.64650843997611 & -1.2413340471230576 \\ 0.41377801570768584 & -1.2413340471230576 & 3.6033950146440785 \end{pmatrix}, \quad (391)$$

with smallest eigenvalue

$$\lambda_* := \lambda_{\min}(J_f^{(4)}) \approx 3.5212959. \quad (392)$$

After extracting the full local Jacobian

$$J_f := DG(\delta_f),$$

the only remaining local matrix uncertainty is

$$R_f := J_f - J_f^{(4)}. \quad (393)$$

The purpose of this section is to provide a fully entry-level certificate for R_f , so that the next refinement branch can be chosen without any spectral decomposition.

CXXXV. 1. ENTRY NOTATION

Write

$$R_f = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{pmatrix}. \quad (394)$$

Define the total Frobenius energy

$$S_f := \|R_f\|_F^2 = r_{11}^2 + r_{22}^2 + r_{33}^2 + r_{12}^2 + r_{13}^2 + r_{21}^2 + r_{23}^2 + r_{31}^2 + r_{32}^2. \quad (395)$$

If $S_f = 0$, then $R_f = 0$ exactly and the local Jacobian is already equal to the quartic baseline. So from now on assume

$$S_f > 0.$$

CXXXVI. 2. A NO-EIGENSOLVER SIZE PARAMETER

The exact perturbative size parameter is

$$\alpha_f = \|R_f\|_2 / \lambda_*.$$

Since

$$\|R_f\|_2 \leq \|R_f\|_F = \sqrt{S_f},$$

we obtain the computable upper bound

$$\alpha_f \leq \hat{\alpha}_f := \frac{\sqrt{S_f}}{\lambda_*}. \quad (396)$$

This gives the exact sufficient invertibility certificate:

Theorem CXXXVI.1 (No-eigensolver invertibility certificate). *If*

$$\hat{\alpha}_f < 1, \quad (397)$$

then automatically $\alpha_f < 1$, hence J_f is invertible.

Proof. Immediate from $\|R_f\|_2 \leq \|R_f\|_F$. □

Numerically,

$$\hat{\alpha}_f < 1 \iff S_f < \lambda_*^2 \approx 12.39952282135081. \quad (398)$$

Likewise, the strong and moderate Jacobian-dominance regimes admit the entry-only sufficient conditions

$$\hat{\alpha}_f \leq 0.2 \iff S_f \leq 0.04 \lambda_*^2 \approx 0.4959809128540324, \quad (399)$$

$$\hat{\alpha}_f \leq 0.5 \iff S_f \leq 0.25 \lambda_*^2 \approx 3.0998807053377026. \quad (400)$$

CXXXVII. 3. EXACT COARSE-BASIS DECOMPOSITION

Define the diagonal part

$$D_f := \text{diag}(r_{11}, r_{22}, r_{33}). \quad (401)$$

Define the symmetric off-diagonal part

$$C_f := \frac{1}{2}(R_f + R_f^T) - D_f = \begin{pmatrix} 0 & \frac{r_{12}+r_{21}}{2} & \frac{r_{13}+r_{31}}{2} \\ \frac{r_{12}+r_{21}}{2} & 0 & \frac{r_{23}+r_{32}}{2} \\ \frac{r_{13}+r_{31}}{2} & \frac{r_{23}+r_{32}}{2} & 0 \end{pmatrix}. \quad (402)$$

Define the antisymmetric part

$$A_f := \frac{1}{2}(R_f - R_f^T) = \begin{pmatrix} 0 & \frac{r_{12}-r_{21}}{2} & \frac{r_{13}-r_{31}}{2} \\ -\frac{r_{12}-r_{21}}{2} & 0 & \frac{r_{23}-r_{32}}{2} \\ -\frac{r_{13}-r_{31}}{2} & -\frac{r_{23}-r_{32}}{2} & 0 \end{pmatrix}. \quad (403)$$

Then

$$R_f = D_f + C_f + A_f,$$

and these three subspaces are Frobenius-orthogonal.

Therefore

$$S_f = \|D_f\|_F^2 + \|C_f\|_F^2 + \|A_f\|_F^2. \quad (404)$$

CXXXVIII. 4. CLOSED ENTRY FORMULAS FOR THE CLASSIFICATION FRACTIONS

The diagonal-fit defect is

$$\varepsilon_{\text{diag}}^2 = \frac{r_{12}^2 + r_{13}^2 + r_{21}^2 + r_{23}^2 + r_{31}^2 + r_{32}^2}{S_f}. \quad (405)$$

The skew fraction is

$$\varepsilon_{\text{sym}}^2 = \frac{(r_{12} - r_{21})^2 + (r_{13} - r_{31})^2 + (r_{23} - r_{32})^2}{2S_f}. \quad (406)$$

The pure symmetric off-diagonal fraction is

$$\varepsilon_{\text{off}}^2 = \frac{(r_{12} + r_{21})^2 + (r_{13} + r_{31})^2 + (r_{23} + r_{32})^2}{2S_f}. \quad (407)$$

The diagonal energy fraction is

$$\varepsilon_D^2 = \frac{r_{11}^2 + r_{22}^2 + r_{33}^2}{S_f}. \quad (408)$$

Hence the exact partition identity is

$$1 = \varepsilon_D^2 + \varepsilon_{\text{off}}^2 + \varepsilon_{\text{sym}}^2. \quad (409)$$

a. Interpretation.

- $\varepsilon_{\text{diag}}$ measures how far R_f is from a diagonal coarse correction.
- ε_{sym} measures how far R_f is from a symmetric coarse correction.
- ε_{off} measures the genuinely symmetric off-diagonal coarse structure.

CXXXIX. 5. EXACT ADAPTED-BASIS FORMULAS

Use the orthonormal corrected local basis

$$t = \frac{1}{\sqrt{11}}(1, -3, 1), \quad n_1 = \frac{1}{\sqrt{2}}(1, 0, -1), \quad n_2 = \frac{1}{\sqrt{22}}(3, 2, 3), \quad (410)$$

and define

$$B = (t \mid n_1 \mid n_2) \in O(3).$$

Then

$$\tilde{R}_f := B^T R_f B = (\tilde{r}_{ij})_{i,j=1}^3. \quad (411)$$

The four entries relevant for tangent-transverse mixing are:

$$\tilde{r}_{12} = \frac{(r_{11} - r_{13}) - 3(r_{21} - r_{23}) + (r_{31} - r_{33})}{\sqrt{22}}, \quad (412)$$

$$\tilde{r}_{21} = \frac{(r_{11} - 3r_{12} + r_{13}) - (r_{31} - 3r_{32} + r_{33})}{\sqrt{22}}, \quad (413)$$

$$\tilde{r}_{13} = \frac{(3r_{11} + 2r_{12} + 3r_{13}) - 3(3r_{21} + 2r_{22} + 3r_{23}) + (3r_{31} + 2r_{32} + 3r_{33})}{\sqrt{242}}, \quad (414)$$

$$\tilde{r}_{31} = \frac{(3r_{11} - 9r_{12} + 3r_{13}) + 2(3r_{21} - 9r_{22} + 3r_{23}) + (3r_{31} - 9r_{32} + 3r_{33})}{\sqrt{242}}. \quad (415)$$

Therefore the tangent-transverse mixing fraction is

$$\varepsilon_{\text{mix}}^2 = \frac{\tilde{r}_{12}^2 + \tilde{r}_{13}^2 + \tilde{r}_{21}^2 + \tilde{r}_{31}^2}{S_f}. \quad (416)$$

This uses only the entries r_{ij} .

CXL. 6. THRESHOLD MEANING FOR $\tau_{\text{cert}} = 0.2$

Fix

$$\tau_{\text{cert}} = 0.2. \quad (417)$$

Then:

a. Diagonal coarse quadratic layer sufficient

$$\varepsilon_{\text{diag}} \leq 0.2$$

is equivalent to

$$r_{12}^2 + r_{13}^2 + r_{21}^2 + r_{23}^2 + r_{31}^2 + r_{32}^2 \leq 0.04 S_f. \quad (418)$$

So at least 96% of the remainder energy lies on the diagonal.

b. Full symmetric coarse 3×3 layer sufficient

$$\varepsilon_{\text{sym}} \leq 0.2$$

is equivalent to

$$(r_{12} - r_{21})^2 + (r_{13} - r_{31})^2 + (r_{23} - r_{32})^2 \leq 0.08 S_f. \quad (419)$$

So at least 96% of the remainder energy is symmetric.

c. Coarse branch basis survives locally

$$\varepsilon_{\text{mix}} \leq 0.2$$

is equivalent to

$$\tilde{r}_{12}^2 + \tilde{r}_{13}^2 + \tilde{r}_{21}^2 + \tilde{r}_{31}^2 \leq 0.04 S_f. \quad (420)$$

So at least 96% of the remainder energy does not mix the corrected tangent direction with the transverse plane.

CXLI. 7. ONE-LINE DECISION THEOREM

Theorem CXLI.1 (Entry-only one-line decision theorem). *Let J_f be the extracted local residual Jacobian at the corrected seed and define*

$$R_f := J_f - J_f^{(4)}.$$

Compute S_f , $\hat{\alpha}_f$, ε_{sym} , ε_{mix} , and $\varepsilon_{\text{diag}}$ from the entry formulas above.

Then, with $\tau_{\text{cert}} = 0.2$, the correct next reduced refinement is determined as follows:

1. If

$$\hat{\alpha}_f \geq 1,$$

then quartic baseline control of invertibility is no longer guaranteed, so quartic Jacobian dominance must be abandoned immediately.

2. If

$$\hat{\alpha}_f < 1, \quad \varepsilon_{\text{sym}} > 0.2,$$

then the local reduced residual map is genuinely non-gradient at 20% accuracy, so a symmetric reduced free-energy correction is not valid.

3. If

$$\hat{\alpha}_f < 1, \quad \varepsilon_{\text{sym}} \leq 0.2, \quad \varepsilon_{\text{mix}} > 0.2,$$

then the coarse (A, E, O) branch picture fails locally and sign-resolved suborbit refinement is the correct next move.

4. If

$$\hat{\alpha}_f < 1, \quad \varepsilon_{\text{sym}} \leq 0.2, \quad \varepsilon_{\text{mix}} \leq 0.2, \quad \varepsilon_{\text{diag}} \leq 0.2,$$

then an orbitwise diagonal quadratic normalization layer is sufficient.

5. If

$$\hat{\alpha}_f < 1, \quad \varepsilon_{\text{sym}} \leq 0.2, \quad \varepsilon_{\text{mix}} \leq 0.2, \quad \varepsilon_{\text{diag}} > 0.2,$$

then the correct next reduced model is a full symmetric coarse 3×3 quadratic layer.

Proof. All statements follow from the exact orthogonal Frobenius decomposition of R_f , the explicit adapted-basis mixing formulas, and the sufficient invertibility certificate $\hat{\alpha}_f < 1$. $\square$

CXLII. 8. PRACTICAL EVALUATION ORDER

The next extraction should therefore be evaluated in the exact order

$$\boxed{J_f \implies R_f \implies S_f \implies \hat{\alpha}_f \implies \varepsilon_{\text{sym}} \implies \varepsilon_{\text{mix}} \implies \varepsilon_{\text{diag}}.} \quad (421)$$

This order is logically minimal:

1. S_f gives the no-eigensolver size bound.
2. $\hat{\alpha}_f$ decides whether quartic invertibility control survives.
3. ε_{sym} decides whether a symmetric reduced free-energy interpretation survives.
4. ε_{mix} decides whether the coarse branch basis survives.
5. $\varepsilon_{\text{diag}}$ distinguishes diagonal versus full symmetric coarse correction.

CXLIII. FINAL SUMMARY

After extracting J_f , the next branch is no longer ambiguous.
Write

$$R_f = J_f - J_f^{(4)} = (r_{ij})_{i,j=1}^3, \quad S_f = \|R_f\|_F^2.$$

Then compute directly from the entries:

$$\hat{\alpha}_f = \frac{\sqrt{S_f}}{\lambda_*},$$

$$\varepsilon_{\text{sym}}^2 = \frac{(r_{12} - r_{21})^2 + (r_{13} - r_{31})^2 + (r_{23} - r_{32})^2}{2S_f},$$

$$\varepsilon_{\text{diag}}^2 = \frac{r_{12}^2 + r_{13}^2 + r_{21}^2 + r_{23}^2 + r_{31}^2 + r_{32}^2}{S_f},$$

and

$$\varepsilon_{\text{mix}}^2 = \frac{\tilde{r}_{12}^2 + \tilde{r}_{13}^2 + \tilde{r}_{21}^2 + \tilde{r}_{31}^2}{S_f},$$

with $\tilde{r}_{12}, \tilde{r}_{13}, \tilde{r}_{21}, \tilde{r}_{31}$ given explicitly above.

Thus the next local refinement is decided by one line of arithmetic:

$$\boxed{(\hat{\alpha}_f, \varepsilon_{\text{sym}}, \varepsilon_{\text{mix}}, \varepsilon_{\text{diag}})}.$$

CXLIV. FINAL SUMMARY

Current status: once J_f is extracted, the next branch is determined by an entry-only no-eigensolver certificate for $R_f = J_f - J$ (422)

The corrected stage is frozen at

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta_f = (0.0096, -0.0288, 0.0096),$$

and the correct reduced object remains the projected residual map

$$G(\delta) = (G_A(\delta), G_E(\delta), G_O(\delta))^T.$$

The current coefficient sheet must still be interpreted honestly as a residual-based effective reduced coefficient set of the current solver family, not yet as theorem-level coefficients of a fully closed microscopic energy functional.
:contentReference[oaicite:2]index=2

After extracting J_f , form

$$R_f = J_f - J_f^{(4)} = (r_{ij})_{i,j=1}^3, \quad S_f = \|R_f\|_F^2.$$

Then the no-eigensolver sufficient invertibility parameter is

$$\hat{\alpha}_f = \frac{\sqrt{S_f}}{\lambda_*}, \quad \lambda_* = \lambda_{\min}(J_f^{(4)}) \approx 3.5212959.$$

The exact entry-level structural fractions are

$$\varepsilon_{\text{sym}}^2 = \frac{(r_{12} - r_{21})^2 + (r_{13} - r_{31})^2 + (r_{23} - r_{32})^2}{2S_f},$$

$$\varepsilon_{\text{diag}}^2 = \frac{r_{12}^2 + r_{13}^2 + r_{21}^2 + r_{23}^2 + r_{31}^2 + r_{32}^2}{S_f},$$

and in the adapted basis,

$$\varepsilon_{\text{mix}}^2 = \frac{\tilde{r}_{12}^2 + \tilde{r}_{13}^2 + \tilde{r}_{21}^2 + \tilde{r}_{31}^2}{S_f}.$$

With $\tau_{\text{cert}} = 0.2$, the exact one-line decision is:

$$\hat{\alpha}_f \geq 1 \implies \text{quartic baseline loses invertibility control};$$

$$\hat{\alpha}_f < 1, \varepsilon_{\text{sym}} > 0.2 \implies \text{locally non-gradient residual map};$$

$$\hat{\alpha}_f < 1, \varepsilon_{\text{sym}} \leq 0.2, \varepsilon_{\text{mix}} > 0.2 \implies \text{sign-resolved suborbit refinement};$$

$$\hat{\alpha}_f < 1, \varepsilon_{\text{sym}} \leq 0.2, \varepsilon_{\text{mix}} \leq 0.2, \varepsilon_{\text{diag}} \leq 0.2 \implies \text{diagonal coarse quadratic layer};$$

$$\hat{\alpha}_f < 1, \varepsilon_{\text{sym}} \leq 0.2, \varepsilon_{\text{mix}} \leq 0.2, \varepsilon_{\text{diag}} > 0.2 \implies \text{full symmetric coarse } 3 \times 3 \text{ quadratic layer}.$$

Goal C: extract J_f and form $R_f = J_f - J_f^{(4)}$.

(423)

Goal M: compute the entry-only certificate $(\hat{\alpha}_f, \varepsilon_{\text{sym}}, \varepsilon_{\text{mix}}, \varepsilon_{\text{diag}})$.

(424)

Goal S: branch uniquely to diagonal coarse correction, full symmetric coarse correction, non-gradient local residual dynamics, or

(425)

CXLV. IMMEDIATE NEXT STEP: REFINED FOUR-VARIABLE RESIDUAL CLOSURE

CXLVI. WHY WE STOP CHASING THE UNAVAILABLE JACOBIAN

At the current corrected residual stage, the true bottleneck is no longer the carrier-side witness and no longer the failed (u, v, κ) shell compression. The files already fix the correct reduced object as the projected residual map

$$G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix},$$

and also show that Newton/damped-Newton updates are not geometry-preserving on the BCC branch. Therefore the correct next move is not to keep chasing an unavailable full Jacobian J_f , but to enlarge the residual-level orbit basis itself. :contentReference[oaicite:6]index=6

CXLVII. REFINED ORBIT SPLIT

The minimal nonredundant refinement beyond the coarse (A, E, O) split is

$$(A, E, O_+, O_-),$$

where

$$A = \{\pm(1, 1, 0)/\sqrt{2}\}, \quad E = \{\pm(1, -1, 0)/\sqrt{2}\},$$

$$O_+ = \{\pm(1, 0, 1)/\sqrt{2}, \pm(0, 1, 1)/\sqrt{2}\}, \quad O_- = \{\pm(1, 0, -1)/\sqrt{2}, \pm(0, 1, -1)/\sqrt{2}\}.$$

The group sizes are

$$n_A = 2, \quad n_E = 2, \quad n_{O_+} = 4, \quad n_{O_-} = 4.$$

CXLVIII. EXACT REFINED QUARTIC MATRIX

Using the already extracted corrected four-class shell coefficients

$$u_d = 2493.1087739398513, \quad u_{\text{opp}} = 14874.171297437068, \quad u_{\perp} = 9027.17290917827, \quad u_{\text{obl}} = 2244.8894311890813,$$

the normalized refined quartic matrix is exactly

$$\bar{M}^{(4)} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5464.26473343877 & 5636.031170183676 \\ 2244.8894311890813 & 2244.8894311890813 & 5636.031170183676 & 5464.26473343877 \end{pmatrix}.$$

CXLIX. REFINED CORRECTED SEED AND QUARTIC LOAD

Lift the corrected coarse seed to the refined symmetric point

$$x_f = (a, e, o_+, o_-) = (0.0096, -0.0288, 0.0096, 0.0096).$$

Then

$$Q^{(4)} = \bar{M}^{(4)}(a^2, e^2, o_+^2, o_-^2)^T$$

is

$$Q^{(4)} = \begin{pmatrix} 8.701560583434645 \\ 8.448280666468076 \\ 3.091893370261702 \\ 3.091893370261702 \end{pmatrix}.$$

Hence the quartic residual vector is

$$G_{(4\text{var})}^{(4)}(x_f) = x_f \odot Q^{(4)}.$$

CL. REFINED QUARTIC JACOBIAN

The quartic Jacobian in the refined four-variable model is

$$J_{IJ}^{(4,4)}(x) = \delta_{IJ} Q_I(x) + 2x_I \bar{M}_{IJ}^{(4)} x_J.$$

At the lifted corrected seed x_f , this gives

$$J_f^{(4,4)} = \begin{pmatrix} 10.302129114812741 & -4.991665531859216 & 0.4137780199567714 & 0.4137780199567714 \\ -4.991665531859216 & 22.853397448870943 & -1.2413340598703142 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 4.099066645929136 & 1.038833265288255 \\ 0.4137780199567714 & -1.2413340598703142 & 1.038833265288255 & 4.099066645929136 \end{pmatrix}.$$

The antisymmetric oblique-splitting mode

$$v_{O\text{split}} = (0, 0, 1, -1)$$

is an exact eigenvector with eigenvalue

$$\lambda_{O\text{split}}^{(4)} = 3.060233380640881 > 0.$$

So the $O_+ - O_-$ splitting channel is real and dynamically meaningful already at quartic level.

CLI. IMMEDIATE RESEARCH DECISION

Therefore the correct next step is now fixed:

$$\boxed{\text{stop chasing the unavailable } J_f, \quad \text{upgrade immediately from coarse } (A, E, O) \text{ to refined } (A, E, O_+, O_-).}$$

The next reduced object is

$$\boxed{G^{(4\text{var})}(a, e, o_+, o_-) = \begin{pmatrix} G_A \\ G_E \\ G_{O_+} \\ G_{O_-} \end{pmatrix},}$$

with quartic part

$$\boxed{G_I^{(4)}(x) = x_I \sum_J \bar{M}_{IJ}^{(4)} x_J^2.}$$

The correct solver is not Newton. It is geometry-constrained residual minimization inside the admissible BCC shell-retaining basin.

CLII. REFINED FOUR-VARIABLE QUARTIC RESIDUAL CLOSURE AND THE OBLIQUE SPLITTING MODE

CLIII. SHORT EXPLANATION

The current corrected stage is frozen at

$$\boxed{(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096).} \quad (426)$$

The correct reduced object is the projected residual map

$$\boxed{G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix},} \quad (427)$$

and the simple (u, v, κ) quartic truncation is already ruled out. The next exact enlarged quartic kernel is the four-class shell kernel

$$(u_d, u_{\text{opp}}, u_{\perp}, u_{\text{obl}}),$$

with corrected extracted values

$$\boxed{u_d = 2493.1087739398513, \quad u_{\text{opp}} = 14874.171297437068, \quad u_{\perp} = 9027.17290917827, \quad u_{\text{obl}} = 2244.8894311890813.} \quad (428)$$

This is exactly the current corrected quartic frontier. :contentReference[oaicite:2]index=2 :contentReference[oaicite:3]index=3

However, the transverse witness does not close on (A, E, O) alone, because the relevant sign-sensitive observables change sign inside the $\mathcal{E}$ and $\mathcal{O}$ orbits. Therefore the correct next minimal refinement is a sign-resolved orbit split. :contentReference[oaicite:4]index=4

CLIV. 1. MINIMAL REFINED ORBIT SPLIT

Fix the selected shell axis

$$\hat{n} = \frac{(1, 1, 0)}{\sqrt{2}}.$$

Define the refined four-group split

$$A = \left\{ \pm \frac{(1, 1, 0)}{\sqrt{2}} \right\}, \quad E = \left\{ \pm \frac{(1, -1, 0)}{\sqrt{2}} \right\}, \quad (429)$$

$$O_+ = \left\{ \pm \frac{(1, 0, 1)}{\sqrt{2}}, \pm \frac{(0, 1, 1)}{\sqrt{2}} \right\}, \quad O_- = \left\{ \pm \frac{(1, 0, -1)}{\sqrt{2}}, \pm \frac{(0, 1, -1)}{\sqrt{2}} \right\}. \quad (430)$$

Thus

$$\boxed{n_A = 2, \quad n_E = 2, \quad n_{O_+} = 4, \quad n_{O_-} = 4.} \quad (431)$$

CLV. 2. EXACT REFINED NORMALIZED QUARTIC MATRIX

Using the same four-class shell coefficients

$$(u_d, u_{\text{opp}}, u_{\perp}, u_{\text{obl}}),$$

the normalized refined quartic matrix in the basis

$$x = (a, e, o_+, o_-)^T$$

is

$$\bar{M}^{(4)} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5464.26473343877 & 5636.031170183676 \\ 2244.8894311890813 & 2244.8894311890813 & 5636.031170183676 & 5464.26473343877 \end{pmatrix}. \quad (432)$$

Therefore the refined quartic residual system is

$$\boxed{G_I^{(4)}(x) = x_I \sum_{J \in \{A, E, O_+, O_-\}} \bar{M}_{IJ}^{(4)} x_J^2, \quad I \in \{A, E, O_+, O_-\}.} \quad (433)$$

Explicitly,

$$G_A^{(4)} = a \left(8683.640035688459 a^2 + 9027.17290917827 e^2 + 2244.8894311890813 (o_+^2 + o_-^2) \right), \quad (434)$$

$$G_E^{(4)} = e \left(9027.17290917827 a^2 + 8683.640035688459 e^2 + 2244.8894311890813 (o_+^2 + o_-^2) \right), \quad (435)$$

$$G_{O_+}^{(4)} = o_+ \left(2244.8894311890813 (a^2 + e^2) + 5464.26473343877 o_+^2 + 5636.031170183676 o_-^2 \right), \quad (436)$$

$$G_{O_-}^{(4)} = o_- \left(2244.8894311890813 (a^2 + e^2) + 5636.031170183676 o_+^2 + 5464.26473343877 o_-^2 \right). \quad (437)$$

CLVI. 3. CORRECTED SYMMETRIC LIFT

Lift the corrected coarse seed to the refined symmetric point

$$\boxed{x_f = (a, e, o_+, o_-) = (0.0096, -0.0288, 0.0096, 0.0096).} \quad (438)$$

Then

$$a^2 = o_+^2 = o_-^2 = (0.0096)^2 = 9.216 \times 10^{-5}, \quad e^2 = (0.0288)^2 = 8.2944 \times 10^{-4}.$$

The quartic load vector

$$Q^{(4)} := \bar{M}^{(4)}(a^2, e^2, o_+^2, o_-^2)^T$$

is therefore

$$Q^{(4)} = \begin{pmatrix} 8.701560583434645 \\ 8.448280666468076 \\ 3.091893370261702 \\ 3.091893370261702 \end{pmatrix}. \quad (439)$$

So the refined quartic residual vector becomes

$$G_{(4\text{var})}^{(4)}(x_f) = x_f \odot Q^{(4)} = \begin{pmatrix} 0.08353498160097259 \\ -0.2433104831942806 \\ 0.02968217635451234 \\ 0.02968217635451234 \end{pmatrix}. \quad (440)$$

CLVII. 4. REFINED QUARTIC JACOBIAN

The refined quartic Jacobian is

$$J_{IJ}^{(4,4)}(x) = \delta_{IJ} Q_I(x) + 2x_I \bar{M}_{IJ}^{(4)} x_J. \quad (441)$$

At the lifted corrected seed x_f ,

$$J_f^{(4,4)} = \begin{pmatrix} 10.302129114812741 & -4.991665531859216 & 0.4137780199567714 & 0.4137780199567714 \\ -4.991665531859216 & 22.853397448870943 & -1.2413340598703142 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 4.099066645929136 & 1.038833265288255 \\ 0.4137780199567714 & -1.2413340598703142 & 1.038833265288255 & 4.099066645929136 \end{pmatrix}. \quad (442)$$

CLVIII. 5. SYMMETRIC / ANTISYMMETRIC OBLIQUE DECOMPOSITION

Introduce

$$s := \frac{o_+ + o_-}{2}, \quad d := \frac{o_+ - o_-}{2}, \quad o_+ = s + d, \quad o_- = s - d. \quad (443)$$

Then the oblique quartic loads become

$$Q_{O_+} = m_{AO}(a^2 + e^2) + (m_d + m_c)(s^2 + d^2) + 2(m_d - m_c)sd, \quad (444)$$

$$Q_{O_-} = m_{AO}(a^2 + e^2) + (m_d + m_c)(s^2 + d^2) - 2(m_d - m_c)sd, \quad (445)$$

where

$$m_{AO} = 2244.8894311890813, \quad m_d = 5464.26473343877, \quad m_c = 5636.031170183676.$$

Define the symmetric and antisymmetric residual combinations

$$G_s := \frac{G_{O_+} + G_{O_-}}{2}, \quad G_d := \frac{G_{O_+} - G_{O_-}}{2}.$$

A direct expansion gives

$$G_s = s \left(m_{AO}(a^2 + e^2) + (m_d + m_c)s^2 + (3m_d - m_c)d^2 \right), \quad (446)$$

$$G_d = d \left(m_{AO}(a^2 + e^2) + (m_d + m_c)d^2 + (3m_d - m_c)s^2 \right). \quad (447)$$

Since

$$m_d + m_c = 11100.295903622446, \quad 3m_d - m_c = 10756.763030132635,$$

both channels are strictly cubic-stiff at quartic level.

CLIX. 6. CORRECTED SYMMETRIC POINT AND THE OBLIQUE-SPLITTING MODE

At the lifted corrected seed,

$$s = 0.0096, \quad d = 0,$$

so automatically

$$\boxed{G_d(x_f) = 0.} \quad (448)$$

Moreover the linearization of the antisymmetric mode is

$$\boxed{\left. \frac{\partial G_d}{\partial d} \right|_{x_f} = m_{AO}(a^2 + e^2) + (3m_d - m_c)s^2.} \quad (449)$$

Substituting the corrected values gives

$$\boxed{\lambda_d^{(4)} = 3.060233380640881 > 0.} \quad (450)$$

Equivalently, in the 4×4 Jacobian, the vector

$$v_d = (0, 0, 1, -1)$$

is an exact eigenvector:

$$\boxed{J_f^{(4,4)} v_d = 3.060233380640881 v_d.} \quad (451)$$

So the oblique-splitting mode is real and dynamically meaningful, but it is *stabilized* at the symmetric corrected point.

CLX. 7. SHARP CONSEQUENCE

This calculation closes the quartic part of the refined orbit problem.

Theorem CLX.1 (Quartic refined no-splitting theorem at the corrected symmetric point). *In the refined four-variable quartic residual model*

$$x = (a, e, o_+, o_-),$$

the lifted corrected symmetric point

$$x_f = (0.0096, -0.0288, 0.0096, 0.0096)$$

has vanishing antisymmetric oblique residual

$$G_d(x_f) = 0,$$

and the local quartic stiffness in the oblique-splitting direction is strictly positive:

$$\lambda_d^{(4)} = 3.060233380640881 > 0.$$

Therefore the quartic refined shell kernel by itself does not drive $O_+ - O_-$ splitting away from the symmetric corrected branch.

Proof. Immediate from the explicit formula for G_d and the linearized coefficient above. □

CLXI. 8. PHYSICAL MEANING

This is the decisive next-step conclusion.

The refined quartic enlargement is *not* useless: it proves that the antisymmetric oblique mode exists and has a clean local stiffness. But it also proves that the quartic pair-class sector alone does not generate the required sign-sensitive oblique splitting.

Therefore the next nonredundant frontier is not another quartic pair-class enlargement. It is the sign-sensitive nonquartic / transverse sector already identified in the corrected-branch notes. :contentReference[oaicite:5]index=5

Equivalently:

$$\boxed{\text{quartic refined closure: now closed;}} \quad (452)$$

$$\boxed{\text{next open slot: sign-resolved transverse / nonquartic insertion.}} \quad (453)$$

CLXII. FINAL SUMMARY

The correct immediate upgrade beyond the failed coarse (A, E, O) quartic model is the refined four-variable quartic residual model

$$(A, E, O_+, O_-).$$

Its exact normalized quartic matrix is

$$\bar{M}^{(4)} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5464.26473343877 & 5636.031170183676 \\ 2244.8894311890813 & 2244.8894311890813 & 5636.031170183676 & 5464.26473343877 \end{pmatrix}.$$

At the lifted corrected seed

$$x_f = (0.0096, -0.0288, 0.0096, 0.0096),$$

the quartic residual vector is

$$G_{(4\text{var})}^{(4)}(x_f) = \begin{pmatrix} 0.08353498160097259 \\ -0.2433104831942806 \\ 0.02968217635451234 \\ 0.02968217635451234 \end{pmatrix},$$

and the quartic Jacobian is

$$J_f^{(4,4)} = \begin{pmatrix} 10.302129114812741 & -4.991665531859216 & 0.4137780199567714 & 0.4137780199567714 \\ -4.991665531859216 & 22.853397448870943 & -1.2413340598703142 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 4.099066645929136 & 1.038833265288255 \\ 0.4137780199567714 & -1.2413340598703142 & 1.038833265288255 & 4.099066645929136 \end{pmatrix}.$$

The antisymmetric oblique mode

$$v_d = (0, 0, 1, -1)$$

is an exact eigenvector with

$$\lambda_d^{(4)} = 3.060233380640881 > 0.$$

Therefore the refined quartic sector already closes the oblique-splitting channel at quartic level: the splitting mode exists, but the symmetric corrected point is locally stabilized against it.

So the next true frontier is no longer quartic pair-class compression. It is the sign-sensitive transverse / nonquartic sector.

CLXIII. FINAL SUMMARY

Current status: the quartic refined orbit upgrade (A, E, O_+, O_-) is now explicitly closed.

(454)

The corrected stage remains frozen at

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096),$$

and the correct reduced object remains the projected residual map

$$G(\delta) = (G_A(\delta), G_E(\delta), G_O(\delta))^T.$$

The current coefficient sheet must still be interpreted honestly as a residual-based effective reduced coefficient set of the current solver family, not yet as theorem-level coefficients of a fully closed microscopic energy functional.

Using the corrected four-class shell coefficients

$$u_d = 2493.1087739398513, \quad u_{\text{opp}} = 14874.171297437068, \quad u_{\perp} = 9027.17290917827, \quad u_{\text{obl}} = 2244.8894311890813,$$

the exact refined normalized quartic matrix in the basis

$$(A, E, O_+, O_-)$$

is

$$\bar{M}^{(4)} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5464.26473343877 & 5636.031170183676 \\ 2244.8894311890813 & 2244.8894311890813 & 5636.031170183676 & 5464.26473343877 \end{pmatrix}.$$

At the lifted corrected seed

$$x_f = (0.0096, -0.0288, 0.0096, 0.0096),$$

the quartic residual vector is

$$G_{(4\text{var})}^{(4)}(x_f) = \begin{pmatrix} 0.08353498160097259 \\ -0.2433104831942806 \\ 0.02968217635451234 \\ 0.02968217635451234 \end{pmatrix},$$

and the quartic Jacobian is

$$J_f^{(4,4)} = \begin{pmatrix} 10.302129114812741 & -4.991665531859216 & 0.4137780199567714 & 0.4137780199567714 \\ -4.991665531859216 & 22.853397448870943 & -1.2413340598703142 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 4.099066645929136 & 1.038833265288255 \\ 0.4137780199567714 & -1.2413340598703142 & 1.038833265288255 & 4.099066645929136 \end{pmatrix}.$$

The antisymmetric oblique splitting mode

$$v_d = (0, 0, 1, -1)$$

is an exact eigenvector with eigenvalue

$$\lambda_d^{(4)} = 3.060233380640881 > 0.$$

Therefore the refined quartic enlargement proves two things at once:

1. the oblique splitting channel is real and dynamically meaningful;
2. the symmetric corrected point is locally stabilized against pure quartic $O_+ - O_-$ splitting.

Hence the next nonredundant frontier is no longer another quartic pair-class enlargement. It is the sign-sensitive transverse / nonquartic sector already identified in the corrected-branch notes. :contentReference[oaicite:7]index=7

$$\boxed{\text{Goal C: keep the refined quartic } (A, E, O_+, O_-) \text{ closure as the new residual baseline.}} \quad (455)$$

$$\boxed{\text{Goal M: insert the sign-sensitive transverse / nonquartic response on top of this refined baseline.}} \quad (456)$$

$$\boxed{\text{Goal S: decide whether the first active symmetry-breaking correction enters through } O_+ - O_- \text{ splitting or through the transverse}} \quad (457)$$

CLXIV. SIGN-SENSITIVE TRANSVERSE / NONQUARTIC INSERTION ON TOP OF THE REFINED QUARTIC BASELINE

CLXV. SHORT EXPLANATION

The quartic refined orbit upgrade

$$(A, E, O) \rightarrow (A, E, O_+, O_-)$$

is already closed. What remains is the first sign-sensitive transverse correction on top of that quartic baseline.

The key point is that the correct reduced object is the projected residual map, not a naive reduced free-energy critical point, and the current corrected coefficients are residual-based effective reduced coefficients of the present solver family. :contentReference[oaicite:4]index=4

At the first-shell sign-resolved level, the frozen corrected seed produces no linear shift in the transverse witness rows. Therefore the odd transverse channel cannot be activated at linear order by the present orbit-symmetric corrected seed alone. :contentReference[oaicite:5]index=5 :contentReference[oaicite:6]index=6

CLXVI. 1. MINIMAL SIGN-RESOLVED FIRST-SHELL BASIS

Choose the shell axis

$$\hat{n} = \frac{(1, 1, 0)}{\sqrt{2}}, \quad e_1 = \frac{(1, -1, 0)}{\sqrt{2}}, \quad e_2 = (0, 0, 1).$$

The minimal sign-resolved first-shell vector is

$$\delta_{\text{sub}} = \begin{pmatrix} \delta_{A+} \\ \delta_{A-} \\ \delta_{E+} \\ \delta_{E-} \\ \delta_{O++} \\ \delta_{O+-} \\ \delta_{O-+} \\ \delta_{O--} \end{pmatrix}. \quad (458)$$

The frozen corrected seed is orbit-symmetric, so its lift is

$$\delta_{\text{sub}}^{\text{corr}} = \begin{pmatrix} 0.0096 \\ 0.0096 \\ -0.0288 \\ -0.0288 \\ 0.0096 \\ 0.0096 \\ 0.0096 \\ 0.0096 \end{pmatrix}. \quad (459)$$

CLXVII. 2. EXACT TRANSVERSE RESPONSE ROWS

At first-shell level the sign-sensitive transverse source depends only on the products

$$c_\alpha t_{\alpha 1}, \quad c_\alpha t_{\alpha 2}.$$

The exact suborbit response rows are

$$r_{\text{sub}}^{(ct_1)} = \left(0, 0, 0, 0, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4} \right), \quad (460)$$

$$r_{\text{sub}}^{(ct_2)} = \left(0, 0, 0, 0, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}} \right). \quad (461)$$

Now contract them with the corrected lifted seed:

$$r_{\text{sub}}^{(ct_1)} \cdot \delta_{\text{sub}}^{\text{corr}} = \frac{1}{4}(0.0096 + 0.0096 - 0.0096 - 0.0096) = 0, \quad (462)$$

$$r_{\text{sub}}^{(ct_2)} \cdot \delta_{\text{sub}}^{\text{corr}} = \frac{1}{2\sqrt{2}}(0.0096 - 0.0096 + 0.0096 - 0.0096) = 0. \quad (463)$$

Therefore

$$r_{\text{sub}}^{(ct_1)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0, \quad r_{\text{sub}}^{(ct_2)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0. \quad (464)$$

Equivalently, if

$$X_\alpha(\delta) = X_\alpha^{(0)} + \ell_\alpha \cdot \delta + \frac{1}{2}\delta^T H^{(\alpha)} \delta + O(\delta^3),$$

then on the present orbit-symmetric corrected seed,

$$\ell_\alpha \cdot \delta_{\text{corr}} = 0, \quad \alpha = 1, 2. \quad (465)$$

So the sign-sensitive transverse factor has no linear corrected shift. :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8 :contentReference[oaicite:9]index=9

CLXVIII. 3. BASE-BRANCH VERDICT ON THE EXPLICIT CANONICAL PURE-EVEN BRANCH

On the explicit canonical pure-even branch,

$$T_\star = T^1, \quad \eta = z_0, \quad \Lambda_O^{(0)} = 0, \quad C_\parallel = C_E = C_O = 1. \quad (466)$$

The reduced coefficients on this branch are

$$\lambda_\parallel = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3, \quad (467)$$

$$\alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad \beta = 0. \quad (468)$$

Hence

$$\alpha \neq 0, \quad \beta = 0,$$

as long as

$$\beta_X \neq 0, \quad \theta_0 \kappa \neq 0.$$

Because

$$\hat{T}_{(II)}^\alpha(G_*) \propto iA_{G_*}(C_\alpha \cdot G_*),$$

this means

$$\boxed{X_1^{(0)} \neq 0, \quad X_2^{(0)} = 0} \quad (469)$$

on the explicit canonical pure-even branch. :contentReference[oaicite:10]index=10

CLXIX. 4. CORRECTED TRANSVERSE EXPANSION

Write

$$A_{G_*}(\delta) = A_0 + a \cdot \delta + Q_A(\delta) + O(\delta^3), \quad (470)$$

and

$$X_\alpha(\delta) = X_\alpha^{(0)} + \ell_\alpha \cdot \delta + Q_\alpha(\delta) + O(\delta^3), \quad Q_\alpha(\delta) = \frac{1}{2} \delta^T H^{(\alpha)} \delta. \quad (471)$$

Since

$$\ell_\alpha \cdot \delta_{\text{corr}} = 0,$$

the total corrected transverse witness is

$$\hat{T}_{(II),\text{corr}}^\alpha(G_*) = i \left[A_0 X_\alpha^{(0)} + (a \cdot \delta_{\text{corr}}) X_\alpha^{(0)} + A_0 Q_\alpha(\delta_{\text{corr}}) + X_\alpha^{(0)} Q_A(\delta_{\text{corr}}) \right] + O(\delta^3). \quad (472)$$

A. Even transverse coefficient

For $\alpha = 1$, because $X_1^{(0)} \neq 0$,

$$\boxed{\alpha^{\text{corr}} = \alpha^{(0)} \left[1 + \frac{a \cdot \delta_{\text{corr}}}{A_0} + \frac{Q_A(\delta_{\text{corr}})}{A_0} + \frac{Q_1(\delta_{\text{corr}})}{X_1^{(0)}} \right] + O(\delta^3).} \quad (473)$$

So α^{corr} generally receives a linear corrected shift through the selected harmonic A_{G_*} . :contentReference[oaicite:11]index=11
:contentReference[oaicite:12]index=12

B. Odd transverse coefficient

For $\alpha = 2$, because $X_2^{(0)} = 0$,

$$\boxed{\hat{T}_{(II),\text{corr}}^2(G_*) = iA_0 Q_2(\delta_{\text{corr}}) + O(\delta^3),} \quad (474)$$

hence

$$\boxed{\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3).} \quad (475)$$

Therefore on the explicit canonical pure-even branch,

$$\boxed{\beta^{\text{corr}} = O(\delta^2).} \quad (476)$$

So the odd transverse channel starts only at quadratic order for the present orbit-symmetric corrected seed. :contentReference[oaicite:13]index=13

CLXX. 5. MINIMAL RIGID-INTERNAL FIRST-SHELL TRUNCATION

In the minimal rigid-internal first-shell truncation,

$$K_A = K_E = K_O = 0.$$

Then the full fast six-row coefficients vanish identically:

$$\boxed{H_{AA}^{(2),\text{fast}} = H_{EE}^{(2),\text{fast}} = H_{OO}^{(2),\text{fast}} = H_{AE}^{(2),\text{fast}} = H_{AO}^{(2),\text{fast}} = H_{EO}^{(2),\text{fast}} = 0.} \quad (477)$$

Consequently,

$$\boxed{Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0 \quad \text{in the minimal rigid-internal first-shell truncation.}} \quad (478)$$

Thus orbit-amplitude corrections alone cannot generate the odd transverse response. :contentReference[oaicite:14]index=14

CLXXI. 6. COMBINED THEOREM

Theorem CLXXI.1 (First active sign-sensitive transverse correction on the corrected branch). *On the frozen corrected seed*

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096),$$

the sign-sensitive first-shell transverse factor has no linear corrected shift:

$$\ell_\alpha \cdot \delta_{\text{corr}} = 0.$$

On the explicit canonical pure-even branch:

1. *the even transverse channel survives already at base level,*

$$X_1^{(0)} \neq 0,$$

so α^{corr} can shift linearly through the selected harmonic A_{G} ;*

2. *the odd transverse channel vanishes at base level,*

$$X_2^{(0)} = 0,$$

so β^{corr} starts at $O(\delta^2)$;

3. *in the minimal rigid-internal first-shell truncation, even this quadratic odd correction vanishes:*

$$Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0.$$

Therefore the first genuinely active source of odd transverse transport must come from additional microscopic structure beyond orbit-amplitude corrections alone.

Proof. The vanishing of the linear transverse shift follows from the exact transverse row contractions with the lifted corrected seed. The base-branch verdict follows from the explicit canonical pure-even coefficient sheet. The minimal-truncation vanishing of Q_2 follows from the identically vanishing six fast coefficients. $\square$

CLXXII. 7. PHYSICAL MEANING

This closes the local sign-sensitive transverse logic.

The refined quartic orbit enlargement was necessary, but it is not sufficient to activate the odd transverse witness. The current corrected branch is too symmetric at first-shell orbit-amplitude level.

Hence the next nonredundant frontier is not another quartic pair-class enlargement. It is one of the genuinely microscopic sources already isolated in the notes:

1. non-rigid internal / projector texture so that P is no longer constant;
2. defect background with $\partial_i \bar{q}^a \neq 0$;
3. completion-level slots beyond the minimal fast branch;
4. higher-shell or non-first-shell contributions.

Equivalently, the next executable microscopic step is the explicit $TD + DT$ one-defect Hessian audit for

$$\Lambda_E, \quad \Lambda_O,$$

not further squeezing of the rigid first-shell orbit-amplitude ansatz. :contentReference[oaicite:15]index=15 :contentReference[oaicite:16]index=16

CLXXIII. FINAL SUMMARY

The exact sign-sensitive first-shell transverse rows satisfy

$$r_{\text{sub}}^{(ct_1)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0, \quad r_{\text{sub}}^{(ct_2)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0.$$

So the frozen orbit-symmetric corrected seed produces no linear first-shell correction to the transverse witnesses. On the explicit canonical pure-even branch,

$$X_1^{(0)} \neq 0, \quad X_2^{(0)} = 0,$$

hence

$$\alpha^{\text{corr}} = \alpha^{(0)} [1 + O(\delta) + O(\delta^2)], \quad \beta^{\text{corr}} = O(\delta^2).$$

In the minimal rigid-internal first-shell truncation one has

$$Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0.$$

Therefore the next true frontier is no longer quartic orbit closure. It is the explicit sign-sensitive nonquartic / transverse microscopic insertion.

CLXXIV. FINAL SUMMARY

Current status: the refined quartic baseline is closed, and the first active sign-sensitive transverse correction has now been located. (479)

The corrected stage remains frozen at

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096),$$

and the correct reduced object remains the projected residual map

$$G(\delta) = (G_A(\delta), G_E(\delta), G_O(\delta))^T.$$

The current coefficient sheet must still be interpreted honestly as a residual-based effective reduced coefficient set of the current solver family, not yet as theorem-level coefficients of a fully closed microscopic energy functional. :contentReference[oaicite:17]index=17

At first-shell sign-resolved level, the frozen corrected seed lifts to

$$\delta_{\text{sub}}^{\text{corr}} = (0.0096, 0.0096, -0.0288, -0.0288, 0.0096, 0.0096, 0.0096, 0.0096)^T,$$

and the exact transverse response rows give

$$r_{\text{sub}}^{(ct_1)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0, \quad r_{\text{sub}}^{(ct_2)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0.$$

So the sign-sensitive transverse factor has no linear corrected shift. :contentReference[oaicite:18]index=18 :contentReference[oaicite:19]index=19

On the explicit canonical pure-even branch one has

$$\alpha \neq 0, \quad \beta = 0,$$

hence

$$X_1^{(0)} \neq 0, \quad X_2^{(0)} = 0.$$

Therefore

$$\alpha^{\text{corr}}$$

can shift linearly through the selected harmonic A_{G_*} , while

$$\beta^{\text{corr}}$$

starts only at $O(\delta^2)$. :contentReference[oaicite:20]index=20 :contentReference[oaicite:21]index=21

Moreover, in the minimal rigid-internal first-shell truncation,

$$Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0.$$

So orbit-amplitude corrections alone do not generate the odd transverse witness. :contentReference[oaicite:22]index=22

Hence the next nonredundant frontier is now sharply fixed:

explicit sign-sensitive nonquartic / transverse microscopic insertion.

Goal C: keep the refined quartic orbit closure as the fixed residual baseline. (480)

Goal M: evaluate the $TD + DT$ one-defect Hessian projections giving Λ_E, Λ_O . (481)

Goal S: determine which genuinely microscopic slot first activates the odd transverse channel beyond the rigid first-shell orbit-a (482)

CLXXV. SOURCE-BY-SOURCE MICROSCOPIC CLOSURE OF THE $TD + DT$ TRANSVERSE AUDIT

CLXXVI. SHORT EXPLANATION

At the present sharpest explicit working-branch level, the coefficient-routing problem is already structurally closed. The longitudinal coefficient is extracted from the OO -block:

$$\mathcal{H}_{OO} \rightarrow \lambda_c,$$

while the transverse even/odd pair is extracted from the mixed transverse block:

$$\mathcal{H}_{TD} + \mathcal{H}_{DT} \rightarrow \Lambda_E, \Lambda_O.$$

More sharply, on the current explicit Class II witness route,

$$JJ \rightarrow \lambda_c, \quad JK + KJ \rightarrow (\Lambda_E, \Lambda_O), \quad KK \rightarrow \text{same-shell residual transverse corrections.}$$

Therefore the remaining task is not conceptual theorem-building. It is the literal evaluation of the background-linearized first-order Hessian kernels

$$\mathcal{V}_{OO}^\mu(\xi), \quad \mathcal{V}_{TD}^\mu(\xi) + \mathcal{V}_{DT}^\mu(\xi),$$

on the frozen corrected branch. :contentReference[oaicite:4]index=4 :contentReference[oaicite:5]index=5 :contentReference[oaicite:6]index=6

CLXXVII. 1. EXACT SOURCE SPLIT AFTER SCHUR REDUCTION

After integrating out the heavy mediator, the quadratic correction has the exact split

$$\Delta_X^{(2)} = -\frac{\alpha_X^2}{M_X^2} \delta J \delta J^\dagger - \frac{\alpha_X \beta_X}{M_X^2} (\delta J \delta K^\dagger + \delta K \delta J^\dagger) - \frac{\beta_X^2}{M_X^2} \delta K \delta K^\dagger. \quad (483)$$

Thus the microscopic source sectors are

$$JJ, \quad JK + KJ, \quad KK.$$

The current branch-level routing is:

$$JJ \leftrightarrow OO\text{-longitudinal audit}, \quad JK + KJ \leftrightarrow TD + DT\text{-transverse audit}. \quad (484)$$

At principal order, the KK -sector is not the primitive longitudinal source and should be treated as a transverse residual/completion correction sector. :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8 :contentReference[oaicite:9]index=9

CLXXVIII. 2. LONGITUDINAL SIDE IS ALREADY ISOLATED

On the intended Class II-compatible branch,

$$\Lambda_{\parallel}^{(0)} = 0, \quad (485)$$

so the practical longitudinal coefficient is Branch B:

$$\lambda_c = 2 \lim_{\theta' \rightarrow 0} \frac{\mathfrak{P}_{\parallel}[V_{OO}]}{\theta'(\xi)^3}, \quad V_{OO}^{\mu} := \Pi_Q \mathcal{V}_{OO}^{\mu} \Pi_Q. \quad (486)$$

Therefore the longitudinal routing is not the current ambiguity.

The current ambiguity is the transverse pair

$$\Lambda_E, \quad \Lambda_O.$$

:contentReference[oaicite:10]index=10 :contentReference[oaicite:11]index=11

CLXXIX. 3. EXACT $TD + DT$ TRANSVERSE CARRIER

Define the mixed first-order transverse carrier

$$\mathcal{M}_{TD+DT}^{\mu} := \Pi_Q (\mathcal{V}_{TD}^{\mu} + \mathcal{V}_{DT}^{\mu}) \Pi_Q. \quad (487)$$

The leading even/odd coefficient point is extracted by

$$\Lambda_E = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}]. \quad (488)$$

Equivalently, in the leading mixed-block truncation,

$$e_0^{(1)} = \Lambda_E, \quad o_0^{(1)} = \Lambda_O. \quad (489)$$

At the branch-summary level, the leading transverse kernel has the invariant form

$$\mathcal{M}_{TD+DT}^{\mu} = \theta'(\xi) \left[\Lambda_E E^{\mu}_a \Sigma_a + \Lambda_O J^{\mu}_a \Sigma_a \right] + \mathcal{M}_{\text{corr}}^{\mu}, \quad (490)$$

where $\mathcal{M}_{\text{corr}}^{\mu}$ starts at higher order and/or completion channels such as

$$OD, \quad DO, \quad DD.$$

:contentReference[oaicite:12]index=12 :contentReference[oaicite:13]index=13

CLXXX. 4. WHY THE NAIVE LOCAL JK STATEMENT IS TOO STRONG

The crucial caveat is that a naive local real Hermitian JK bilinear does not automatically generate the transverse pair.

In the shell-adapted Pauli basis,

$$J \sim j \Sigma_3, \quad K \sim k_E \Sigma_1 + k_O \Sigma_2.$$

Then the simplest local real Hermitian bilinear gives

$$\Sigma_3(k_E \Sigma_1 + k_O \Sigma_2) + (k_E \Sigma_1 + k_O \Sigma_2) \Sigma_3 = k_E \{\Sigma_3, \Sigma_1\} + k_O \{\Sigma_3, \Sigma_2\} \quad (491)$$

$$= 0. \quad (492)$$

Hence

$$\boxed{\text{the naive local real Hermitian } JK \text{ bilinear vanishes.}} \quad (493)$$

So the statement

$$JK \text{ alone automatically gives } (\Lambda_E, \Lambda_O)$$

is too strong.

What is actually load-bearing is the derivative-resolved shell audit

$$\boxed{\partial_{p_\nu} \hat{\Sigma}_{JK}(G_*; p)|_{p=0}, \quad \partial_{p_\nu} \hat{\Sigma}_{KK}(G_*; p)|_{p=0},} \quad (494)$$

or, in the explicit folded-Bragg Class II bookkeeping,

$$\boxed{\hat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \hat{J}_i^A(Q) \hat{K}_{i,\alpha}^A(Q').} \quad (495)$$

Thus the real transverse frontier is the explicit differentiated kernel, not a local algebraic JK shortcut. :contentReference[oaicite:14]index=14

CLXXXI. 5. EXACT PROJECTOR FORMULAS FOR THE MATRIX ENTRIES

Let the projected Q_B -valued mixed carrier be represented in the Pauli basis as

$$\mathcal{M}_{TD+DT}^\mu = m_0^\mu \mathbf{1}_2 + m_a^\mu \Sigma_a. \quad (496)$$

Then the invariant projectors read

$$\boxed{\mathfrak{P}_E[\mathcal{M}] = \frac{1}{4} E^\mu{}_a \text{Tr}(\Sigma_a \mathcal{M}_\mu), \quad \mathfrak{P}_O[\mathcal{M}] = \frac{1}{4} J^\mu{}_a \text{Tr}(\Sigma_a \mathcal{M}_\mu).} \quad (497)$$

If one writes the Q_B -matrix entries as

$$r_{11}, \quad r_{22}, \quad r_{12},$$

then the Pauli components are

$$\boxed{s_3 = \frac{r_{11} - r_{22}}{2}, \quad s_1 = \Re r_{12}, \quad s_2 = -\Im r_{12}.} \quad (498)$$

The longitudinal OO -projection reads s_3 , whereas the transverse $TD + DT$ -projection reads s_1, s_2 , i.e.

$$\Lambda_E, \quad \Lambda_O.$$

:contentReference[oaicite:15]index=15

CLXXXII. 6. REDUCED COEFFICIENT DICTIONARY

Once $\lambda_c, \Lambda_E, \Lambda_O$ are extracted, the reduced Branch-B coefficients are

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3} \lambda_c, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O. \quad (499)$$

With the explicit overlaps

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad (500)$$

this becomes

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \quad (501)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \Lambda_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \Lambda_O \theta_0 \kappa. \quad (502)$$

Therefore the exact microscopic chain is

$$\mathcal{H}_{OO} \rightarrow \lambda_c \rightarrow \lambda_{\parallel}, \quad \mathcal{H}_{TD} + \mathcal{H}_{DT} \rightarrow (\Lambda_E, \Lambda_O) \rightarrow (\alpha, \beta). \quad (503)$$

:contentReference[oaicite:16]index=16 :contentReference[oaicite:17]index=17

CLXXXIII. 7. WHAT IS ACTUALLY CLOSED AND WHAT IS STILL OPEN

What is already closed:

$$\text{routing logic is closed: } OO \rightarrow \lambda_c, \quad TD + DT \rightarrow (\Lambda_E, \Lambda_O). \quad (504)$$

What is also closed:

$$JJ \text{ is the unique principal longitudinal source, } JK + KJ \text{ is the principal transverse source,} \quad (505)$$

while

$$KK \text{ is a transverse residual/completion sector, not the principal carrier source.} \quad (506)$$

What remains open is not conceptual structure, but the literal evaluation of

$$\mathcal{V}_{OO}^{\mu}(\xi), \quad \mathcal{V}_{TD}^{\mu}(\xi) + \mathcal{V}_{DT}^{\mu}(\xi) \quad (507)$$

on the frozen corrected branch. :contentReference[oaicite:18]index=18 :contentReference[oaicite:19]index=19

CLXXXIV. 8. EXECUTABLE NEXT MICROSCOPIC STEP

The next executable calculation is therefore uniquely fixed:

1. write the explicit background-linearized kernels

$$\delta J, \quad \delta K$$

on the frozen corrected branch;

2. evaluate the OO -projection

$$V_{OO}^\mu = \Pi_Q \mathcal{V}_{OO}^\mu \Pi_Q$$

and extract

$$\lambda_c = 2 \lim_{\theta' \rightarrow 0} \frac{\mathfrak{P}_{||}[V_{OO}]}{\theta'(\xi)^3};$$

3. evaluate the mixed projection

$$\mathcal{M}_{TD+DT}^\mu = \Pi_Q (\mathcal{V}_{TD}^\mu + \mathcal{V}_{DT}^\mu) \Pi_Q$$

and extract

$$\Lambda_E = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}];$$

4. only after that, return to the residual-level enlarged-kernel rerun.

So the next frontier is a literal source-by-source Hessian differentiation and projection computation, not another abstract reduction theorem. :contentReference[oaicite:20]index=20 :contentReference[oaicite:21]index=21

CLXXXV. FINAL SUMMARY

The remaining coefficient point on the current explicit Class II working branch is

$$\boxed{\lambda_c, \quad \Lambda_E, \quad \Lambda_O.}$$

The exact routing is

$$\boxed{JJ \rightarrow \lambda_c, \quad JK + KJ \rightarrow (\Lambda_E, \Lambda_O), \quad KK \rightarrow \text{transverse residual corrections.}}$$

Equivalently, at block level,

$$\boxed{\mathcal{H}_{OO} \rightarrow \lambda_c, \quad \mathcal{H}_{TD} + \mathcal{H}_{DT} \rightarrow \Lambda_E, \Lambda_O.}$$

The naive local real Hermitian JK bilinear vanishes, so a nonzero transverse pair requires derivative-resolved and/or phase-resolved microscopic structure.

Thus the next exact microscopic task is not structural. It is the explicit evaluation of

$$\mathcal{V}_{OO}^\mu(\xi), \quad \mathcal{V}_{TD}^\mu(\xi) + \mathcal{V}_{DT}^\mu(\xi)$$

on the frozen corrected branch and the corresponding projector traces.

CLXXXVI. FINAL SUMMARY

$$\boxed{\text{Current stage: } TD + DT \text{ projected Hessian audit.}}$$

(508)

The current explicit Class II working branch has the exact coefficient-routing table

$$\boxed{JJ \rightarrow \lambda_c, \quad JK + KJ \rightarrow (\Lambda_E, \Lambda_O), \quad KK \rightarrow \text{transverse residual corrections.}}$$

Equivalently,

$$\boxed{\mathcal{H}_{OO} \rightarrow \lambda_c, \quad \mathcal{H}_{TD} + \mathcal{H}_{DT} \rightarrow \Lambda_E, \Lambda_O.}$$

The longitudinal coefficient is

$$\lambda_c = 2 \lim_{\theta' \rightarrow 0} \frac{\mathfrak{P}_{\parallel}[V_{OO}]}{\theta'(\xi)^3}, \quad V_{OO}^{\mu} = \Pi_Q \mathcal{V}_{OO}^{\mu} \Pi_Q,$$

while the transverse pair is

$$\Lambda_E = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \mathcal{M}_{TD+DT}^{\mu} = \Pi_Q(\mathcal{V}_{TD}^{\mu} + \mathcal{V}_{DT}^{\mu})\Pi_Q.$$

After overlap insertion,

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512}\lambda_c\theta_0^3\kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16}\Lambda_E\theta_0\kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16}\Lambda_O\theta_0\kappa.$$

The naive local real Hermitian JK bilinear vanishes, so the transverse pair cannot be claimed from a local algebraic shortcut. A genuinely nonzero (Λ_E, Λ_O) requires derivative-resolved and/or phase-resolved microscopic structure. :contentReference[oaicite:22]index=22

Therefore the next exact task is already fixed:

write $\delta J, \delta K$ on the frozen corrected branch, evaluate $\mathcal{V}_{OO}^{\mu}$, $\mathcal{V}_{TD}^{\mu} + \mathcal{V}_{DT}^{\mu}$, and apply the projector traces.

Goal C: evaluate the actual one-defect Hessian kernels on the frozen corrected branch. (509)

Goal M: extract $\lambda_c, \Lambda_E, \Lambda_O$ source-by-source. (510)

Goal S: only after that, return to the enlarged residual-kernel rerun. (511)

CLXXXVII. LOAD-BEARING $\delta J, \delta K$ AUDIT ON THE FROZEN CORRECTED BRANCH

CLXXXVIII. GOAL

At the current sharpest explicit working-branch level, the routing problem is already closed:

$$\mathcal{H}_{OO} \rightarrow \lambda_c, \quad \mathcal{H}_{TD} + \mathcal{H}_{DT} \rightarrow (\Lambda_E, \Lambda_O).$$

So the next task is no longer abstract theorem-building. It is the literal background-linearized evaluation of the load-bearing currents

$$\delta J, \quad \delta K$$

on the frozen corrected branch and the extraction of the first active transverse odd source.

CLXXXIX. 1. FROZEN WORKING BRANCH AND EXACT SOURCE SPLIT

We keep fixed the explicit Class II working branch

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \Psi]. \quad (512)$$

The heavy mediator is algebraic:

$$X_i^a = -\frac{1}{M_X^2} \left(\alpha_X J_i^a + \beta_X K_i^a \right). \quad (513)$$

Substituting back gives the exact tree-level effective interaction

$$L_{\text{eff}}^{(X)} = -\frac{\alpha_X^2}{2M_X^2} J_i^a J_i^a - \frac{\alpha_X \beta_X}{M_X^2} J_i^a K_i^a - \frac{\beta_X^2}{2M_X^2} K_i^a K_i^a. \quad (514)$$

Thus the exact source decomposition is

$$\boxed{JJ, \quad JK + KJ, \quad KK.} \quad (515)$$

At the sharp current routing level,

$$\boxed{JJ \rightarrow \lambda_c, \quad JK + KJ \rightarrow (\Lambda_E, \Lambda_O), \quad KK \rightarrow \text{transverse residual corrections.}} \quad (516)$$

CXC. 2. DEFECT-BACKGROUND CURRENTS

Take the locked one-defect background

$$\boxed{\Psi_B(\xi, x_{\parallel}) = \phi_0(x_{\parallel}) z_B(\xi), \quad P_B(\xi) = z_B(\xi) z_B^{\dagger}(\xi), \quad z_B^{\dagger} z_B = 1.} \quad (517)$$

The microscopic currents are

$$\boxed{J_i^a[\Psi] = (\partial_i \Psi^{\dagger}) T_a \Psi + \Psi^{\dagger} T_a (\partial_i \Psi),} \quad (518)$$

$$\boxed{K_i^a[P, \psi] = \text{Tr}(T_a \partial_i P) \mathcal{O}[\psi].} \quad (519)$$

On the defect background one has

$$\boxed{J_i^a[\Psi_B] = 2(\partial_i \phi_0) \phi_0 m_a(\theta) + \delta_{i\xi} \phi_0^2 \mathcal{A}_{\xi}^a(\theta, \theta'),} \quad (520)$$

with

$$m_a(\theta) = z_B^{\dagger} T_a z_B, \quad \mathcal{A}_{\xi}^a = (\partial_{\xi} z_B^{\dagger}) T_a z_B + z_B^{\dagger} T_a (\partial_{\xi} z_B), \quad (521)$$

and

$$\boxed{K_i^a[P_B, \psi] = \delta_{i\xi} \theta'(\xi) k_a(\theta) \mathcal{O}[\psi], \quad k_a(\theta) := \text{Tr}[T_a (e_1 z_B^{\dagger} + z_B e_1^{\dagger})].} \quad (522)$$

So the background J -sector is already linear in the shell gradient $\partial_i \phi_0$, whereas the K -sector is linear in the defect/projector derivative $\theta'(\xi)$.

CXCI. 3. LOAD-BEARING LINEARIZED KERNELS

Around the locked background, the lowest-derivative load-bearing parts can be written in the factorized form

$$\boxed{\delta J_i^a[\Psi_0; \delta \Psi]_{\text{lb}} = m^a(P_0) (\partial_i \phi_0) \mathcal{U}_{jr}{}^b (-i \partial_j) s_b \psi_r,} \quad (523)$$

$$\boxed{\delta K_i^a[P_0; \delta \Psi]_{\text{lb}} = (\partial_i \xi^a) \mathcal{V}_{jr}{}^b (-i \partial_j) s_b \psi_r.} \quad (524)$$

Equivalently, before factorization, the general lowest-order load-bearing pieces are

$$\boxed{\delta J_i^a[\Psi_0; \delta \Psi]_{\text{lb}} = \mathcal{B}_{jr}^a (\partial_i \phi_0) (-i \partial_j) \psi_r,} \quad (525)$$

$$\boxed{\delta K_i^a[P_0; \delta \Psi]_{\text{lb}} = \mathcal{F}_{ijr}^{ab} (\partial_i \xi^a) (-i \partial_j) s_b \psi_r.} \quad (526)$$

These are exactly the (i, a) -carrier structures needed for the one-mediator Schur witness route.

CXCII. 4. PROJECTED TRANSPORT KERNELS

The first-order one-defect Hessian blocks admit the expansion

$$\mathcal{H}_{AB} = \mathcal{H}_{AB}^{(0)}(\xi) - i \mathcal{V}_{AB}^\mu(\xi) \partial_{y^\mu} + \cdots. \quad (527)$$

The longitudinal coefficient is extracted from the OO -projection:

$$V_{OO}^\mu = \Pi_Q \mathcal{V}_{OO}^\mu \Pi_Q, \quad \lambda_c = 2 \lim_{\theta' \rightarrow 0} \frac{\mathfrak{P}_\parallel[V_{OO}]}{\theta'(\xi)^3}. \quad (528)$$

The transverse pair is extracted from the mixed $TD + DT$ -projection:

$$\mathcal{M}_{TD+DT}^\mu = \Pi_Q (\mathcal{V}_{TD}^\mu + \mathcal{V}_{DT}^\mu) \Pi_Q, \quad (529)$$

$$\Lambda_E = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}]. \quad (530)$$

So after overlap insertion the reduced coefficients are

$$\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O. \quad (531)$$

CXCIII. 5. FIRST ACTIVE ODD TRANSVERSE SOURCE

At the exact transport-extraction level, the odd transverse source is

$$X_2(\delta) = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}(\delta)]. \quad (532)$$

Therefore the corrected odd transverse Hessian response is defined by

$$H_{rs}^{(2)} := \partial_{\delta_r} \partial_{\delta_s} X_2(\delta) \big|_{\delta=0} = \partial_{\delta_r} \partial_{\delta_s} \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}(\delta)]_{\delta=0}. \quad (533)$$

On the current refined locked + Class II working branch, the leading mixed-channel realization is

$$H_{rs}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\partial_{rs} (\delta J \delta K^\dagger + \delta K \delta J^\dagger)_{\delta=0} \right] + \text{subleading completion terms}. \quad (534)$$

This is the decisive result: the first active odd transverse response on the frozen corrected branch is the quadratic Hessian response of the mixed $JK + KJ$ channel.

CXCIV. 6. WHY NO LINEAR ODD TRANSVERSE ACTIVATION OCCURS AT THE FROZEN CORRECTED SEED

On the sign-resolved first-shell lift of the frozen corrected seed, the exact transverse rows satisfy

$$r_{\text{sub}}^{(ct_1)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0, \quad r_{\text{sub}}^{(ct_2)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0.$$

Hence the sign-sensitive first-shell transverse factor has no linear corrected shift at the frozen corrected branch.

Therefore the odd transverse coefficient does not turn on linearly in the present orbit-symmetric corrected seed. Instead, the first genuine odd source is the quadratic response (534).

CXCV. 7. HONEST BRANCH-LEVEL CONCLUSION

The $TD + DT$ audit is now sharp enough to separate what is already closed from what is still open. What is closed:

$$\boxed{JJ \rightarrow \lambda_c, \quad JK + KJ \rightarrow (\Lambda_E, \Lambda_O),} \quad (535)$$

and

$$\boxed{\Lambda_O \text{ first appears through } H_{rs}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\partial_{rs} (\delta J \delta K^\dagger + \delta K \delta J^\dagger)_{\delta=0} \right] + \dots} \quad (536)$$

What is still open is the literal evaluation of the background-dependent tensors

$$\mathcal{U}_{jr}{}^b, \quad \mathcal{V}_{jr}{}^b, \quad m^a(P_0), \quad \partial_i \xi^a,$$

on the frozen corrected branch.

So the next executable microscopic task is:

$$\boxed{\text{compute the actual matrix } H_{rs}^{(2)} \text{ from the frozen corrected branch harmonics,}} \quad (537)$$

and then project it by $\mathfrak{P}_O$ to decide whether $\Lambda_O \neq 0$ beyond the rigid first-shell truncation.

CXCVI. FINAL SUMMARY

The background currents are explicit:

$$J_i^a[\Psi_B] = 2(\partial_i \phi_0) \phi_0 m_a(\theta) + \delta_{i\xi} \phi_0^2 \mathcal{A}_\xi^a(\theta, \theta'),$$

$$K_i^a[P_B, \psi] = \delta_{i\xi} \theta'(\xi) k_a(\theta) \mathcal{O}[\psi].$$

Their load-bearing linearized kernels are

$$\delta J_i^a[\Psi_0; \delta \Psi]_{\text{lb}} = m^a(P_0) (\partial_i \phi_0) \mathcal{U}_{jr}{}^b (-i \partial_j) s_b \psi_r,$$

$$\delta K_i^a[P_0; \delta \Psi]_{\text{lb}} = (\partial_i \xi^a) \mathcal{V}_{jr}{}^b (-i \partial_j) s_b \psi_r.$$

The exact extraction problem is

$$V_{OO} \rightarrow \lambda_c, \quad \mathcal{M}_{TD+DT} \rightarrow (\Lambda_E, \Lambda_O).$$

The first active odd transverse source is not linear in the frozen corrected seed. It is the quadratic response

$$H_{rs}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\partial_{rs} (\delta J \delta K^\dagger + \delta K \delta J^\dagger)_{\delta=0} \right] + \dots$$

Thus the next actual microscopic computation is now fixed: evaluate the matrix $H_{rs}^{(2)}$ on the frozen corrected branch and decide whether the odd transverse channel is really activated.

CXCVII. FINAL SUMMARY

$$\boxed{\text{Current stage: } TD + DT \text{ projected Hessian audit on the frozen corrected branch.}} \quad (538)$$

The strongest explicit Class II working branch is

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \Psi],$$

and after eliminating the heavy mediator,

$$L_{\text{eff}}^{(X)} = -\frac{\alpha_X^2}{2M_X^2} J_i^a J_i^a - \frac{\alpha_X \beta_X}{M_X^2} J_i^a K_i^a - \frac{\beta_X^2}{2M_X^2} K_i^a K_i^a.$$

The exact routing is

$$JJ \rightarrow \lambda_c, \quad JK + KJ \rightarrow (\Lambda_E, \Lambda_O), \quad KK \rightarrow \text{transverse corrections}.$$

On the defect background,

$$J_i^a[\Psi_B] = 2(\partial_i \phi_0) \phi_0 m_a(\theta) + \delta_{i\xi} \phi_0^2 \mathcal{A}_\xi^a(\theta, \theta'),$$

$$K_i^a[P_B, \psi] = \delta_{i\xi} \theta'(\xi) k_a(\theta) \mathcal{O}[\psi].$$

Hence the load-bearing linearized kernels are of the form

$$\delta J_i^a[\Psi_0; \delta\Psi]_{\text{lb}} = m^a(P_0) (\partial_i \phi_0) \mathcal{U}_{jr}{}^b (-i\partial_j) s_b \psi_r,$$

$$\delta K_i^a[P_0; \delta\Psi]_{\text{lb}} = (\partial_i \xi^a) \mathcal{V}_{jr}{}^b (-i\partial_j) s_b \psi_r.$$

The exact extraction problem is

$$V_{OO} \rightarrow \lambda_c, \quad \mathcal{M}_{TD+DT} \rightarrow (\Lambda_E, \Lambda_O).$$

The first active odd transverse source is

$$H_{rs}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\partial_{rs} (\delta J \delta K^\dagger + \delta K \delta J^\dagger)_{\delta=0} \right] + \text{subleading completion terms}.$$

So the next actual microscopic step is no longer abstract:

$$\boxed{\text{compute } H_{rs}^{(2)} \text{ on the frozen corrected branch and decide whether } \Lambda_O \neq 0.}$$

$$\boxed{\text{Goal C: evaluate the actual background-linearized } OO \text{ and } TD + DT \text{ kernels.}} \quad (539)$$

$$\boxed{\text{Goal M: compute } H_{rs}^{(2)} \text{ and extract the first nonzero odd transverse source.}} \quad (540)$$

$$\boxed{\text{Goal S: decide whether the rigid first-shell truncation must be upgraded by non-rigid projector/internal structure or higher-shell}} \quad (541)$$

CXCVIII. ACTUAL COMPUTATION OF THE FIRST FROZEN-BRANCH ODD TRANSVERSE RESPONSE

CXCIX. SHORT EXPLANATION

We now compute the odd transverse correction as far as the currently available explicit formulas allow.

The working branch is frozen at

$$\delta_{\text{corr}} = (0.0096, -0.0288, 0.0096)^T,$$

and the corrected odd transverse coefficient is

$$\beta^{\text{corr}} \approx -\frac{\sqrt{3}\pi}{16}\theta_0\kappa A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3), \quad (542)$$

with

$$Q_2(\delta_{\text{corr}}) = 4.608 \times 10^{-5} H_{AA}^{(2)} + 4.1472 \times 10^{-4} H_{EE}^{(2)} + 4.608 \times 10^{-5} H_{OO}^{(2)} - 2.7648 \times 10^{-4} H_{AE}^{(2)} + 9.216 \times 10^{-5} H_{AO}^{(2)} - 2.7648 \times 10^{-4} H_{EO}^{(2)} \quad (543)$$

:contentReference[oaicite:3]index=3

CC. 1. FAST SIX-ROW FORMULA

Under the fast practical approximation

$$J_{rs} \approx 0, \quad K_{rs} \approx 0, \quad \Delta_{rs}^{\text{comp}} \approx 0,$$

all six rows reduce to

$$H_{rs}^{(2),\text{fast}} \approx -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_r K_s^\dagger + J_s K_r^\dagger + K_r J_s^\dagger + K_s J_r^\dagger]. \quad (544)$$

So the problem is reduced to the three orbit-current pairs

$$(J_A, K_A), \quad (J_E, K_E), \quad (J_O, K_O).$$

:contentReference[oaicite:4]index=4 :contentReference[oaicite:5]index=5

CCI. 2. EXPLICIT ORBIT-CURRENT PAIRS

The completed orbit currents are

$$J_{r,i}^a = q_0^a \partial_i N_r + (\partial_i \bar{\rho})(\eta_r c^a + \bar{\eta}_r \bar{c}^a) + \bar{\rho}(c^a \partial_i \eta_r + \bar{c}^a \partial_i \bar{\eta}_r), \quad (545)$$

$$K_{r,i}^a = (c^a \partial_i \eta_r + \bar{c}^a \partial_i \bar{\eta}_r) \bar{\chi}. \quad (546)$$

Thus

$$(J_A, K_A), \quad (J_E, K_E), \quad (J_O, K_O)$$

are explicit in terms of the orbit data (N_r, η_r) , the background $(\bar{\rho}, \bar{\chi})$, and the internal coefficients c^a . :contentReference[oaicite:6]index=6

CCII. 3. MINIMAL RIGID-INTERNAL FIRST-SHELL TRUNCATION

In the minimal rigid-internal first-shell truncation one has

$$Q_r^a = 0 \quad \forall r \in \{A, E, O\}, \quad (547)$$

hence

$$K_A = K_E = K_O = 0. \quad (548)$$

Therefore every fast six-row coefficient vanishes identically:

$$\boxed{H_{AA}^{(2),\text{fast}} = H_{EE}^{(2),\text{fast}} = H_{OO}^{(2),\text{fast}} = H_{AE}^{(2),\text{fast}} = H_{AO}^{(2),\text{fast}} = H_{EO}^{(2),\text{fast}} = 0.} \quad (549)$$

Substituting into (543),

$$\boxed{Q_2(\delta_{\text{corr}}) = 0.} \quad (550)$$

Hence

$$\boxed{\beta^{\text{corr}} = 0 \quad \text{in the minimal rigid-internal first-shell truncation.}} \quad (551)$$

:contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8

CCIII. 4. WHAT IS THE FIRST NONZERO SOURCE BEYOND THE RIGID TRUNCATION?

Once K_A, K_E, K_O cease to vanish, the corrected odd coefficient is controlled by the weighted linear combination (543). The absolute weights are:

$$4.1472 \times 10^{-4}, \quad 2.7648 \times 10^{-4}, \quad 2.7648 \times 10^{-4}, \quad 9.216 \times 10^{-5}, \quad 4.608 \times 10^{-5}, \quad 4.608 \times 10^{-5}.$$

Therefore the relative hierarchy is exactly

$$\boxed{|EE| : |AE| : |EO| : |AO| : |AA| : |OO| = 9 : 6 : 6 : 2 : 1 : 1.} \quad (552)$$

So if the six rows are of comparable microscopic size after the rigid truncation is lifted, then the leading corrected odd response is dominated first by

$$H_{EE}^{(2)},$$

next by

$$H_{AE}^{(2)}, \quad H_{EO}^{(2)},$$

and only then by

$$H_{AO}^{(2)}, \quad H_{AA}^{(2)}, \quad H_{OO}^{(2)}.$$

CCIV. 5. PRACTICAL NEXT COMPUTATION ORDER

Therefore the correct source-by-source priority order is

$$\boxed{(J_E, K_E) \text{ first,} \quad (J_A, K_E) \text{ and } (J_O, K_E) \text{ next,}} \quad (553)$$

equivalently

$$\boxed{H_{EE}^{(2)} \text{ first,} \quad H_{AE}^{(2)}, H_{EO}^{(2)} \text{ second.}} \quad (554)$$

This is the first genuinely nonredundant microscopic direction beyond the rigid first-shell ansatz.

CCV. FINAL SUMMARY

The fast six-row system is

$$H_{rs}^{(2),\text{fast}} \approx -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_r K_s^\dagger + J_s K_r^\dagger + K_r J_s^\dagger + K_s J_r^\dagger].$$

In the minimal rigid-internal first-shell truncation,

$$K_A = K_E = K_O = 0,$$

so

$$H_{AA}^{(2),\text{fast}} = H_{EE}^{(2),\text{fast}} = H_{OO}^{(2),\text{fast}} = H_{AE}^{(2),\text{fast}} = H_{AO}^{(2),\text{fast}} = H_{EO}^{(2),\text{fast}} = 0,$$

and therefore

$$\boxed{Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0.}$$

Beyond the rigid truncation, the corrected odd transverse coefficient is weighted by

$$Q_2(\delta_{\text{corr}}) = 4.608 \times 10^{-5} H_{AA}^{(2)} + 4.1472 \times 10^{-4} H_{EE}^{(2)} + 4.608 \times 10^{-5} H_{OO}^{(2)} - 2.7648 \times 10^{-4} H_{AE}^{(2)} + 9.216 \times 10^{-5} H_{AO}^{(2)} - 2.7648 \times 10^{-4} H_{EO}^{(2)}.$$

Thus the first likely nonzero odd source is dominated, in order, by

$$\boxed{H_{EE}^{(2)}, \quad H_{AE}^{(2)}, \quad H_{EO}^{(2)}, \quad H_{AO}^{(2)}, \quad H_{AA}^{(2)}, \quad H_{OO}^{(2)}}.$$

So the next actual microscopic calculation should start from the E -sector current pair (J_E, K_E) .

CCVI. FINAL SUMMARY

$$\boxed{\text{Current stage: first actual computation of the odd transverse source on the frozen corrected branch.}} \quad (555)$$

The corrected odd transverse coefficient is

$$\beta^{\text{corr}} \approx -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3),$$

with

$$Q_2(\delta_{\text{corr}}) = 4.608 \times 10^{-5} H_{AA}^{(2)} + 4.1472 \times 10^{-4} H_{EE}^{(2)} + 4.608 \times 10^{-5} H_{OO}^{(2)} - 2.7648 \times 10^{-4} H_{AE}^{(2)} + 9.216 \times 10^{-5} H_{AO}^{(2)} - 2.7648 \times 10^{-4} H_{EO}^{(2)}.$$

:contentReference[oaicite:9]index=9

Under the fast working approximation,

$$H_{rs}^{(2),\text{fast}} \approx -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_r K_s^\dagger + J_s K_r^\dagger + K_r J_s^\dagger + K_s J_r^\dagger],$$

so the problem is reduced to the three orbit-current pairs

$$(J_A, K_A), \quad (J_E, K_E), \quad (J_O, K_O).$$

In the minimal rigid-internal first-shell truncation,

$$K_A = K_E = K_O = 0,$$

hence all six fast coefficients vanish:

$$H_{AA}^{(2),\text{fast}} = H_{EE}^{(2),\text{fast}} = H_{OO}^{(2),\text{fast}} = H_{AE}^{(2),\text{fast}} = H_{AO}^{(2),\text{fast}} = H_{EO}^{(2),\text{fast}} = 0.$$

Therefore

$$Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0.$$

Once the rigid truncation is lifted, the weight hierarchy in the corrected odd shift is

$$|EE| : |AE| : |EO| : |AO| : |AA| : |OO| = 9 : 6 : 6 : 2 : 1 : 1.$$

So the first actual microscopic priority order is

$$H_{EE}^{(2)} \text{ first, } H_{AE}^{(2)}, H_{EO}^{(2)} \text{ second.}$$

$$\text{Goal C: leave the rigid first-shell truncation, because there } \beta^{\text{corr}} = 0 \text{ identically.} \quad (556)$$

$$\text{Goal M: compute the non-rigid } E\text{-sector current pair } (J_E, K_E) \text{ first.} \quad (557)$$

$$\text{Goal S: use } H_{EE}^{(2)} \text{ as the first decisive test for whether } \Lambda_O \neq 0 \text{ beyond the rigid truncation.} \quad (558)$$

CCVII. ACTUAL COMPUTATION OF THE E -SECTOR SOURCE FOR THE FIRST NONZERO ODD TRANSVERSE RESPONSE

CCVIII. SHORT EXPLANATION

The six-row fast practical approximation is already explicit:

$$H_{rs}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_r K_s^\dagger + J_s K_r^\dagger + K_r J_s^\dagger + K_s J_r^\dagger]. \quad (559)$$

The corrected odd transverse coefficient is then

$$\beta^{\text{corr}} \approx -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 \left[4.608 \times 10^{-5} H_{AA}^{(2)} + 4.1472 \times 10^{-4} H_{EE}^{(2)} + 4.608 \times 10^{-5} H_{OO}^{(2)} - 2.7648 \times 10^{-4} H_{AE}^{(2)} + 9.216 \times 10^{-5} H_{AO}^{(2)} - 2 \right] \quad (560)$$

Therefore the E -sector source $H_{EE}^{(2)}$ is the single most heavily weighted fast coefficient in the odd corrected shift.

CCIX. 1. EXACT E -SECTOR CURRENTS

From the completed orbit-current formulas,

$$N_E = \bar{\Psi}^\dagger \Phi_E + \Phi_E^\dagger \bar{\Psi}, \quad M_E^a = \bar{\Psi}^\dagger T^a \Phi_E + \Phi_E^\dagger T^a \bar{\Psi}, \quad (561)$$

$$J_{E,i}^a = \partial_i M_E^a, \quad (562)$$

$$K_{E,i}^a = \partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} + (\partial_i \bar{q}^a) \chi_E. \quad (563)$$

It is convenient to define

$$F_E^a := \frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}}. \quad (564)$$

Then

$$K_{E,i}^a = (\partial_i F_E^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_E. \quad (565)$$

:contentReference[oaicite:4]index=4

:contentReference[oaicite:5]index=5

CCX. 2. EXACT FAST FORMULA FOR $H_{EE}^{(2)}$

Set $r = s = E$ in (559). Since all four terms coincide pairwise,

$$H_{EE}^{(2),\text{fast}} = -\frac{2\alpha_X\beta_X}{M_X^2}\mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger]. \quad (566)$$

Substitute (562) and (565):

$$H_{EE}^{(2),\text{fast}} = -\frac{2\alpha_X\beta_X}{M_X^2}\mathfrak{P}_O\left[(\partial_i M_E^a)((\partial_i F_E^a)\bar{\chi} + (\partial_i \bar{q}^a)\chi_E)^\dagger + ((\partial_i F_E^a)\bar{\chi} + (\partial_i \bar{q}^a)\chi_E)(\partial_i M_E^a)^\dagger\right]. \quad (567)$$

The sums over i and a are implicit.

This is already the exact closed E -sector formula needed for the fast practical audit.

CCXI. 3. SOURCE DECOMPOSITION OF $H_{EE}^{(2)}$

Equation (567) splits naturally into two parts:

$$H_{EE}^{(2),\text{fast}} = H_{EE,\text{proj}}^{(2)} + H_{EE,\text{qgrad}}^{(2)}, \quad (568)$$

with

$$H_{EE,\text{proj}}^{(2)} = -\frac{2\alpha_X\beta_X}{M_X^2}\mathfrak{P}_O\left[(\partial_i M_E^a)((\partial_i F_E^a)\bar{\chi})^\dagger + ((\partial_i F_E^a)\bar{\chi})(\partial_i M_E^a)^\dagger\right], \quad (569)$$

$$H_{EE,\text{qgrad}}^{(2)} = -\frac{2\alpha_X\beta_X}{M_X^2}\mathfrak{P}_O\left[(\partial_i M_E^a)((\partial_i \bar{q}^a)\chi_E)^\dagger + ((\partial_i \bar{q}^a)\chi_E)(\partial_i M_E^a)^\dagger\right]. \quad (570)$$

Thus the first nonzero odd transverse response can arise from either

$$\partial_i F_E^a \neq 0 \quad (\text{projector/internal texture source}) \quad (571)$$

or

$$\partial_i \bar{q}^a \neq 0 \quad (\text{background charge-gradient source}). \quad (572)$$

CCXII. 4. FURTHER EXPLICIT EXPANSION OF THE PROJECTOR/INTERNAL TEXTURE SOURCE

Differentiate (564):

$$\begin{aligned} \partial_i F_E^a &= \partial_i \left(\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right) \\ &= \frac{\partial_i M_E^a - (\partial_i \bar{q}^a) N_E - \bar{q}^a \partial_i N_E}{\bar{\rho}} - \frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}^2} \partial_i \bar{\rho}. \end{aligned} \quad (573)$$

Hence

$$\partial_i F_E^a = \frac{\partial_i M_E^a - \bar{q}^a \partial_i N_E}{\bar{\rho}} - \frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}^2} \partial_i \bar{\rho} - \frac{(\partial_i \bar{q}^a) N_E}{\bar{\rho}}. \quad (574)$$

So the projector/internal-texture contribution already contains three distinct microscopic slots:

$$\boxed{\partial_i M_E^a \quad (\text{orbit-current gradient}),} \quad (575)$$

$$\boxed{\bar{q}^a \partial_i N_E, (M_E^a - \bar{q}^a N_E) \partial_i \bar{\rho} \quad (\text{density-modulated projector source}),} \quad (576)$$

$$\boxed{(\partial_i \bar{q}^a) N_E \quad (\text{charge-gradient leakage into the projector channel}).} \quad (577)$$

Thus $H_{EE,\text{proj}}^{(2)}$ is already a sum of several microscopic non-rigid mechanisms.

CCXIII. 5. RIGID TRUNCATION CHECK

In the minimal rigid-internal first-shell truncation, the notes prove

$$\boxed{Q_r^a = 0, \quad K_A = K_E = K_O = 0.} \quad (578)$$

Equivalently,

$$F_E^a = 0, \quad \partial_i F_E^a = 0, \quad \partial_i \bar{q}^a = 0.$$

Therefore both source pieces vanish:

$$H_{EE,\text{proj}}^{(2)} = 0, \quad H_{EE,\text{qgrad}}^{(2)} = 0.$$

Hence

$$\boxed{H_{EE}^{(2),\text{fast}} = 0 \quad \text{in the minimal rigid-internal first-shell truncation.}} \quad (579)$$

This reproduces the previously established result that

$$Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0$$

in that truncation. :contentReference[oaicite:6]index=6

CCXIV. 6. FIRST NONZERO CRITERION FOR $H_{EE}^{(2)}$

From (568), a sufficient criterion for the first odd source to be nonzero is:

$$\boxed{\mathfrak{P}_O \left[(\partial_i M_E^a) ((\partial_i F_E^a) \bar{\chi})^\dagger + ((\partial_i F_E^a) \bar{\chi}) (\partial_i M_E^a)^\dagger \right] \neq 0} \quad (580)$$

or

$$\boxed{\mathfrak{P}_O \left[(\partial_i M_E^a) ((\partial_i \bar{q}^a) \chi_E)^\dagger + ((\partial_i \bar{q}^a) \chi_E) (\partial_i M_E^a)^\dagger \right] \neq 0.} \quad (581)$$

Equivalently, the three minimal requirements are:

1. $K_E \neq 0$, i.e. some non-rigid internal/projector or charge-gradient structure must exist;
2. the resulting $J_E K_E^\dagger + K_E J_E^\dagger$ must carry a nonzero Σ_2 component;
3. that Σ_2 component must survive the odd projector $\mathfrak{P}_O$.

So the real obstruction is not the absence of a formula. It is whether the frozen corrected branch contains a genuinely odd-projectable phase-resolved E -sector mixed current.

CCXV. 7. DOMINANT CONTRIBUTION TO β^{corr}

Since the corrected odd shift is

$$\beta^{\text{corr}} \approx -\frac{\sqrt{3}\pi}{16}\theta_0\kappa A_0 Q_2(\delta_{\text{corr}}),$$

and the largest weight in Q_2 is the EE -entry,

$$4.1472 \times 10^{-4},$$

the first nonzero β^{corr} beyond the rigid truncation is dominated, unless strong cancellations occur, by

$$\boxed{H_{EE}^{(2)}} \quad (582)$$

More precisely, if

$$|H_{EE}^{(2)}| \gg \max \left\{ \frac{2.7648}{4.1472} |H_{AE}^{(2)}|, \frac{2.7648}{4.1472} |H_{EO}^{(2)}|, \frac{0.9216}{4.1472} |H_{AO}^{(2)}|, \frac{0.4608}{4.1472} |H_{AA}^{(2)}|, \frac{0.4608}{4.1472} |H_{OO}^{(2)}| \right\},$$

then

$$\boxed{\beta^{\text{corr}} \approx -\frac{\sqrt{3}\pi}{16}\theta_0\kappa A_0 (4.1472 \times 10^{-4}) H_{EE}^{(2)}} \quad (583)$$

Substituting (568), this becomes

$$\boxed{\beta^{\text{corr}} \approx \frac{\sqrt{3}\pi}{16}\theta_0\kappa A_0 (4.1472 \times 10^{-4}) \frac{2\alpha_X\beta_X}{M_X^2} [\mathfrak{P}_O(\cdots)_{\text{proj}} + \mathfrak{P}_O(\cdots)_{\text{qgrad}}]} \quad (584)$$

So the sign and size of the first odd transverse activation are controlled directly by the odd-projected Hermitian mixed current in the E -sector.

CCXVI. FINAL SUMMARY

The exact fast E -sector coefficient is

$$H_{EE}^{(2),\text{fast}} = -\frac{2\alpha_X\beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger],$$

with

$$J_{E,i}^a = \partial_i M_E^a, \quad K_{E,i}^a = \partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} + (\partial_i \bar{q}^a) \chi_E.$$

Therefore the first nonzero odd source splits exactly into

$$H_{EE,\text{proj}}^{(2)} + H_{EE,\text{qgrad}}^{(2)},$$

namely projector/internal texture and background charge-gradient contributions.

In the minimal rigid-internal first-shell truncation,

$$F_E^a = 0, \quad \partial_i F_E^a = 0, \quad \partial_i \bar{q}^a = 0,$$

so

$$\boxed{H_{EE}^{(2),\text{fast}} = 0, \quad Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0.}$$

Beyond that rigid truncation, the first nonzero odd transverse activation is controlled first by the odd-projectable E -sector mixed current $J_E K_E^\dagger + K_E J_E^\dagger$.

CCXVII. FINAL SUMMARY

$$\boxed{\text{Current stage: actual computation of the leading } E\text{-sector odd transverse source.}} \quad (585)$$

The exact E -sector orbit-current pair is

$$J_{E,i}^a = \partial_i M_E^a, \quad K_{E,i}^a = \partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} + (\partial_i \bar{q}^a) \chi_E.$$

:contentReference[oaicite:7]index=7

Hence the fast six-row coefficient is

$$H_{EE}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger].$$

This splits exactly into

$$H_{EE,\text{proj}}^{(2)} + H_{EE,\text{qgrad}}^{(2)},$$

where the first comes from

$$\partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi}$$

and the second from

$$(\partial_i \bar{q}^a) \chi_E.$$

Therefore the first nonzero odd transverse source beyond the rigid truncation can come from either:

$$\boxed{\partial_i F_E^a \neq 0 \quad (\text{non-rigid internal/projector texture}),}$$

or

$$\boxed{\partial_i \bar{q}^a \neq 0 \quad (\text{background charge-gradient source}).}$$

In the minimal rigid-internal first-shell truncation,

$$F_E^a = 0, \quad \partial_i F_E^a = 0, \quad \partial_i \bar{q}^a = 0,$$

so

$$\boxed{H_{EE}^{(2),\text{fast}} = 0, \quad Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0.}$$

:contentReference[oaicite:8]index=8

Since the largest weight in the corrected odd shift

$$Q_2(\delta_{\text{corr}})$$

sits in front of $H_{EE}^{(2)}$, the E -sector remains the first decisive microscopic test for whether $\Lambda_O \neq 0$ beyond the rigid truncation. :contentReference[oaicite:9]index=9

$$\boxed{\text{Goal C: evaluate } \partial_i F_E^a \text{ and } \partial_i \bar{q}^a \text{ on the frozen corrected branch.}} \quad (586)$$

$$\boxed{\text{Goal M: decide whether } H_{EE,\text{proj}}^{(2)} \text{ or } H_{EE,\text{qgrad}}^{(2)} \text{ is the first nonzero odd source.}} \quad (587)$$

$$\boxed{\text{Goal S: once } H_{EE}^{(2)} \neq 0 \text{ is confirmed, use it as the leading source for } \beta^{\text{corr}}.} \quad (588)$$

CCXVIII. ACTUAL DECISION BETWEEN THE TWO E -SECTOR SOURCES: $\partial_i F_E^a$ VERSUS $\partial_i \bar{q}^a$

CCXIX. SHORT EXPLANATION

The completed orbit-current formulas are

$$\boxed{J_{r,i}^a = \partial_i M_r^a, \quad K_{r,i}^a = (\partial_i Q_r^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_r,} \quad (589)$$

with

$$Q_r^a = \frac{M_r^a - \bar{q}^a N_r}{\bar{\rho}}.$$

Therefore the fast E -sector coefficient is

$$\boxed{H_{EE}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O [J_E K_E^\dagger + K_E J_E^\dagger].} \quad (590)$$

The question is: which part of

$$K_E = (\partial_i Q_E^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_E$$

is the first unavoidable nonzero source on the frozen corrected branch?

CCXX. 1. EXACT SPLIT OF $H_{EE}^{(2)}$

Substitute (589) into (590):

$$\boxed{H_{EE}^{(2),\text{fast}} = H_{EE,\text{proj}}^{(2)} + H_{EE,\text{qgrad}}^{(2)},} \quad (591)$$

where

$$\boxed{H_{EE,\text{proj}}^{(2)} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O [(\partial_i M_E^a)((\partial_i Q_E^a) \bar{\chi})^\dagger + ((\partial_i Q_E^a) \bar{\chi})(\partial_i M_E^a)^\dagger],} \quad (592)$$

$$\boxed{H_{EE,\text{qgrad}}^{(2)} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O [(\partial_i M_E^a)((\partial_i \bar{q}^a) \chi_E)^\dagger + ((\partial_i \bar{q}^a) \chi_E)(\partial_i M_E^a)^\dagger].} \quad (593)$$

CCXXI. 2. RIGID TRUNCATION CHECK

In the minimal rigid-internal first-shell truncation one has

$$Q_E^a = 0, \quad \partial_i P = 0, \quad \partial_i \bar{q}^a = 0.$$

Hence both pieces vanish:

$$\boxed{H_{EE,\text{proj}}^{(2)} = 0, \quad H_{EE,\text{qgrad}}^{(2)} = 0, \quad H_{EE}^{(2),\text{fast}} = 0.} \quad (594)$$

Therefore

$$Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0$$

in that truncation.

CCXXII. 3. DEFECT-BACKGROUND EVALUATION OF $\partial_i \bar{q}^a$

For the locked one-defect background

$$\boxed{\Psi_B(\xi, x_{\parallel}) = \phi_0(x_{\parallel}) z_B(\xi), \quad \bar{\rho} = |\phi_0|^2,} \quad (595)$$

the background charge density is

$$\boxed{\bar{q}^a = \frac{\bar{\Psi}^\dagger T^a \bar{\Psi}}{\bar{\rho}} = z_B^\dagger T^a z_B =: m_a(\theta).} \quad (596)$$

Therefore

$$\boxed{\partial_i \bar{q}^a = \delta_{i\xi} \theta'(\xi) \partial_\theta m_a(\theta).} \quad (597)$$

So on a genuine defect background, the q-gradient source is automatically present unless

$$\partial_\theta m_a(\theta) \equiv 0$$

for the relevant internal component.

CCXXIII. 4. EXPLICIT Q-GRADIENT FORMULA FOR THE FIRST UNAVOIDABLE SOURCE

Substituting (597) into (593), only the ξ -component survives:

$$\boxed{H_{EE, \text{qgrad}}^{(2)} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[(\partial_\xi M_E^a) (\theta'(\xi) \partial_\theta m_a(\theta) \chi_E)^\dagger + (\theta'(\xi) \partial_\theta m_a(\theta) \chi_E) (\partial_\xi M_E^a)^\dagger \right].} \quad (598)$$

This is the first unavoidable odd-source formula beyond the rigid truncation.

CCXXIV. 5. WHY THE Q-GRADIENT SOURCE IS THE FIRST GENERIC CANDIDATE

The projector/internal-texture contribution $H_{EE, \text{proj}}^{(2)}$ requires

$$\partial_i Q_E^a \neq 0,$$

which in turn needs a non-rigid projector/internal completion beyond the locked first-shell orbit ansatz.

By contrast, the q-gradient piece requires only:

$$\boxed{\theta'(\xi) \neq 0, \quad \partial_\theta m_a(\theta) \neq 0,} \quad (599)$$

which are already natural on a genuine one-defect background.

Therefore the first generic candidate for a nonzero $H_{EE}^{(2)}$ on the frozen corrected branch is

$$\boxed{H_{EE, \text{qgrad}}^{(2)},} \quad (600)$$

while

$$H_{EE, \text{proj}}^{(2)}$$

is the first optional completion-level correction.

CCXXV. 6. NECESSARY AND SUFFICIENT ODD-PROJECTION CONDITION

A nonzero q-gradient source still requires survival of the odd projector. From (598), the exact condition is

$$\boxed{\mathfrak{P}_O \left[(\partial_\xi M_E^a) \chi_E^\dagger + \chi_E (\partial_\xi M_E^a)^\dagger \right] \neq 0} \quad (601)$$

on a set of nonzero ξ -measure where

$$\theta'(\xi) \partial_\theta m_a(\theta) \neq 0.$$

So the first nonzero odd transverse activation is controlled by:

1. the defect profile gradient $\theta'(\xi)$,
2. the internal rotation rate $\partial_\theta m_a(\theta)$,
3. the odd-projected phase-resolved overlap between $\partial_\xi M_E^a$ and χ_E .

CCXXVI. 7. DOMINANT CORRECTED ODD SHIFT

Because the corrected odd shift uses

$$Q_2(\delta_{\text{corr}}) = 4.608 \times 10^{-5} H_{AA}^{(2)} + 4.1472 \times 10^{-4} H_{EE}^{(2)} + 4.608 \times 10^{-5} H_{OO}^{(2)} - 2.7648 \times 10^{-4} H_{AE}^{(2)} + 9.216 \times 10^{-5} H_{AO}^{(2)} - 2.7648 \times 10^{-4} H_{EO}^{(2)},$$

the EE -slot is the most heavily weighted one.

Hence, once (601) is satisfied and no large cancellation occurs, the leading corrected odd coefficient is

$$\boxed{\beta^{\text{corr}} \approx -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 (4.1472 \times 10^{-4}) H_{EE, \text{qgrad}}^{(2)}} \quad (602)$$

CCXXVII. FINAL SUMMARY

The exact E -sector odd source is

$$H_{EE}^{(2), \text{fast}} = H_{EE, \text{proj}}^{(2)} + H_{EE, \text{qgrad}}^{(2)}.$$

The projector/internal-texture piece is

$$H_{EE, \text{proj}}^{(2)} \propto \mathfrak{P}_O \left[(\partial_i M_E^a) ((\partial_i Q_E^a) \bar{\chi})^\dagger + \text{h.c.} \right],$$

while the q-gradient piece is

$$H_{EE, \text{qgrad}}^{(2)} \propto \mathfrak{P}_O \left[(\partial_\xi M_E^a) (\theta'(\xi) \partial_\theta m_a(\theta) \chi_E)^\dagger + \text{h.c.} \right].$$

In the rigid first-shell truncation both vanish, so

$$H_{EE}^{(2)} = 0, \quad \beta^{\text{corr}} = 0.$$

Beyond that truncation, the first generic candidate is the q-gradient source, because it is already carried by the genuine defect background. Therefore the first decisive microscopic test is:

$$\boxed{\mathfrak{P}_O \left[(\partial_\xi M_E^a) \chi_E^\dagger + \chi_E (\partial_\xi M_E^a)^\dagger \right] \stackrel{?}{\neq} 0.}$$

CCXXVIII. FINAL SUMMARY

$$\boxed{\text{Current stage: first actual decision between the two } E\text{-sector odd sources.}} \quad (603)$$

The exact completed orbit-current formulas are

$$J_{E,i}^a = \partial_i M_E^a, \quad K_{E,i}^a = (\partial_i Q_E^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_E.$$

So

$$H_{EE}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger]$$

splits exactly into

$$H_{EE,\text{proj}}^{(2)} + H_{EE,\text{qgrad}}^{(2)}.$$

For the defect background

$$\bar{q}^a = z_B^\dagger T^a z_B =: m_a(\theta),$$

one has

$$\partial_i \bar{q}^a = \delta_{i\xi} \theta'(\xi) \partial_\theta m_a(\theta).$$

Therefore the first unavoidable post-rigid source is

$$\boxed{H_{EE,\text{qgrad}}^{(2)} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[(\partial_\xi M_E^a)(\theta'(\xi) \partial_\theta m_a(\theta) \chi_E)^\dagger + \text{h.c.}].}$$

In the minimal rigid-internal first-shell truncation,

$$Q_E^a = 0, \quad \partial_i \bar{q}^a = 0,$$

so

$$H_{EE}^{(2)} = 0, \quad \beta^{\text{corr}} = 0.$$

Hence the first generic candidate for $\Lambda_O \neq 0$ beyond the rigid truncation is not the projector/internal-texture source but the background q-gradient source.

$$\boxed{\text{Goal C: evaluate } \partial_\theta m_a(\theta) \text{ and the odd-projected overlap } \mathfrak{P}_O[(\partial_\xi M_E^a) \chi_E^\dagger + \chi_E (\partial_\xi M_E^a)^\dagger].} \quad (604)$$

$$\boxed{\text{Goal M: decide whether } H_{EE,\text{qgrad}}^{(2)} \neq 0 \text{ already on the frozen corrected branch.}} \quad (605)$$

$$\boxed{\text{Goal S: if yes, use } H_{EE}^{(2)} \text{ as the leading source for } \beta^{\text{corr}}; \text{ if not, move to } H_{EE,\text{proj}}^{(2)}.} \quad (606)$$

CCXXIX. ACTUAL COMPUTATION: THE q -GRADIENT SOURCE IS THE FIRST GENERIC ODD TRANSVERSE ACTIVATION

CCXXX. SHORT EXPLANATION

The completed E -sector currents are

$$\boxed{J_{E,i}^a = \partial_i M_E^a, \quad K_{E,i}^a = (\partial_i Q_E^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_E.} \quad (607)$$

Hence

$$\boxed{H_{EE}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger] = H_{EE,\text{proj}}^{(2)} + H_{EE,\text{qgrad}}^{(2)}.} \quad (608)$$

We now show that, on the explicit embedded $SU(2)_{(13)}$ defect branch, the q -gradient source is the first generic nonzero candidate beyond the rigid truncation.

CCXXXI. 1. EXPLICIT DEFECT BRANCH DATA

Take the embedded $SU(2)_{(13)}$ generators

$$T_1 = \frac{1}{2} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad T_2 = \frac{1}{2} \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad T_3 = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (609)$$

Take the real defect representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}. \quad (610)$$

Define

$$m_a(\theta) := z_B^\dagger T_a z_B, \quad k_a(\theta) := \text{Tr} \left[T_a (e_1 z_B^\dagger + z_B e_1^\dagger) \right], \quad A_a(\theta) := e_1^\dagger T_a z_B + z_B^\dagger T_a e_1.$$

Direct evaluation gives

$$m_1 = \frac{1}{2} \sin 2\theta, \quad m_2 = 0, \quad m_3 = \frac{1}{2} \cos 2\theta, \quad (611)$$

$$k_1 = \cos 2\theta, \quad k_2 = 0, \quad k_3 = -\sin 2\theta, \quad (612)$$

and

$$A_a(\theta) = k_a(\theta). \quad (613)$$

:contentReference[oaicite:2]index=2

CCXXXII. 2. EXACT DERIVATIVE IDENTITY

Differentiate (611) with respect to θ :

$$\partial_\theta m_1 = \cos 2\theta, \quad \partial_\theta m_2 = 0, \quad \partial_\theta m_3 = -\sin 2\theta.$$

Comparing with (612), we obtain the exact identity

$$\partial_\theta m_a(\theta) = k_a(\theta) = A_a(\theta). \quad (614)$$

Now on the defect background

$$\bar{q}^a = z_B^\dagger T_a z_B = m_a(\theta),$$

so

$$\partial_i \bar{q}^a = \delta_{i\xi} \theta'(\xi) \partial_\theta m_a(\theta) = \delta_{i\xi} \theta'(\xi) k_a(\theta). \quad (615)$$

This is the key exact computation: the q -gradient source is aligned with the tangent adjoint vector k_a .

CCXXXIII. 3. EXPLICIT Q-GRADIENT CONTRIBUTION TO $H_{EE}^{(2)}$

Substitute (615) into the q -gradient piece:

$$H_{EE,\text{qgrad}}^{(2)} = -\frac{2\alpha_X\beta_X}{M_X^2}\mathfrak{P}_O\left[(\partial_i M_E^a)((\partial_i \bar{q}^a)\chi_E)^\dagger + ((\partial_i \bar{q}^a)\chi_E)(\partial_i M_E^a)^\dagger\right]. \quad (616)$$

Only the normal component survives, so

$$H_{EE,\text{qgrad}}^{(2)} = -\frac{2\alpha_X\beta_X}{M_X^2}\mathfrak{P}_O\left[(\partial_\xi M_E^a)(\theta'(\xi)k_a(\theta)\chi_E)^\dagger + (\theta'(\xi)k_a(\theta)\chi_E)(\partial_\xi M_E^a)^\dagger\right]. \quad (617)$$

Using $A_a = k_a$, one may also write

$$H_{EE,\text{qgrad}}^{(2)} = -\frac{2\alpha_X\beta_X}{M_X^2}\mathfrak{P}_O\left[(\partial_\xi M_E^a)(\theta'(\xi)A_a(\theta)\chi_E)^\dagger + (\theta'(\xi)A_a(\theta)\chi_E)(\partial_\xi M_E^a)^\dagger\right]. \quad (618)$$

Thus the first generic odd source is carried by the same tangent internal direction that already appeared in the branch geometry.

CCXXXIV. 4. PARITY CHECK

For the reflection-symmetric defect profile,

$$\theta(-\xi) = \theta(\xi),$$

one has

$$\theta'(-\xi) = -\theta'(\xi).$$

Since $m_a(\theta)$, $k_a(\theta)$, and $A_a(\theta)$ are functions of θ only, they are even in ξ . Therefore

$$\partial_i \bar{q}^a = \delta_{i\xi} \theta'(\xi) k_a(\theta)$$

is odd in ξ .

This matches the general odd-kernel requirement for the transport operator built from defect gradients. So the q -gradient source is not parity-forbidden. :contentReference[oaicite:3]index=3

CCXXXV. 5. WHY THE PROJECTOR/INTERNAL SOURCE IS SUBLEADING GENERICALLY

The projector/internal source is

$$H_{EE,\text{proj}}^{(2)} = -\frac{2\alpha_X\beta_X}{M_X^2}\mathfrak{P}_O\left[(\partial_i M_E^a)((\partial_i Q_E^a)\bar{\chi})^\dagger + ((\partial_i Q_E^a)\bar{\chi})(\partial_i M_E^a)^\dagger\right]. \quad (619)$$

This requires

$$\partial_i Q_E^a \neq 0,$$

which is absent in the minimal rigid-internal first-shell truncation and only appears after a non-rigid projector/internal completion.

By contrast, the q -gradient piece requires only

$$\theta'(\xi) \neq 0 \quad \text{and} \quad k_a(\theta) \neq 0,$$

both of which are already present on the genuine defect background.

Therefore the first generic candidate beyond the rigid truncation is

$$H_{EE,\text{qgrad}}^{(2)}, \quad (620)$$

not $H_{EE,\text{proj}}^{(2)}$.

CCXXXVI. 6. EXPLICIT NONZERO CRITERION

From (617), a sufficient condition for a nonzero odd E -sector source is

$$\boxed{\Re_O \left[(\partial_\xi M_E^a) (k_a \chi_E)^\dagger + (k_a \chi_E) (\partial_\xi M_E^a)^\dagger \right] \neq 0} \quad (621)$$

on a set of nonzero ξ -measure where $\theta'(\xi) \neq 0$.

Because the corrected odd shift weights $H_{EE}^{(2)}$ most strongly, once (621) holds and no accidental cancellation occurs, the leading corrected odd coefficient is

$$\boxed{\beta^{\text{corr}} \approx -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 (4.1472 \times 10^{-4}) H_{EE, \text{qgrad}}^{(2)}} \quad (622)$$

CCXXXVII. 7. FINAL THEOREM

Theorem CCXXXVII.1 (First generic odd source on the frozen corrected branch). *On the embedded $SU(2)_{(13)}$ defect branch one has*

$$\partial_\theta m_a(\theta) = k_a(\theta) = A_a(\theta),$$

hence

$$\partial_i \bar{q}^a = \delta_{i\xi} \theta'(\xi) k_a(\theta).$$

Therefore, beyond the rigid first-shell truncation, the first generic candidate for a nonzero odd transverse response in the dominant E -sector slot is

$$H_{EE, \text{qgrad}}^{(2)},$$

not $H_{EE, \text{proj}}^{(2)}$.

A sufficient condition for activation is

$$\Re_O \left[(\partial_\xi M_E^a) (k_a \chi_E)^\dagger + (k_a \chi_E) (\partial_\xi M_E^a)^\dagger \right] \neq 0$$

on a region where $\theta'(\xi) \neq 0$.

Proof. The derivative identity follows by direct differentiation of the explicit $m_a(\theta)$. Substituting it into the completed E -sector current formula gives the explicit q-gradient source. The projector/internal source requires an additional non-rigid completion $\partial_i Q_E^a \neq 0$, whereas the q-gradient source is already carried by the defect background itself. $\square$

CCXXXVIII. FINAL SUMMARY

The exact embedded defect branch satisfies

$$m_1 = \frac{1}{2} \sin 2\theta, \quad m_2 = 0, \quad m_3 = \frac{1}{2} \cos 2\theta,$$

$$k_1 = \cos 2\theta, \quad k_2 = 0, \quad k_3 = -\sin 2\theta,$$

so

$$\boxed{\partial_\theta m_a = k_a = A_a.}$$

Hence on the defect background

$$\boxed{\partial_i \bar{q}^a = \delta_{i\xi} \theta'(\xi) k_a(\theta),}$$

and the dominant E -sector odd source becomes

$$H_{EE, \text{qgrad}}^{(2)} = -\frac{2\alpha_X \beta_X}{M_X^2} \Re_O \left[(\partial_\xi M_E^a) (\theta' k_a \chi_E)^\dagger + (\theta' k_a \chi_E) (\partial_\xi M_E^a)^\dagger \right].$$

Therefore the first generic nonzero source beyond the rigid truncation is the q-gradient source, not the projector/internal one.

CCXXXIX. FINAL SUMMARY

Current stage: the first generic odd transverse source has now been identified explicitly. (623)

The completed E -sector currents are

$$J_{E,i}^a = \partial_i M_E^a, \quad K_{E,i}^a = (\partial_i Q_E^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_E.$$

On the embedded $SU(2)_{(13)}$ defect branch,

$$m_1 = \frac{1}{2} \sin 2\theta, \quad m_2 = 0, \quad m_3 = \frac{1}{2} \cos 2\theta,$$

$$k_1 = \cos 2\theta, \quad k_2 = 0, \quad k_3 = -\sin 2\theta,$$

and therefore

$$\partial_\theta m_a = k_a = A_a.$$

:contentReference[oaicite:4]index=4

Hence the background charge gradient is

$$\partial_i \bar{q}^a = \delta_{i\xi} \theta'(\xi) k_a(\theta),$$

so the q-gradient source is already present on a genuine defect background and is not parity-forbidden.

The dominant E -sector odd source is therefore

$$H_{EE,\text{qgrad}}^{(2)} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[(\partial_\xi M_E^a) (\theta' k_a \chi_E)^\dagger + (\theta' k_a \chi_E) (\partial_\xi M_E^a)^\dagger \right].$$

So beyond the rigid truncation, the first generic nonzero candidate is the q-gradient source, not the projector/internal source.

$$\text{Goal C: compute the odd-projected overlap } \mathfrak{P}_O \left[(\partial_\xi M_E^a) (k_a \chi_E)^\dagger + (k_a \chi_E) (\partial_\xi M_E^a)^\dagger \right]. \quad (624)$$

$$\text{Goal M: decide whether } H_{EE,\text{qgrad}}^{(2)} \neq 0 \text{ already on the frozen corrected branch.} \quad (625)$$

$$\text{Goal S: if yes, use } H_{EE}^{(2)} \text{ as the leading source for } \beta^{\text{corr}}; \text{ only if not, move to } \partial_i Q_E^a. \quad (626)$$

CCXL. ACTUAL FOURIER-SHELL FORMULA FOR THE FIRST GENERIC ODD SOURCE

CCXLI. SHORT EXPLANATION

At this point the only genuinely missing ingredient is not a new theorem. It is the finite harmonic overlap evaluation on the frozen corrected branch.

The completed Fourier-space orbit currents are already explicit:

$$\widehat{M}_r^a(Q) = \sum_G \left(\bar{A}_G^\dagger T^a B_{G+Q}^{(r)} + B_G^{(r)\dagger} T^a \bar{A}_{G+Q} \right), \quad (627)$$

$$\widehat{J}_{r,i}^a(Q) = i Q_i \widehat{M}_r^a(Q), \quad (628)$$

$$\widehat{K}_{r,i}^a(Q) = i Q_i \widehat{Q}_r^a(Q) \bar{\chi} + (\partial_i \bar{q}^a) \widehat{\chi}_r(Q). \quad (629)$$

So the remaining odd-source audit is a literal finite overlap computation. :contentReference[oaicite:2]index=2

CCXLII. 1. EXACT E -SECTOR Q-GRADIENT CONTRIBUTION IN FOURIER SPACE

The fast six-row coefficient is

$$H_{rs}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\sum_Q \hat{J}_{r,i}^a(Q) \hat{K}_{s,i}^a(-Q)^\dagger + \sum_Q \hat{J}_{s,i}^a(Q) \hat{K}_{r,i}^a(-Q)^\dagger + \text{h.c.} \right]. \quad (630)$$

For the E -sector q-gradient source, keep only the $(\partial_i \bar{q}^a) \hat{\chi}_E$ term in $\hat{K}_E$ and set $r = s = E$. Then

$$H_{EE,\text{qgrad}}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\sum_Q \hat{J}_{E,i}^a(Q) ((\partial_i \bar{q}^a) \hat{\chi}_E(-Q))^\dagger + \text{h.c.} \right]. \quad (631)$$

Substituting (628),

$$H_{EE,\text{qgrad}}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\sum_Q iQ_i \widehat{M}_E^a(Q) ((\partial_i \bar{q}^a) \hat{\chi}_E(-Q))^\dagger + \text{h.c.} \right]. \quad (632)$$

CCXLIII. 2. DEFECT-BACKGROUND SPECIALIZATION

On the embedded one-defect background,

$$\partial_i \bar{q}^a = \delta_{i\xi} \theta'(\xi) k_a(\theta). \quad (633)$$

Therefore only the normal component survives:

$$H_{EE,\text{qgrad}}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\sum_Q iQ_\xi \widehat{M}_E^a(Q) (\theta'(\xi) k_a(\theta) \hat{\chi}_E(-Q))^\dagger + \text{h.c.} \right]. \quad (634)$$

Assume the background gauge is chosen so that $\theta'(\xi) k_a(\theta)$ is real. Then for each Q ,

$$iQ_\xi \widehat{M}_E^a(Q) (\theta' k_a \hat{\chi}_E(-Q))^\dagger + \text{h.c.} = -2Q_\xi \theta' k_a \Im \left(\widehat{M}_E^a(Q) \widehat{\chi}_E(-Q) \right).$$

Hence

$$H_{EE,\text{qgrad}}^{(2),\text{fast}} = \frac{4\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\sum_Q Q_\xi \theta'(\xi) k_a(\theta) \Im \left(\widehat{M}_E^a(Q) \widehat{\chi}_E(-Q) \right) \right]. \quad (635)$$

This is the first genuine finite shell-overlap formula for the odd source.

CCXLIV. 3. EXACT NONZERO CRITERION

Equation (635) shows that a sufficient condition for a nonzero odd E -sector source is the existence of at least one shell momentum Q such that

$$Q_\xi \neq 0, \quad \Im \left(\widehat{M}_E^a(Q) \widehat{\chi}_E(-Q) \right) \neq 0, \quad \mathfrak{P}_O \text{ does not kill the corresponding component.} \quad (636)$$

Equivalently, the first generic odd transverse activation requires:

1. a shell harmonic with nonzero normal momentum component;

2. a genuine phase misalignment between the E -sector orbit current $\widehat{M}_E^a(Q)$ and the worldvolume derivative mode $\widehat{\chi}_E(Q)$;
3. survival of that component under the odd projector.

Thus the first generic source is not an arbitrary background scalar. It is the odd-projected *imaginary shell overlap* between the E -sector current and the E -sector test-mode derivative.

CCXLIV. 4. DOMINANT FIRST-SHELL APPROXIMATION

If the frozen corrected branch is sharply peaked on the selected first shell and the dominant contribution comes from the two opposite selected shell modes $Q = \pm G_*$, then using the reality relations

$$\widehat{M}_E^a(-G_*) = \overline{\widehat{M}_E^a(G_*)}, \quad \widehat{\chi}_E(-G_*) = \overline{\widehat{\chi}_E(G_*)},$$

the two terms combine and give

$$H_{EE, \text{qgrad}}^{(2), \text{fast}} \approx \frac{8\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[G_{*\xi} \theta'(\xi) k_a(\theta) \Im \left(\widehat{M}_E^a(G_*) \overline{\widehat{\chi}_E(G_*)} \right) \right]. \quad (637)$$

So in the first-shell-dominant regime, the whole problem collapses to a *single shell-phase overlap test*.

CCXLVI. 5. LEADING CORRECTED ODD COEFFICIENT

The corrected odd shift is

$$\beta^{\text{corr}} \approx -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 \left[4.608 \times 10^{-5} H_{AA}^{(2), \text{fast}} + 4.1472 \times 10^{-4} H_{EE}^{(2), \text{fast}} + 4.608 \times 10^{-5} H_{OO}^{(2), \text{fast}} - 2.7648 \times 10^{-4} H_{AE}^{(2), \text{fast}} + 9.216 \times 10^{-5} H_{OE}^{(2), \text{fast}} \right]. \quad (638)$$

Since the EE -slot has the largest weight, once (636) holds and no accidental cancellation occurs,

$$\beta^{\text{corr}} \approx -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 (4.1472 \times 10^{-4}) H_{EE, \text{qgrad}}^{(2), \text{fast}}. \quad (639)$$

Substituting (635),

$$\beta^{\text{corr}} \approx -\frac{\sqrt{3}\pi}{4} \frac{\alpha_X \beta_X}{M_X^2} \theta_0 \kappa A_0 (4.1472 \times 10^{-4}) \mathfrak{P}_O \left[\sum_Q Q_\xi \theta'(\xi) k_a(\theta) \Im \left(\widehat{M}_E^a(Q) \overline{\widehat{\chi}_E(-Q)} \right) \right]. \quad (640)$$

CCXLVII. 6. WHAT IS COMPUTED AND WHAT REMAINS

What is now computed exactly:

$$\text{the first generic odd source is the } E\text{-sector } q\text{-gradient shell overlap} \quad (641)$$

and its exact finite formula is (635).

What remains is a literal finite harmonic evaluation: insert the actual frozen-branch harmonics

$$\bar{A}_G, \quad B_G^{(E)}$$

into

$$\widehat{M}_E^a(Q), \quad \widehat{\chi}_E(Q),$$

and test whether

$$\Im \left(\widehat{M}_E^a(Q) \overline{\widehat{\chi}_E(-Q)} \right)$$

survives the odd projector.

CCXLVIII. FINAL SUMMARY

The completed Fourier-space currents are already explicit:

$$\widehat{M}_r^a(Q), \quad \widehat{J}_{r,i}^a(Q) = iQ_i \widehat{M}_r^a(Q), \quad \widehat{K}_{r,i}^a(Q) = iQ_i \widehat{Q}_r^a(Q) \bar{\chi} + (\partial_i \bar{q}^a) \widehat{\chi}_r(Q).$$

On the defect background

$$\partial_i \bar{q}^a = \delta_{i\xi} \theta'(\xi) k_a(\theta),$$

the first generic odd E -sector source is

$$H_{EE,\text{qgrad}}^{(2),\text{fast}} = \frac{4\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\sum_Q Q_\xi \theta'(\xi) k_a(\theta) \Im \left(\widehat{M}_E^a(Q) \overline{\widehat{\chi}_E(-Q)} \right) \right].$$

So the first actual activation condition for $\Lambda_O \neq 0$ is not vague anymore. It is the finite shell-phase criterion

$$\exists Q : Q_\xi \neq 0, \Im \left(\widehat{M}_E^a(Q) \overline{\widehat{\chi}_E(-Q)} \right) \neq 0, \text{ and the odd projector survives.}$$

In the first-shell-dominant regime this reduces to the single overlap test

$$\Im \left(\widehat{M}_E^a(G_*) \overline{\widehat{\chi}_E(G_*)} \right) \stackrel{?}{\neq} 0.$$

CCXLIX. FINAL SUMMARY

Current stage: the first generic odd source has now been reduced to an explicit finite shell-overlap computation. (642)

The relevant completed Fourier currents are

$$\widehat{M}_r^a(Q) = \sum_G \left(\bar{A}_G^\dagger T^a B_{G+Q}^{(r)} + B_G^{(r)\dagger} T^a \bar{A}_{G+Q} \right),$$

$$\widehat{J}_{r,i}^a(Q) = iQ_i \widehat{M}_r^a(Q), \quad \widehat{K}_{r,i}^a(Q) = iQ_i \widehat{Q}_r^a(Q) \bar{\chi} + (\partial_i \bar{q}^a) \widehat{\chi}_r(Q).$$

:contentReference[oaicite:3]index=3

On the embedded defect background,

$$\partial_i \bar{q}^a = \delta_{i\xi} \theta'(\xi) k_a(\theta),$$

so the first generic E -sector odd source is

$$H_{EE,\text{qgrad}}^{(2),\text{fast}} = \frac{4\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\sum_Q Q_\xi \theta'(\xi) k_a(\theta) \Im \left(\widehat{M}_E^a(Q) \overline{\widehat{\chi}_E(-Q)} \right) \right].$$

Therefore the first actual activation condition for $\Lambda_O \neq 0$ is the finite shell-phase criterion

$$\exists Q : Q_\xi \neq 0, \Im \left(\widehat{M}_E^a(Q) \overline{\widehat{\chi}_E(-Q)} \right) \neq 0, \text{ and the odd projector survives.}$$

In the first-shell-dominant regime this collapses to the single test

$$\Im \left(\widehat{M}_E^a(G_*) \overline{\widehat{\chi}_E(G_*)} \right) \stackrel{?}{\neq} 0.$$

Goal C: insert the actual frozen-branch harmonics $\bar{A}_G, B_G^{(E)}$ into $\widehat{M}_E^a, \widehat{\chi}_E$.

(643)

Goal M: evaluate $\Im(\widehat{M}_E^a(G_*)\widehat{\chi}_E(G_*))$ and the odd projector.

(644)

Goal S: if nonzero, this is the first explicit microscopic activation of $\Lambda_O \neq 0$ on the frozen corrected branch.

(645)

CCL. IMMEDIATE FORWARD ROUTE: EXPLICIT ODD EXTRA CARRIER INSTEAD OF THE UNAVAILABLE FROZEN-HARMONIC JK AUDIT

CCLI. SHORT EXPLANATION

The JK -route q-gradient audit has been reduced as far as possible using the currently available files. Its remaining final step is the literal finite harmonic evaluation of

$$\Im(\widehat{M}_E^a(G_*)\widehat{\chi}_E(G_*)),$$

which requires the actual frozen-branch harmonics

$$\bar{A}_G, \quad B_G^{(E)}.$$

These harmonics are not numerically or symbolically instantiated in the present conversation context. :contentReference[oaicite:4]index=4 :contentReference[oaicite:5]index=5

So the correct non-stalling move is to switch to the already explicit lowest-order complete realization route, where the odd transverse carrier is fully written and directly nonzero on the defect background. :contentReference[oaicite:6]index=6

CCLII. 1. EXPLICIT ODD EXTRA CARRIER

On the embedded $SU(2)_{(13)}$ defect branch, define

$M_O := i(e_2 z_B^\dagger - z_B e_2^\dagger),$

(646)

and the odd generator

$S_O := \frac{1}{2} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$

(647)

Then

$q_O(\theta) := \text{Tr}(S_O M_O) = \cos \theta.$

(648)

Therefore the explicit odd extra slot is

$\mathcal{I}_O = X_i^{(O)} \text{Tr}(S_O M_O) J_j D_j \psi = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi.$

(649)

So whenever

$$\cos \theta \neq 0,$$

the odd extra carrier is locally nonzero. :contentReference[oaicite:7]index=7

CCLIII. 2. INDUCED TANGENT TRANSPORT BLOCK

The induced tangent transport block from the explicit extra E/O slots is

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right]. \quad (650)$$

Therefore

$$a_3^{(EO)} = 0, \quad b_E^{(EO)} = \tilde{\Lambda}_E \theta'(\xi) \cos \theta(\xi), \quad b_O^{(EO)} = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi). \quad (651)$$

Thus the odd transport density is explicitly present at lowest order:

$$b_O^{(EO)} \neq 0 \quad \text{whenever} \quad \tilde{\Lambda}_O \neq 0, \theta'(\xi) \neq 0, \cos \theta(\xi) \neq 0. \quad (652)$$

:contentReference[oaicite:8]index=8

CCLIV. 3. CONSEQUENCE FOR THE REDUCED COEFFICIENTS

This route gives a direct explicit odd transverse carrier without needing the unavailable frozen-harmonic shell-overlap data.

In the reduced coefficient language, this means that the odd transverse coefficient can now be fed through the explicit extra-slot branch:

$$\beta \sim I_\theta \tilde{\Lambda}_O, \quad (653)$$

with local density proportional to

$$\theta'(\xi) \cos \theta(\xi).$$

So at the current stage, the honest forward statement is:

$$\boxed{\text{the odd transverse channel is explicitly realizable on the complete lowest-order carrier route.}} \quad (654)$$

CCLV. 4. WHY THIS IS THE CORRECT NEXT MOVE

The JK -route remains valuable, but its final sign decision requires actual frozen branch harmonics:

$$\bar{A}_G, \quad B_G^{(E)}.$$

Those are not instantiated here.

By contrast, the explicit extra-carrier route already computes

$$q_O(\theta) = \cos \theta$$

and therefore gives a genuine nonzero odd channel directly on the defect background. This avoids stalling and advances the branch-level coefficient closure immediately. :contentReference[oaicite:9]index=9 :contentReference[oaicite:10]index=10

CCLVI. FINAL SUMMARY

The JK -route q-gradient audit has been reduced to one finite shell-overlap test, but that test still requires unavailable explicit frozen harmonics.

To move forward now, the correct branch is the explicit extra transverse carrier route.

Its exact odd carrier is

$$\mathcal{I}_O = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi,$$

and its induced transport block is

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right].$$

Therefore the odd transverse density is explicitly nonzero whenever

$$\tilde{\Lambda}_O \neq 0, \quad \theta'(\xi) \neq 0, \quad \cos \theta(\xi) \neq 0.$$

This is the first actual forward-computable odd channel on the present data.

CCLVII. FINAL SUMMARY

Current status: stop stalling on the unavailable frozen-harmonic JK overlap and move to the explicit odd extra-carrier route. (655)

The JK -route q-gradient audit has been reduced to the single overlap test

$$\Im \left(\widehat{M}_E^a(G_*) \overline{\widehat{\chi}_E(G_*)} \right) \stackrel{?}{\neq} 0,$$

but that literal evaluation still requires the actual frozen harmonics

$$\bar{A}_G, \quad B_G^{(E)},$$

which are not instantiated in the present conversation context. :contentReference[oaicite:11]index=11

By contrast, the explicit odd extra-carrier route is already completely concrete.

On the embedded defect branch,

$$q_O(\theta) = \text{Tr}(S_O M_O) = \cos \theta,$$

so

$$\mathcal{I}_O = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi$$

is locally nonzero whenever $\cos \theta \neq 0$. :contentReference[oaicite:12]index=12

The induced tangent transport block is

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right],$$

so in particular

$$b_O^{(EO)} = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi).$$

Hence the odd transport density is explicitly present whenever

$$\tilde{\Lambda}_O \neq 0, \quad \theta'(\xi) \neq 0, \quad \cos \theta(\xi) \neq 0.$$

:contentReference[oaicite:13]index=13

Goal C: adopt the explicit extra odd carrier as the active forward route. (656)

Goal M: insert $\mathcal{I}_O$ into the branch-level coefficient sheet and read off the resulting odd transverse coefficient. (657)

Goal S: return to the unavailable JK shell-overlap audit only after the actual frozen harmonics

$(\bar{A}_G, B_G^{(E)})$ are instantiated. (658)

CCLVIII. INSERTION OF THE EXPLICIT ODD EXTRA CARRIER INTO THE REDUCED COEFFICIENT SHEET

CCLIX. SHORT EXPLANATION

At the present stage, the correct move is no longer to search for a new microscopic source. The explicit odd transverse carrier is already available in the complete lowest-order realization.

The relevant tangent transport block is

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right]. \quad (659)$$

Hence the local transport densities contributed by the EO -slot are

$$a_3^{(EO)} = 0, \quad b_E^{(EO)} = \tilde{\Lambda}_E \theta'(\xi) \cos \theta(\xi), \quad b_O^{(EO)} = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi). \quad (660)$$

So the explicit odd channel is already present at the local tangent-transport level. :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8

CCLX. 1. PROJECTION TO THE BLOCH ZERO MODES

Let $u_a^\pm(\xi) = f_\pm(\xi) e_a(\xi)$ be the opposite-valley low modes. Projection gives the reduced transport matrix

$$(M^\mu)_{ab} = \int d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^\mu(\xi) e_b(\xi). \quad (661)$$

Define the overlap integrals

$$I_3 := \int d\xi f_+ f_- \theta'(\xi)^3, \quad I_c := \int d\xi f_+ f_- \theta'(\xi) \cos \theta(\xi). \quad (662)$$

Then the reduced coefficients are

$$\lambda_\parallel = \frac{\lambda_c}{2} I_3, \quad \alpha = \tilde{\Lambda}_E I_c, \quad \beta = \tilde{\Lambda}_O I_c. \quad (663)$$

Equivalently, with the notation $I_{\theta, \cos}$,

$$\alpha = \tilde{\Lambda}_E I_{\theta, \cos}, \quad \beta = \tilde{\Lambda}_O I_{\theta, \cos}. \quad (664)$$

:contentReference[oaicite:9]index=9 :contentReference[oaicite:10]index=10 :contentReference[oaicite:11]index=11

CCLXI. 2. EXPLICIT REDUCED PDE

Therefore the solved lowest-order Branch-B reduced PDE becomes

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_\parallel) - I_c \tilde{\Lambda}_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - I_c \tilde{\Lambda}_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0. \quad (665)$$

Thus the odd transverse coefficient is not schematic anymore. It is exactly

$$\beta = \tilde{\Lambda}_O I_c. \quad (666)$$

:contentReference[oaicite:12]index=12 :contentReference[oaicite:13]index=13

CCLXII. 3. MINIMAL NONZERO CONDITION FOR THE ODD CHANNEL

From (666), the odd transverse channel is nonzero if and only if

$$\boxed{\tilde{\Lambda}_O \neq 0 \quad \text{and} \quad I_c \neq 0.} \quad (667)$$

Here

$$I_c = \int d\xi f_+ f_- \theta'(\xi) \cos \theta(\xi)$$

is the geometric overlap. So the odd channel requires:

1. a nonzero microscopic odd transverse carrier coefficient $\tilde{\Lambda}_O$,
2. a nonvanishing defect/zero-mode overlap I_c .

This is the exact branch-level odd-channel reading.

CCLXIII. 4. RELATION TO THE SOURCE TABLE

At principal order, the source table is:

$$\boxed{a_3^{(\text{prim})} = \frac{\lambda_c}{2} \theta'(\xi)^3, \quad a_3^{(K)} = 0, \quad a_3^{(EO)} = 0.} \quad (668)$$

So the longitudinal source is uniquely primitive, while both K -sector and EO -slot are purely transverse. Therefore the odd channel does not compete with the longitudinal one; it simply enters the transverse reduced coefficient sheet through $\tilde{\Lambda}_O$. :contentReference[oaicite:14]index=14 :contentReference[oaicite:15]index=15 :contentReference[oaicite:16]index=16

CCLXIV. 5. EXACT BRANCH-LEVEL DELIVERABLE

The exact current deliverable is now:

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = \tilde{\Lambda}_E I_c, \quad \beta = \tilde{\Lambda}_O I_c.} \quad (669)$$

So the actual microscopic frontier is reduced to determining the coefficient point

$$\lambda_c, \quad \tilde{\Lambda}_E, \quad \tilde{\Lambda}_O,$$

with the odd channel already read explicitly through $\tilde{\Lambda}_O$.

CCLXV. 6. HONEST CURRENT STATUS

What is now genuinely closed is the branch-level coefficient reading of the explicit odd carrier:

$$\boxed{\mathcal{I}_O \implies b_O(\xi) = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi) \implies \beta = \tilde{\Lambda}_O I_c.}$$

What is still open is not the reduced reading itself, but the actual microscopic value/sign of

$$\tilde{\Lambda}_O.$$

That is the remaining coefficient problem.

CCLXVI. FINAL SUMMARY

The explicit extra odd carrier route is now fully inserted into the reduced coefficient sheet.
Its local transport density is

$$b_O(\xi) = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi),$$

and after Bloch zero-mode projection the reduced odd transverse coefficient is

$$\beta = \tilde{\Lambda}_O I_c, \quad I_c = \int d\xi f_+ f_- \theta'(\xi) \cos \theta(\xi).$$

Therefore the odd transverse channel is nonzero exactly when

$$\tilde{\Lambda}_O \neq 0 \quad \text{and} \quad I_c \neq 0.$$

This is the first fully explicit branch-level odd-channel formula available on the present data.

CCLXVII. FINAL SUMMARY

Current stage: the explicit odd extra carrier has now been fully inserted into the reduced coefficient sheet. (670)

The explicit EO -slot tangent transport block is

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right],$$

so its local densities are

$$a_3^{(EO)} = 0, \quad b_E^{(EO)} = \tilde{\Lambda}_E \theta'(\xi) \cos \theta(\xi), \quad b_O^{(EO)} = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi).$$

:contentReference[oaicite:17]index=17 :contentReference[oaicite:18]index=18

Projecting to the Bloch zero modes with

$$I_3 = \int d\xi f_+ f_- \theta'^3, \quad I_c = \int d\xi f_+ f_- \theta'(\xi) \cos \theta(\xi),$$

gives the reduced coefficients

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = \tilde{\Lambda}_E I_c, \quad \beta = \tilde{\Lambda}_O I_c.$$

:contentReference[oaicite:19]index=19 :contentReference[oaicite:20]index=20 :contentReference[oaicite:21]index=21

Therefore the odd channel is nonzero exactly when

$$\tilde{\Lambda}_O \neq 0 \quad \text{and} \quad I_c \neq 0.$$

So the reduced reading problem is closed. The remaining microscopic frontier is only the actual value/sign extraction of

$$\tilde{\Lambda}_O.$$

Goal C: keep the explicit odd carrier route as the active forward route. (671)

Goal M: determine the microscopic value/sign of $\tilde{\Lambda}_O$. (672)

Goal S: once $\tilde{\Lambda}_O \neq 0$ is fixed, the odd transverse coefficient is immediately $\beta = \tilde{\Lambda}_O I_c$. (673)

**CCLXVIII. ACTUAL READING OF THE ODD COEFFICIENT ON THE EXPLICIT
EXTRA-CARRIER ROUTE**

CCLXIX. GOAL

The local odd carrier has already been identified explicitly. The remaining task is to read its reduced coefficient and decide whether the present explicit branch can actually activate the odd channel.

CCLXX. EXACT REDUCED COEFFICIENT SHEET

For the solved lowest-order Branch-B realization, the reduced symbol is

$$A_1(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_{\parallel} \sigma_3 + I_{\theta, \cos} \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta, \cos} \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2). \quad (674)$$

Therefore

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta, \cos} \tilde{\Lambda}_E, \quad \beta = I_{\theta, \cos} \tilde{\Lambda}_O. \quad (675)$$

CCLXXI. MICROSCOPIC ONE-MEDIATOR COEFFICIENT POINT

At operator level, the one-mediator coefficient audit gives

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1. \quad (676)$$

If the shell basis is canonically normalized so that

$$C_{\parallel} = C_E = C_O = 1,$$

then

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} J_1. \quad (677)$$

In the transport-side canonical explicit branch notation, the same normalization layer is absorbed into the coefficient point, so one writes

$$\tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2}. \quad (678)$$

CCLXXII. EXPLICIT ODD COEFFICIENT

Substituting (676) into (675),

$$\beta = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1 I_{\theta, \cos}. \quad (679)$$

In the canonical explicit normalization this reduces to

$$\beta = -\frac{\beta_X \gamma_O}{M_X^2} I_{\theta, \cos} \quad (680)$$

if the transport-side normalization absorbs J_1 , or

$$\beta = -\frac{\beta_X \gamma_O}{M_X^2} J_1 I_{\theta, \cos} \quad (681)$$

if J_1 is kept explicit.

CCLXXIII. MINIMAL NONZERO CRITERION

Since

$$I_{\theta,\cos} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \cos \theta(\xi)$$

is generically nonzero for a monotone defect profile with nontrivial θ -variation and nonvanishing even-odd overlap, the odd channel is nonzero exactly when

$$\boxed{\beta_X \neq 0, \quad \gamma_O \neq 0, \quad I_{\theta,\cos} \neq 0,} \quad (682)$$

and, in the unabsorbed operator normalization,

$$\boxed{C_O J_1 \neq 0.} \quad (683)$$

CCLXXIV. WHY THE CURRENT EXPLICIT WITNESS BRANCH CANNOT ACTIVATE THE ODD CHANNEL

The explicit pure-even witness branch fixes

$$\boxed{T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0.} \quad (684)$$

Therefore

$$\boxed{\tilde{\Lambda}_O = 0, \quad \beta = 0.} \quad (685)$$

So the present explicit pure-even witness branch can never be used to read the odd coefficient.

CCLXXV. IMMEDIATE NEXT BRANCH DECISION

To activate the odd channel, one must move from the pure-even witness branch to a mixed transverse branch with

$$\boxed{\gamma_O \neq 0.} \quad (686)$$

Then the odd coefficient becomes

$$\boxed{\beta = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1 I_{\theta,\cos},} \quad (687)$$

or, in the transport-side canonical notation,

$$\boxed{\beta = -\frac{\beta_X \gamma_O}{M_X^2} I_{\theta,\cos}.} \quad (688)$$

Thus the odd-channel problem is no longer structural. It is now reduced to choosing the mixed branch and fixing the sign of γ_O relative to β_X and the overlap orientation.

CCLXXVI. FINAL SUMMARY

The explicit extra odd carrier route is already sufficient to read the odd coefficient. Its reduced coefficient is

$$\boxed{\beta = I_{\theta,\cos} \tilde{\Lambda}_O.}$$

At operator level,

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1,$$

so, in canonical explicit normalization,

$$\beta = -\frac{\beta_X \gamma_O}{M_X^2} I_{\theta, \cos}$$

up to the bookkeeping convention for J_1 .

Therefore the current pure-even explicit witness branch with $\gamma_O = 0$ necessarily has

$$\beta = 0.$$

To proceed, the next branch must satisfy

$$\gamma_O \neq 0.$$

CCLXXVII. IMMEDIATE UPGRADE TO THE MIXED TRANSVERSE BRANCH

CCLXXVIII. SHORT EXPLANATION

The explicit pure-even canonical witness branch is no longer useful for the odd coefficient. On that branch,

$$\gamma_E = 1, \quad \gamma_O = 0,$$

so

$$\beta = 0$$

is structurally forced. Therefore the next mathematically correct move is to leave the pure-even branch and move to the mixed transverse branch.

CCLXXIX. 1. MIXED-BRANCH MASTER SYMBOL

The exact mixed-branch reduced symbol is

$$A_1(p) = \frac{1}{2} \alpha_X L_* I_{\theta'^3} p_{\parallel} \sigma_3 + \beta_X I_{\theta} T_E (p_1 \sigma_1 + p_2 \sigma_2) + \beta_X I_{\theta} T_O (-p_2 \sigma_1 + p_1 \sigma_2). \quad (689)$$

Here

$$L_* := m_{\parallel} u_{\parallel}, \quad T_* := (T_E, T_O), \quad R_{\perp} := \sqrt{T_E^2 + T_O^2}. \quad (690)$$

The explicit overlap factors are

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3. \quad (691)$$

:contentReference[oaicite:4]index=4

CCLXXX. 2. EXACT COEFFICIENT SHEET ON THE MIXED BRANCH

Therefore the exact reduced coefficients are

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad (692)$$

$$\alpha = \beta_X I_{\theta} T_E = -\frac{\sqrt{3}\pi}{16} \beta_X T_E \theta_0 \kappa, \quad (693)$$

$$\beta = \beta_X I_{\theta} T_O = -\frac{\sqrt{3}\pi}{16} \beta_X T_O \theta_0 \kappa. \quad (694)$$

Hence the exact nondegeneracy condition is simply

$$L_* \neq 0, \quad R_{\perp} > 0. \quad (695)$$

Equivalently,

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (696)$$

:contentReference[oaicite:5]index=5

CCLXXXI. 3. POLAR TRANSVERSE PARAMETRIZATION

Write the transverse vector in polar form:

$$T_E = R_{\perp} \cos \varphi, \quad T_O = R_{\perp} \sin \varphi. \quad (697)$$

Then

$$\alpha = -\frac{\sqrt{3}\pi}{16} \beta_X R_{\perp} \cos \varphi \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \beta_X R_{\perp} \sin \varphi \theta_0 \kappa. \quad (698)$$

So the odd/even ratio is exactly

$$\frac{\beta}{\alpha} = \tan \varphi \quad (\cos \varphi \neq 0). \quad (699)$$

This is the cleanest branch-level reading of the odd coefficient:

$$\varphi = 0 \implies \beta = 0, \quad \varphi \neq 0 \implies \beta \neq 0$$

provided $\beta_X \theta_0 \kappa R_{\perp} \neq 0$.

CCLXXXII. 4. CANONICAL MIXED SAMPLE

The smallest nontrivial mixed sample is

$$\delta = c = u_3 = m_{\parallel} = 1, \quad T_E = 1, \quad T_O = 1. \quad (700)$$

Then

$$L_* = 1, \quad R_{\perp} = \sqrt{2}. \quad (701)$$

Hence

$$\alpha = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa. \quad (702)$$

So the canonical mixed sample gives

$$\alpha = \beta \quad (703)$$

up to the common overall sign $\text{sgn}(\beta_X \theta_0 \kappa)$.

Moreover the mixed-branch gain factor is

$$\frac{R_\perp}{|L_*|} = \sqrt{2}, \quad \frac{|L_*|}{R_\perp} = \frac{1}{\sqrt{2}}. \quad (704)$$

For the minimal real ansatz the Bloch-overlap scalar is

$$\Delta_{00} = \frac{\delta}{\sqrt{\delta^2 + c^2}} = \frac{1}{\sqrt{2}}. \quad (705)$$

So the longitudinal overlap remains order one while the transverse norm is enlarged by the $\sqrt{2}$ factor. :contentReference[oaicite:6]index=6

CCLXXXIII. 5. LEADING ANISOTROPY RATIO ON THE MIXED BRANCH

The exact leading anisotropy ratio is

$$\rho_{\text{LO}} = \frac{3}{32} \frac{\alpha_X}{\beta_X} \frac{L_*}{R_\perp} (\theta_0 \kappa)^2. \quad (706)$$

In the canonical mixed sample

$$L_* = 1, \quad R_\perp = \sqrt{2},$$

so

$$\rho_{\text{LO}} = \frac{3}{32\sqrt{2}} \frac{\alpha_X}{\beta_X} (\theta_0 \kappa)^2. \quad (707)$$

Therefore the mixed branch enlarges the transverse support by a factor $\sqrt{2}$, but it does not remove the fundamental small-defect suppression factor $(\theta_0 \kappa)^2$.

CCLXXXIV. 6. IMMEDIATE BRANCH DECISION

The current pure-even branch is

$$\gamma_E = 1, \quad \gamma_O = 0,$$

so

$$\beta = 0.$$

Hence it must be abandoned if the goal is to read the odd coefficient.

The correct next branch is a mixed branch with

$$T_O \neq 0 \quad \text{equivalently} \quad \gamma_O \neq 0. \quad (708)$$

The simplest canonical choice is

$$T_E = T_O = 1,$$

for which $\alpha = \beta$ and $R_\perp = \sqrt{2}$.

CCLXXXV. FINAL SUMMARY

The odd coefficient is now completely readable on the mixed transverse branch:

$$\beta = -\frac{\sqrt{3}\pi}{16} \beta_X T_O \theta_0 \kappa.$$

So the correct forward route is no longer the pure-even witness branch, but the mixed branch

$$T_O \neq 0.$$

The canonical minimal mixed sample

$$T_E = T_O = 1$$

gives

$$\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad R_\perp = \sqrt{2}, \quad \Delta_{00} = \frac{1}{\sqrt{2}}.$$

This is the first explicit branch on which the odd coefficient is genuinely turned on.

CCLXXXVI. FINAL SUMMARY

Current stage: mixed transverse branch upgrade.

(709)

The pure-even branch

$$\gamma_E = 1, \quad \gamma_O = 0$$

necessarily gives

$$\beta = 0,$$

so it is no longer useful for reading the odd coefficient.

On the mixed branch, the exact reduced symbol is

$$A_1(p) = \frac{1}{2} \alpha_X L_* I_{\theta^3} p_{\parallel} \sigma_3 + \beta_X I_{\theta} T_E (p_1 \sigma_1 + p_2 \sigma_2) + \beta_X I_{\theta} T_O (-p_2 \sigma_1 + p_1 \sigma_2),$$

hence

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16} \beta_X T_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \beta_X T_O \theta_0 \kappa.$$

:contentReference[oaicite:9]index=9

Writing

$$T_E = R_\perp \cos \varphi, \quad T_O = R_\perp \sin \varphi,$$

one gets

$$\frac{\beta}{\alpha} = \tan \varphi.$$

So the odd channel is turned on exactly when

$$T_O \neq 0 \quad \text{equivalently} \quad \gamma_O \neq 0.$$

The canonical minimal mixed sample

$$T_E = T_O = 1$$

gives

$$\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad R_\perp = \sqrt{2}, \quad \Delta_{00} = \frac{1}{\sqrt{2}}.$$

Its leading anisotropy ratio is

$$\rho_{\text{LO}} = \frac{3}{32\sqrt{2}} \frac{\alpha_X}{\beta_X} (\theta_0 \kappa)^2.$$

So the mixed branch enlarges the transverse support, but it still does not remove the small-defect suppression factor $(\theta_0 \kappa)^2$.

$$\boxed{\text{Goal C: leave the pure-even witness branch and adopt the mixed transverse branch } T_O \neq 0.} \quad (710)$$

$$\boxed{\text{Goal M: use the canonical mixed sample } T_E = T_O = 1 \text{ as the first explicit odd-active branch.}} \quad (711)$$

$$\boxed{\text{Goal S: after fixing the mixed branch, return to the finite same-shell and support audits with } (\alpha, \beta) \neq (0, 0).} \quad (712)$$

CCLXXXVII. BRANCH-CONSISTENT MIXED-BRANCH SAME-SHELL SUPPORT CRITERION

CCLXXXVIII. SHORT EXPLANATION

The pure-even same-shell benchmark is already fully reduced to the finite criterion

$$\boxed{|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > 2a_K \mathcal{S}_{\text{diag}}^{EE} + \varepsilon_{\text{small-}p, \rho}^{(1)}, \quad a_K = \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2.} \quad (713)$$

Its small- p drift is controlled by

$$a_{+,i}, \quad a_{-,i}, \quad b_i,$$

with

$$a_{+,i}^{\text{dir}} = \delta_{i\parallel} q_* \left(Z - \frac{Y q_*^2}{2} \right), \quad a_{-,i}^{\text{dir}} = -\delta_{i\parallel} q_* \left(Z - \frac{Y q_*^2}{2} \right), \quad b_i^{\text{dir}} = 0.$$

The mixed transverse branch has

$$\boxed{T_E = R_\perp \cos \varphi, \quad T_O = R_\perp \sin \varphi, \quad R_\perp = \sqrt{T_E^2 + T_O^2}.} \quad (714)$$

In the canonical mixed sample,

$$\boxed{T_E = T_O = 1, \quad R_\perp = \sqrt{2}, \quad \varphi = \frac{\pi}{4}.} \quad (715)$$

CCLXXXIX. 1. MINIMAL BRANCH-CONSISTENT UPLIFT OF THE SAME-SHELL RESIDUAL STRENGTH

The pure-even same-shell correction strength is

$$a_K = \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2.$$

For the mixed branch, the minimal branch-consistent uplift is to replace the single pure-even transverse amplitude by the full transverse norm:

$$a_T := \frac{\beta_X^2}{M_X^2} R_\perp^2 = \frac{\beta_X^2}{M_X^2} (T_E^2 + T_O^2). \quad (716)$$

In the canonical mixed sample,

$$a_T = \frac{2\beta_X^2}{M_X^2}. \quad (717)$$

This is the smallest natural mixed-branch uplift of the pure-even KK -driven same-shell correction scale.

CCXC. 2. MIXED TRANSVERSE SAME-SHELL OVERLAP PACKAGE

The pure-even support theorem is written in terms of the EE -overlap family

$$\mathcal{S}_{++}^{EE}(p), \quad \mathcal{S}_{--}^{EE}(p), \quad \mathcal{S}_{+-}^{EE}(p).$$

On the mixed branch, the branch-consistent transverse overlap package is the quadratic form along the mixed direction

$$T_* = (T_E, T_O).$$

Define

$$\mathcal{S}_{ab}^{TT}(p; \varphi) := \cos^2 \varphi \mathcal{S}_{ab}^{EE}(p) + \sin^2 \varphi \mathcal{S}_{ab}^{OO}(p) + 2 \sin \varphi \cos \varphi \mathcal{S}_{ab}^{EO}(p), \quad (718)$$

where $a, b \in \{+, -\}$.

This is the minimal branch-consistent transverse same-shell overlap obtained by projecting the transverse correction sector along the activated mixed branch direction.

For the canonical mixed sample $\varphi = \pi/4$,

$$\mathcal{S}_{ab}^{(\pi/4)}(p) = \frac{1}{2} \mathcal{S}_{ab}^{EE}(p) + \frac{1}{2} \mathcal{S}_{ab}^{OO}(p) + \mathcal{S}_{ab}^{EO}(p). \quad (719)$$

Likewise define the diagonal benchmark quantity

$$\mathcal{S}_{\text{diag}}^{TT} := \mathcal{S}_{++}^{TT}(0) = \mathcal{S}_{--}^{TT}(0) \quad (720)$$

whenever the valley-symmetric diagonal subcase holds.

CCXCI. 3. MIXED-BRANCH SAME-SHELL BLOCK

The pure-even same-shell block was

$$H_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} d_+(p) & m(p) \\ m(p) & d_-(p) \end{pmatrix},$$

with

$$d_{\pm}(p) = E_{z_0}^{\text{dir}}(\pm p) - a_K \mathcal{S}_{\pm\pm}^{EE}(p), \quad m(p) = B_{z_0}(p) - a_K \mathcal{S}_{+-}^{EE}(p).$$

The branch-consistent mixed uplift is therefore

$$\boxed{d_+^{(T)}(p) = E_{z_0}^{\text{dir}}(p) - a_T \mathcal{S}_{++}^{TT}(p)}, \quad (721)$$

$$\boxed{d_-^{(T)}(p) = E_{z_0}^{\text{dir}}(-p) - a_T \mathcal{S}_{--}^{TT}(p)}, \quad (722)$$

$$\boxed{m^{(T)}(p) = B_{z_0}(p) - a_T \mathcal{S}_{+-}^{TT}(p)}. \quad (723)$$

So the mixed-branch same-shell benchmark remains a 2×2 Hermitian problem, but with the EE -overlap family replaced by the mixed TT -overlap family.

CCXCII. 4. MIXED-BRANCH SMALL- p DRIFT COEFFICIENTS

Differentiate at $p = 0$. The direct core piece is unchanged:

$$\boxed{a_{+,i}^{\text{dir}} = \delta_{i\parallel} q_* \left(Z - \frac{Y q_*^2}{2} \right), \quad a_{-,i}^{\text{dir}} = -\delta_{i\parallel} q_* \left(Z - \frac{Y q_*^2}{2} \right), \quad b_i^{\text{dir}} = 0.} \quad (724)$$

Now define the mixed derivative budgets

$$\boxed{\ell_{+,i}^{TT} := \partial_{p_i} \mathcal{S}_{++}^{TT}(p; \varphi)|_{p=0}, \quad \ell_{-,i}^{TT} := \partial_{p_i} \mathcal{S}_{--}^{TT}(p; \varphi)|_{p=0},} \quad (725)$$

$$\boxed{\ell_{0,i}^{TT} := \partial_{p_i} \mathcal{S}_{+-}^{TT}(p; \varphi)|_{p=0}.} \quad (726)$$

Then the full mixed-branch drift coefficients are

$$\boxed{a_{+,i}^{(T)} = \delta_{i\parallel} q_* \left(Z - \frac{Y q_*^2}{2} \right) - a_T \ell_{+,i}^{TT},} \quad (727)$$

$$\boxed{a_{-,i}^{(T)} = -\delta_{i\parallel} q_* \left(Z - \frac{Y q_*^2}{2} \right) - a_T \ell_{-,i}^{TT},} \quad (728)$$

$$\boxed{b_i^{(T)} = -a_T \ell_{0,i}^{TT}.} \quad (729)$$

Define the norm-level budgets

$$\boxed{L_+^{TT} = \left(\sum_i |\ell_{+,i}^{TT}|^2 \right)^{1/2}, \quad L_-^{TT} = \left(\sum_i |\ell_{-,i}^{TT}|^2 \right)^{1/2}, \quad L_0^{TT} = \left(\sum_i |\ell_{0,i}^{TT}|^2 \right)^{1/2}.} \quad (730)$$

Thus

$$\boxed{A_+^{(T)} \leq \left| q_* \left(Z - \frac{Y q_*^2}{2} \right) \right| + a_T L_+^{TT},} \quad (731)$$

$$\boxed{A_-^{(T)} \leq \left| q_* \left(Z - \frac{Y q_*^2}{2} \right) \right| + a_T L_-^{TT},} \quad (732)$$

$$\boxed{B^{(T)} \leq a_T L_0^{TT}.} \quad (733)$$

CCXCIII. 5. MIXED-BRANCH SAME-SHELL SUPPORT CRITERION

Exactly parallel to the pure-even case, define

$$\varepsilon_{\text{small-}p,\rho}^{(1,T)} \leq \left[(\rho A_+^{(T)} + C_+^{TT} \rho^2)^2 + (\rho A_-^{(T)} + C_-^{TT} \rho^2)^2 + 2(\rho B^{(T)} + C_m^{TT} \rho^2)^2 \right]^{1/2}. \quad (734)$$

Then a branch-consistent mixed same-shell sufficient condition is

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > 2a_T \mathcal{S}_{\text{diag}}^{TT} + \varepsilon_{\text{small-}p,\rho}^{(1,T)}. \quad (735)$$

In the minimal scalar-only locked core,

$$\left| -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \right| - 4uA^2 > 2a_T \mathcal{S}_{\text{diag}}^{TT} + \varepsilon_{\text{small-}p,\rho}^{(1,T)}. \quad (736)$$

CCXCIV. 6. CANONICAL MIXED SAMPLE

For

$$T_E = T_O = 1, \quad R_\perp = \sqrt{2}, \quad \varphi = \frac{\pi}{4},$$

we have

$$a_T = \frac{2\beta_X^2}{M_X^2},$$

and

$$\mathcal{S}_{ab}^{(\pi/4)} = \frac{1}{2} \mathcal{S}_{ab}^{EE} + \frac{1}{2} \mathcal{S}_{ab}^{OO} + \mathcal{S}_{ab}^{EO}.$$

Therefore the canonical mixed same-shell sufficient condition becomes

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}. \quad (737)$$

Equivalently, in scalar-only locked-core form,

$$\left| -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \right| - 4uA^2 > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}. \quad (738)$$

CCXCV. 7. HONEST STATUS

The mixed-branch coefficient sheet is exact. :contentReference[oaicite:5]index=5 The pure-even same-shell support theorem is exact.

The formula above is the sharpest honest next move: it is the minimal branch-consistent uplift of the same-shell support inequality from the pure-even branch to the mixed branch.

What is still not inserted explicitly are the mixed transverse same-shell constants

$$\mathcal{S}_{\text{diag}}^{TT}, \quad L_+^{TT}, \quad L_-^{TT}, \quad L_0^{TT}, \quad C_+^{TT}, \quad C_-^{TT}, \quad C_m^{TT}.$$

So the remaining frontier is still finite insertion, not conceptual structure.

CCXCVI. FINAL SUMMARY

The canonical mixed branch

$$T_E = T_O = 1$$

has

$$R_\perp = \sqrt{2}, \quad \alpha = \beta = -\frac{\sqrt{3}\pi}{16}\beta_X\theta_0\kappa,$$

and its branch-consistent same-shell correction strength is

$$a_T = \frac{2\beta_X^2}{M_X^2}.$$

Its mixed same-shell overlap package is

$$\mathcal{S}_{ab}^{(\pi/4)} = \frac{1}{2}\mathcal{S}_{ab}^{EE} + \frac{1}{2}\mathcal{S}_{ab}^{OO} + \mathcal{S}_{ab}^{EO}.$$

Therefore the branch-consistent mixed same-shell sufficient condition is

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}.$$

This is the correct next support-side inequality for the first explicit odd-active branch.

CCXCVII. FINAL SUMMARY

Current stage: same-shell support audit on the first explicit odd-active mixed branch.

(739)

The exact mixed-branch coefficient sheet is

$$\lambda_\parallel = -\frac{3\sqrt{3}\pi}{512}\alpha_X L_* \theta_0^3 \kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16}\beta_X T_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16}\beta_X T_O \theta_0 \kappa,$$

with

$$R_\perp = \sqrt{T_E^2 + T_O^2}.$$

The pure-even same-shell support theorem is already explicit:

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > 2a_K \mathcal{S}_{\text{diag}}^{EE} + \varepsilon_{\text{small-}p,\rho}^{(1)}.$$

The minimal branch-consistent uplift to the mixed branch is:

$$a_T = \frac{\beta_X^2}{M_X^2}(T_E^2 + T_O^2),$$

$$\mathcal{S}_{ab}^{TT} = \cos^2 \varphi \mathcal{S}_{ab}^{EE} + \sin^2 \varphi \mathcal{S}_{ab}^{OO} + 2 \sin \varphi \cos \varphi \mathcal{S}_{ab}^{EO},$$

with

$$T_E = R_\perp \cos \varphi, \quad T_O = R_\perp \sin \varphi.$$

For the canonical mixed sample

$$T_E = T_O = 1, \quad R_\perp = \sqrt{2}, \quad \varphi = \frac{\pi}{4},$$

one gets

$$a_T = \frac{2\beta_X^2}{M_X^2}, \quad \mathcal{S}_{ab}^{(\pi/4)} = \frac{1}{2}\mathcal{S}_{ab}^{EE} + \frac{1}{2}\mathcal{S}_{ab}^{OO} + \mathcal{S}_{ab}^{EO}.$$

Therefore the sharpest honest mixed-branch same-shell sufficient condition currently available is

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p, \rho}^{(1, \pi/4)}.$$

This is not yet the final theorem-level full microscopic insertion. It is the minimal branch-consistent mixed uplift of the explicit pure-even support theorem. The only remaining frontier is the finite insertion of the mixed same-shell constants

$$\mathcal{S}_{\text{diag}}^{(\pi/4)}, \quad L_+^{(\pi/4)}, \quad L_-^{(\pi/4)}, \quad L_0^{(\pi/4)}, \quad C_+^{(\pi/4)}, \quad C_-^{(\pi/4)}, \quad C_m^{(\pi/4)}.$$

$$\boxed{\text{Goal C: keep the mixed branch } T_E = T_O = 1 \text{ as the first explicit odd-active branch.}} \quad (740)$$

$$\boxed{\text{Goal M: compute the mixed same-shell constants } \mathcal{S}_{ab}^{(\pi/4)} \text{ and their small-}p \text{ drift budgets.}} \quad (741)$$

$$\boxed{\text{Goal S: insert these finite constants into the mixed support inequality and decide whether the first explicit odd-active branch p}} \quad (742)$$

CCXCVIII. CORRECTED RESIDUAL QUARTIC CLOSURE ON THE EXACT FOUR-CLASS SHELL KERNEL

CCXCIX. CURRENT STAGE

The project has reached the corrected residual quartic closure stage. The carrier-side working-branch admissibility is already effectively closed, while the actual remaining frontier is the enlarged quartic residual kernel closure. The operational backend and the corrected seed are frozen as

$$\boxed{(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096).} \quad (743)$$

The correct reduced object is the projected residual map

$$\boxed{G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix}}, \quad (744)$$

not a naive reduced free-energy critical point. Moreover the corrected projection extractor has already produced

$$\boxed{\sigma_0 = -109.41971030446824, \quad \eta = -6.502583100163014,} \quad (745)$$

$$\boxed{\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933,} \quad (746)$$

and the quartic class averages

$$\boxed{U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813.} \quad (747)$$

The simple compressed quartic shell kernel (u, v, κ) fails strongly, with exact consistency defect

$$\boxed{\Delta_{\text{ker}} = 32976.132300215555.} \quad (748)$$

Hence the next exact reduced object must be built on the four-class shell kernel rather than the failed three-parameter compression.

CCC. FOUR-CLASS EXACT SHELL KERNEL

The minimal enlarged quartic shell kernel is

$$U_{\alpha\beta}^{(4c)} = u_d P_{\alpha\beta}^{(d)} + u_{\text{opp}} P_{\alpha\beta}^{(\text{opp})} + u_{\perp} P_{\alpha\beta}^{(\perp)} + u_{\text{obl}} P_{\alpha\beta}^{(\text{obl})}, \quad (749)$$

and at the corrected branch one identifies directly

$$u_d = U_d, \quad u_{\text{opp}} = U_{\text{opp}}, \quad u_{\perp} = U_{\perp}, \quad u_{\text{obl}} = U_{\text{obl}}. \quad (750)$$

After orbit reduction to the coarse (A, E, O) basis, the exact four-class quartic reduced matrix is

$$M^{(4c)} = \begin{pmatrix} \frac{U_d + U_{\text{opp}}}{2} & U_{\perp} & U_{\text{obl}} \\ U_{\perp} & \frac{U_d + U_{\text{opp}}}{2} & U_{\text{obl}} \\ U_{\text{obl}} & U_{\text{obl}} & \frac{U_d + U_{\text{opp}} + 2U_{\perp} + 4U_{\text{obl}}}{8} \end{pmatrix}. \quad (751)$$

Substituting the corrected projected quartic data gives

$$M^{(4c)} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}. \quad (752)$$

CCCI. QUARTIC RESIDUAL BASELINE AT THE CORRECTED SEED

Write

$$(a, e, o) = (\delta_A, \delta_E, \delta_O), \quad r_A = a^2, \quad r_E = e^2, \quad r_O = o^2.$$

The quartic reduced residual contribution is

$$G_X^{(4)}(a, e, o) = a_X \sum_{Y \in \{A, E, O\}} M_{XY}^{(4c)} r_Y, \quad X \in \{A, E, O\}, \quad (753)$$

where $a_A = a$, $a_E = e$, $a_O = o$.

At the corrected seed

$$(a, e, o) = (0.0096, -0.0288, 0.0096),$$

one has

$$r_A = r_O = (0.0096)^2 = 9.216 \times 10^{-5}, \quad r_E = (0.0288)^2 = 8.2944 \times 10^{-4}.$$

Therefore

$$Q := M^{(4c)} \begin{pmatrix} r_A \\ r_E \\ r_O \end{pmatrix} = \begin{pmatrix} 8.494671573456258 \\ 8.241391656489691 \\ 2.580391735022780 \end{pmatrix}. \quad (754)$$

Hence the quartic residual baseline vector is

$$G^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} aQ_A \\ eQ_E \\ oQ_O \end{pmatrix} = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}. \quad (755)$$

Its Euclidean norm is

$$\|G^{(4)}(\delta^{\text{final}})\| = 0.25219013527770096. \quad (756)$$

CCCII. MISMATCH WITH THE ACTUAL CORRECTED RESIDUAL

The corrected numerical branch diagnostics give

$$\|G(\delta^{\text{final}})\| \approx 0.03791852545979427. \quad (757)$$

Therefore the nonquartic remainder at the corrected seed obeys the sharp lower bound

$$\|G(\delta^{\text{final}}) - G^{(4)}(\delta^{\text{final}})\| \geq \|G^{(4)}(\delta^{\text{final}})\| - \|G(\delta^{\text{final}})\| = 0.2142716098179067. \quad (758)$$

Thus the corrected local branch requires a large nonquartic completion on top of the exact four-class quartic baseline.

CCCIII. FIRST REDUCED TARGET FOR THE ACTUAL MICROSCOPIC COMPLETION

If one writes the next reduced closure as a minimal diagonal quadratic completion on top of the exact quartic baseline,

$$G_X^{\text{model}} = a_X \left(q_X + \sum_Y M_{XY}^{(4c)} r_Y \right), \quad X \in \{A, E, O\}, \quad (759)$$

then forcing the corrected seed to be stationary in this minimal model requires

$$q_X = -Q_X. \quad (760)$$

So the exact target values are

$$q_A = -8.494671573456258, \quad q_E = -8.241391656489691, \quad q_O = -2.580391735022780. \quad (761)$$

This is the first actual reduced target that the microscopic operator must reproduce.

CCCIV. DIRECT IDENTIFICATION WITH THE EXTRACTED σ -SHEET FAILS

If one naively identified the corrected orbit means with the quadratic completion target,

$$q_X \stackrel{?}{=} \sigma_X,$$

one would obtain

$$\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933,$$

which are dramatically different from (761). The exact mismatches are

$$\sigma_A - q_A = -107.31548638718991, \quad (762)$$

$$\sigma_E - q_E = -97.59062076594143, \quad (763)$$

$$\sigma_O - q_O = -106.13863112591022. \quad (764)$$

Therefore the extracted (σ_0, η) and $(\sigma_A, \sigma_E, \sigma_O)$ must indeed be interpreted only as residual-based effective reduced coefficients of the current solver family, not as the actual microscopic quadratic coefficients of the corrected branch.

CCCV. MAINLINE CONCLUSION

The corrected residual quartic closure has now been pushed to the exact four-class shell-kernel baseline.

The exact four-class quartic reduced matrix $M^{(4c)}$, the corrected-seed quartic baseline $G^{(4)}(\delta^{\text{final}})$, the nonquartic mismatch lower bound, and the first reduced quadratic target $(q_A^{q_E}, q_O^{q_E})$ are all fixed numerically.

Therefore the next mainline microscopic task is now sharply determined:

identify which microscopic completion source reproduces $(q_A^{q_E}, q_O^{q_E}) = (-8.494671573456258, -8.241391656489691, -2.580391735022780)$.

(765)

Equivalently, the next source-by-source microscopic comparison must ask which combination of

$$JJ, \quad JK + KJ, \quad KK$$

or, in block language,

$$OO, \quad TD + DT, \quad KK,$$

can generate the required corrected local completion beyond the exact four-class quartic baseline.

CCCVI. FINAL SUMMARY

Current stage: corrected residual quartic closure on the exact four-class shell kernel.

(766)

The frozen backend and corrected seed are

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096),$$

and the correct reduced object is the projected residual map

$$G(\delta) = (G_A(\delta), G_E(\delta), G_O(\delta))^T.$$

The corrected projected quartic class means are

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813,$$

while the simple (u, v, κ) compression fails strongly with

$$\Delta_{\text{ker}} = 32976.132300215555.$$

Therefore the minimal exact enlarged shell kernel is the four-class kernel, which yields the coarse (A, E, O) reduced quartic matrix

$$M^{(4c)} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}.$$

At the corrected seed

$$(a, e, o) = (0.0096, -0.0288, 0.0096),$$

the corresponding quartic baseline is

$$Q_A = 8.494671573456258, \quad Q_E = 8.241391656489691, \quad Q_O = 2.580391735022780,$$

and hence

$$G^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}, \quad \|G^{(4)}(\delta^{\text{final}})\| = 0.25219013527770096.$$

Since the actual corrected residual norm is

$$\|G(\delta^{\text{final}})\| \approx 0.03791852545979427,$$

the nonquartic mismatch is necessarily large:

$$\|G(\delta^{\text{final}}) - G^{(4)}(\delta^{\text{final}})\| \geq 0.2142716098179067.$$

Therefore the corrected local branch requires a large nonquartic completion beyond the exact four-class quartic baseline.

The first reduced target for the microscopic completion is the minimal diagonal quadratic completion

$$G_X^{\text{model}} = a_X \left(q_X + \sum_Y M_{XY}^{(4c)} r_Y \right),$$

which forces

$$\begin{array}{ccc} q_A = -8.494671573456258, & q_E = -8.241391656489691, & q_O = -2.580391735022780. \end{array}$$

The extracted orbit means

$$\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933$$

cannot be identified directly with these targets, so the σ -sheet must still be interpreted only as an effective residual-based reduced sheet of the current solver family.

Mainline result: the four-class quartic baseline is now fixed numerically, and the next microscopic task is to reproduce $(q_A^{q_E}, q_O^{q_E})$

(767)

from the actual microscopic completion sources.

CCCVII. MAINLINE ONLY: CORRECTED RESIDUAL QUARTIC CLOSURE ON THE FOUR-CLASS EXACT SHELL KERNEL

CCCVIII. CURRENT STAGE

The current stage is not carrier sign polishing. The actual remaining frontier is the corrected residual quartic kernel closure:

$$\text{carrier-side working-branch admissibility: effectively closed,} \quad \text{corrected residual quartic closure: open.} \quad (768)$$

Keep fixed

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096). \quad (769)$$

The primary reduced object is the projected residual map

$$G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix}. \quad (770)$$

CCCIX. EXACT CORRECTED QUARTIC DATA

The corrected projection has already produced

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813. \quad (771)$$

The simple (u, v, κ) compression fails strongly, with

$$\Delta_{\text{ker}} = 32976.132300215555. \quad (772)$$

Therefore the next exact reduced shell kernel must be the four-class kernel

$$U_{\alpha\beta}^{(4c)} = u_d P_{\alpha\beta}^{(d)} + u_{\text{opp}} P_{\alpha\beta}^{(\text{opp})} + u_{\perp} P_{\alpha\beta}^{(\perp)} + u_{\text{obl}} P_{\alpha\beta}^{(\text{obl})}. \quad (773)$$

CCCX. COARSE (A, E, O) REDUCED QUARTIC MATRIX

After orbit reduction, the exact four-class quartic reduced matrix is

$$M^{(4c)} = \begin{pmatrix} \frac{U_d + U_{\text{opp}}}{2} & U_{\perp} & U_{\text{obl}} \\ U_{\perp} & \frac{U_d + U_{\text{opp}}}{2} & U_{\text{obl}} \\ U_{\text{obl}} & U_{\text{obl}} & \frac{U_d + U_{\text{opp}} + 2U_{\perp} + 4U_{\text{obl}}}{8} \end{pmatrix} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}. \quad (774)$$

CCCXI. CORRECTED-SEED QUARTIC RESIDUAL BASELINE

Write

$$(a, e, o) = (0.0096, -0.0288, 0.0096), \quad r_A = a^2, \quad r_E = e^2, \quad r_O = o^2.$$

Then

$$Q = M^{(4c)} \begin{pmatrix} r_A \\ r_E \\ r_O \end{pmatrix} = \begin{pmatrix} 8.494671573456258 \\ 8.241391656489691 \\ 2.580391735022780 \end{pmatrix}. \quad (775)$$

Hence

$$G^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} aQ_A \\ eQ_E \\ oQ_O \end{pmatrix} = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}, \quad (776)$$

with

$$\|G^{(4)}(\delta^{\text{final}})\| = 0.25219013527770096. \quad (777)$$

CCCXII. ACTUAL CORRECTED RESIDUAL MISMATCH

The corrected branch diagnostics give

$$\|G(\delta^{\text{final}})\| \approx 0.03791852545979427. \quad (778)$$

Therefore

$$\|G(\delta^{\text{final}}) - G^{(4)}(\delta^{\text{final}})\| \geq 0.2142716098179067. \quad (779)$$

So the corrected branch requires a large nonquartic completion on top of the exact four-class quartic baseline.

CCCXIII. FIRST REDUCED TARGET

If one writes the minimal corrected local model as

$$G_X^{\text{model}} = a_X \left(q_X + \sum_Y M_{XY}^{(4c)} r_Y \right), \quad (780)$$

then stationarity at the corrected seed requires

$$\begin{array}{ccc} q_A = -8.494671573456258, & q_E = -8.241391656489691, & q_O = -2.580391735022780. \end{array} \quad (781)$$

CCCXIV. MAINLINE NEXT STEP

So the next actual microscopic task is not another branch search and not another carrier discussion. It is:

$$\text{rerun the corrected projection directly on the enlarged four-class shell kernel basis} \quad (782)$$

and then determine which microscopic completion reproduces

$$(q_A^{q_O}) = (-8.494671573456258, -8.241391656489691, -2.580391735022780). \quad (783)$$

CCCXV. ACTUAL MAINLINE RESULT: THE FOUR-CLASS ENLARGED QUARTIC KERNEL STILL DOES NOT CLOSE A FREE-ENERGY STATIONARY BRANCH

CCCXVI. CURRENT STAGE AND CORRECT REDUCED OBJECT

The current project stage is not carrier-sign polishing and not another branch scan. The actual remaining frontier is the corrected residual quartic closure stage, with frozen operational backend

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096). \quad (784)$$

The correct reduced object is the projected residual map

$$G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix}, \quad (785)$$

not a naive reduced free-energy critical point. Therefore the immediate next step must be formulated directly at the residual level, after enlarging the quartic reduced shell kernel beyond the failed (u, v, κ) truncation. :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8 :contentReference[oaicite:9]index=9 :contentReference[oaicite:10]index=10

CCCXVII. CORRECTED PROJECTED QUARTIC DATA

The corrected branch projection gives the orbit means

$$\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933, \quad (786)$$

and the four quartic class averages

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813. \quad (787)$$

The simple three-parameter compression fails strongly, with

$$\boxed{\Delta_{\text{ker}} = 32976.132300215555.} \quad (788)$$

So the minimal mathematically correct enlarged quartic shell kernel is the four-class exact shell kernel

$$U_{\alpha\beta}^{(4c)} = u_d P_{\alpha\beta}^{(d)} + u_{\text{opp}} P_{\alpha\beta}^{(\text{opp})} + u_{\perp} P_{\alpha\beta}^{(\perp)} + u_{\text{obl}} P_{\alpha\beta}^{(\text{obl})}.$$

:contentReference[oaicite:11]index=11 :contentReference[oaicite:12]index=12

CCCXVIII. COARSE (A, E, O) QUARTIC MATRIX

Reducing the four-class exact shell kernel to the coarse (A, E, O) orbit basis gives

$$M^{(4c)} = \begin{pmatrix} \frac{U_d + U_{\text{opp}}}{2} & U_{\perp} & U_{\text{obl}} \\ U_{\perp} & \frac{U_d + U_{\text{opp}}}{2} & U_{\text{obl}} \\ U_{\text{obl}} & U_{\text{obl}} & \frac{U_d + U_{\text{opp}} + 2U_{\perp} + 4U_{\text{obl}}}{8} \end{pmatrix}. \quad (789)$$

Substituting the corrected projected quartic data yields

$$M^{(4c)} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}. \quad (790)$$

CCCXIX. FIRST NUMERICAL STABILITY CHECK OF THE ENLARGED QUARTIC MATRIX

The trace and determinant are

$$\text{Tr } M^{(4c)} \approx 22917.428023188142, \quad \det M^{(4c)} \approx -3.0305980590 \times 10^{10}. \quad (791)$$

Since the determinant is negative, the matrix is not positive definite.

Numerically, its eigenvalues are

$$\boxed{\lambda_1 \approx -343.533, \quad \lambda_2 \approx 4771.217, \quad \lambda_3 \approx 18489.744.} \quad (792)$$

Therefore:

$$\boxed{\text{the enlarged four-class quartic kernel solves the basis problem,}} \quad (793)$$

but

$$\boxed{\text{it is not itself a simple positive quartic free-energy closure on the corrected branch.}} \quad (794)$$

CCCXX. QUARTIC BASELINE AT THE CORRECTED SEED

With

$$(a, e, o) = (0.0096, -0.0288, 0.0096), \quad r_A = a^2, \quad r_E = e^2, \quad r_O = o^2,$$

the quartic load vector is

$$Q = M^{(4c)} \begin{pmatrix} r_A \\ r_E \\ r_O \end{pmatrix} = \begin{pmatrix} 8.494671573456258 \\ 8.241391656489691 \\ 2.580391735022780 \end{pmatrix}. \quad (795)$$

Hence the corrected-seed quartic residual baseline is

$$G^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} aQ_A \\ eQ_E \\ oQ_O \end{pmatrix} = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}, \quad (796)$$

with

$$\|G^{(4)}(\delta^{\text{final}})\| = 0.25219013527770096. \quad (797)$$

The actual corrected residual norm is

$$\|G(\delta^{\text{final}})\| \approx 0.03791852545979427. \quad (798)$$

Therefore the nonquartic mismatch is necessarily large:

$$\|G(\delta^{\text{final}}) - G^{(4)}(\delta^{\text{final}})\| \geq 0.2142716098179067. \quad (799)$$

CCCXXI. NAIVE QUARTIC STATIONARY EQUATIONS STILL FAIL

If one nevertheless tries to build a quartic free-energy-like stationary system using the corrected linear orbit means,

$$\sigma + M^{(4c)}r = 0, \quad r = (r_A, r_E, r_O)^T, \quad \sigma = (\sigma_A, \sigma_E, \sigma_O)^T, \quad (800)$$

then solving (820) gives

$$r_A^{\text{crit}} \approx -1.031725 \times 10^{-2}, \quad r_E^{\text{crit}} \approx 1.872843 \times 10^{-2}, \quad r_O^{\text{crit}} \approx 1.618639 \times 10^{-2}. \quad (801)$$

This candidate is immediately inadmissible because

$$r_A^{\text{crit}} < 0.$$

So the enlarged quartic basis does not restore a viable free-energy stationary interpretation.

CCCXXII. MAINLINE CONCLUSION

The exact four-class enlargement has now accomplished precisely one thing: it removes the false (u, v, κ) compression and provides the correct enlarged quartic shell basis.

But after this enlargement, the actual obstruction remains:

$$\text{the correct next reduced object is the projected residual system } G_A(\delta) = 0, \quad G_E(\delta) = 0, \quad G_O(\delta) = 0, \quad (802)$$

not a quartic free-energy stationary system.

Equivalently,

$$\text{the basis problem is solved, but the residual-closure problem remains.} \quad (803)$$

CCCXXIII. ACTUAL MAINLINE RESULT: THE FOUR-CLASS ENLARGED QUARTIC KERNEL STILL DOES NOT CLOSE A FREE-ENERGY STATIONARY BRANCH

CCCXXIV. CURRENT STAGE AND CORRECT REDUCED OBJECT

The current project stage is not carrier-sign polishing and not another branch scan. The actual remaining frontier is the corrected residual quartic closure stage, with frozen operational backend

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096). \quad (804)$$

The correct reduced object is the projected residual map

$$G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix}, \quad (805)$$

not a naive reduced free-energy critical point. Therefore the immediate next step must be formulated directly at the residual level, after enlarging the quartic reduced shell kernel beyond the failed (u, v, κ) truncation. :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8 :contentReference[oaicite:9]index=9 :contentReference[oaicite:10]index=10

CCCXXV. CORRECTED PROJECTED QUARTIC DATA

The corrected branch projection gives the orbit means

$$\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933, \quad (806)$$

and the four quartic class averages

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813. \quad (807)$$

The simple three-parameter compression fails strongly, with

$$\Delta_{\text{ker}} = 32976.132300215555. \quad (808)$$

So the minimal mathematically correct enlarged quartic shell kernel is the four-class exact shell kernel

$$U_{\alpha\beta}^{(4c)} = u_d P_{\alpha\beta}^{(d)} + u_{\text{opp}} P_{\alpha\beta}^{(\text{opp})} + u_{\perp} P_{\alpha\beta}^{(\perp)} + u_{\text{obl}} P_{\alpha\beta}^{(\text{obl})}.$$

:contentReference[oaicite:11]index=11 :contentReference[oaicite:12]index=12

CCCXXVI. COARSE (A, E, O) QUARTIC MATRIX

Reducing the four-class exact shell kernel to the coarse (A, E, O) orbit basis gives

$$M^{(4c)} = \begin{pmatrix} \frac{U_d + U_{\text{opp}}}{2} & U_{\perp} & U_{\text{obl}} \\ U_{\perp} & \frac{U_d + U_{\text{opp}}}{2} & U_{\text{obl}} \\ U_{\text{obl}} & U_{\text{obl}} & \frac{U_d + U_{\text{opp}} + 2U_{\perp} + 4U_{\text{obl}}}{8} \end{pmatrix}. \quad (809)$$

Substituting the corrected projected quartic data yields

$$M^{(4c)} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}. \quad (810)$$

CCCXXVII. FIRST NUMERICAL STABILITY CHECK OF THE ENLARGED QUARTIC MATRIX

The trace and determinant are

$$\text{Tr } M^{(4c)} \approx 22917.428023188142, \quad \det M^{(4c)} \approx -3.0305980590 \times 10^{10}. \quad (811)$$

Since the determinant is negative, the matrix is not positive definite.

Numerically, its eigenvalues are

$$\lambda_1 \approx -343.533, \quad \lambda_2 \approx 4771.217, \quad \lambda_3 \approx 18489.744. \quad (812)$$

Therefore:

$$\text{the enlarged four-class quartic kernel solves the basis problem,} \quad (813)$$

but

$$\text{it is not itself a simple positive quartic free-energy closure on the corrected branch.} \quad (814)$$

CCCXXVIII. QUARTIC BASELINE AT THE CORRECTED SEED

With

$$(a, e, o) = (0.0096, -0.0288, 0.0096), \quad r_A = a^2, \quad r_E = e^2, \quad r_O = o^2,$$

the quartic load vector is

$$Q = M^{(4c)} \begin{pmatrix} r_A \\ r_E \\ r_O \end{pmatrix} = \begin{pmatrix} 8.494671573456258 \\ 8.241391656489691 \\ 2.580391735022780 \end{pmatrix}. \quad (815)$$

Hence the corrected-seed quartic residual baseline is

$$G^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} aQ_A \\ eQ_E \\ oQ_O \end{pmatrix} = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}, \quad (816)$$

with

$$\|G^{(4)}(\delta^{\text{final}})\| = 0.25219013527770096. \quad (817)$$

The actual corrected residual norm is

$$\|G(\delta^{\text{final}})\| \approx 0.03791852545979427. \quad (818)$$

Therefore the nonquartic mismatch is necessarily large:

$$\|G(\delta^{\text{final}}) - G^{(4)}(\delta^{\text{final}})\| \geq 0.2142716098179067. \quad (819)$$

CCCXXIX. NAIVE QUARTIC STATIONARY EQUATIONS STILL FAIL

If one nevertheless tries to build a quartic free-energy-like stationary system using the corrected linear orbit means,

$$\sigma + M^{(4c)}r = 0, \quad r = (r_A, r_E, r_O)^T, \quad \sigma = (\sigma_A, \sigma_E, \sigma_O)^T, \quad (820)$$

then solving (820) gives

$$r_A^{\text{crit}} \approx -1.031725 \times 10^{-2}, \quad r_E^{\text{crit}} \approx 1.872843 \times 10^{-2}, \quad r_O^{\text{crit}} \approx 1.618639 \times 10^{-2}. \quad (821)$$

This candidate is immediately inadmissible because

$$r_A^{\text{crit}} < 0.$$

So the enlarged quartic basis does not restore a viable free-energy stationary interpretation.

CCCXXX. MAINLINE CONCLUSION

The exact four-class enlargement has now accomplished precisely one thing: it removes the false (u, v, κ) compression and provides the correct enlarged quartic shell basis.

But after this enlargement, the actual obstruction remains:

$$\boxed{\text{the correct next reduced object is the projected residual system } G_A(\delta) = 0, \quad G_E(\delta) = 0, \quad G_O(\delta) = 0,} \quad (822)$$

not a quartic free-energy stationary system.

Equivalently,

$$\boxed{\text{the basis problem is solved, but the residual-closure problem remains.}} \quad (823)$$

CCCXXXI. FINAL SUMMARY

$$\boxed{\text{Current stage: corrected residual quartic closure on the enlarged four-class exact shell kernel.}} \quad (824)$$

The corrected projected quartic data are

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813,$$

with orbit means

$$\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933.$$

The simple (u, v, κ) truncation fails strongly:

$$\Delta_{\text{ker}} = 32976.132300215555.$$

:contentReference[oaicite:13]index=13

The corresponding coarse (A, E, O) four-class quartic matrix is

$$M^{(4c)} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}.$$

At the corrected seed

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096),$$

the quartic residual baseline is

$$G^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}, \quad \|G^{(4)}(\delta^{\text{final}})\| = 0.25219013527770096.$$

Since the actual corrected residual norm is

$$\|G(\delta^{\text{final}})\| \approx 0.03791852545979427,$$

the nonquartic mismatch is necessarily large:

$$\boxed{\|G(\delta^{\text{final}}) - G^{(4)}(\delta^{\text{final}})\| \geq 0.2142716098179067.}$$

Moreover the enlarged quartic matrix is not positive definite:

$$\det M^{(4c)} \approx -3.0305980590 \times 10^{10},$$

and its eigenvalues are approximately

$$-343.533, \quad 4771.217, \quad 18489.744.$$

If one nevertheless solves the naive quartic stationary system

$$\sigma + M^{(4c)}r = 0,$$

one gets

$$r_A^{\text{crit}} \approx -1.031725 \times 10^{-2}, \quad r_E^{\text{crit}} \approx 1.872843 \times 10^{-2}, \quad r_O^{\text{crit}} \approx 1.618639 \times 10^{-2},$$

which is physically inadmissible because $r_A^{\text{crit}} < 0$.

Therefore the exact four-class enlargement solves the quartic-basis problem, but it does not restore a viable free-energy stationary interpretation. The correct next reduced object remains the projected residual system

$$G_A(\delta) = 0, \quad G_E(\delta) = 0, \quad G_O(\delta) = 0.$$

:contentReference[oaicite:14]index=14 :contentReference[oaicite:15]index=15 :contentReference[oaicite:16]index=16

$$\boxed{\text{Goal C: complete the corrected residual branch by an enlarged quartic shell-kernel closure.}} \quad (825)$$

$$\boxed{\text{Goal M: rerun the corrected projection on the four-class exact shell kernel basis.}} \quad (826)$$

$$\boxed{\text{Goal S: solve the projected residual system directly, rather than return to a naive free-energy reconstruction.}} \quad (827)$$

CCCXXXII. ACTUAL MAINLINE RESULT ON THE CORRECTED RESIDUAL QUARTIC CLOSURE STAGE

CCCXXXIII. CURRENT STAGE

The current frontier is no longer carrier-sign polishing. The files state explicitly that carrier-side working-branch admissibility is effectively closed, while the actual remaining obstruction is the corrected residual quartic kernel closure. The correct reduced object is the projected residual map

$$G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix}, \quad (828)$$

with frozen operational backend and corrected seed

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096). \quad (829)$$

CCCXXXIV. CORRECTED PROJECTED QUARTIC DATA

The corrected projection extractor gives

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813, \quad (830)$$

together with orbit means

$$\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933. \quad (831)$$

The simple three-parameter quartic kernel (u, v, κ) is decisively ruled out by the exact defect

$$\boxed{\Delta_{\text{ker}} = 32976.132300215555.} \quad (832)$$

Therefore the minimal exact enlarged quartic shell kernel must be the four-class exact shell kernel

$$\boxed{U_{\alpha\beta}^{(4c)} = u_d P_{\alpha\beta}^{(d)} + u_{\text{opp}} P_{\alpha\beta}^{(\text{opp})} + u_{\perp} P_{\alpha\beta}^{(\perp)} + u_{\text{obl}} P_{\alpha\beta}^{(\text{obl})}.} \quad (833)$$

CCCXXXV. COARSE (A, E, O) FOUR-CLASS QUARTIC MATRIX

After orbit reduction, the exact quartic reduced matrix is

$$\boxed{M^{(4c)} = \begin{pmatrix} \frac{U_d + U_{\text{opp}}}{2} & U_{\perp} & U_{\text{obl}} \\ U_{\perp} & \frac{U_d + U_{\text{opp}}}{2} & U_{\text{obl}} \\ U_{\text{obl}} & U_{\text{obl}} & \frac{U_d + U_{\text{opp}} + 2U_{\perp} + 4U_{\text{obl}}}{8} \end{pmatrix}.} \quad (834)$$

Numerically,

$$\boxed{M^{(4c)} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}.} \quad (835)$$

CCCXXXVI. FIRST STRUCTURAL CHECK: THE ENLARGED QUARTIC MATRIX IS NOT POSITIVE DEFINITE

The determinant is

$$\boxed{\det M^{(4c)} \approx -3.030598059046301 \times 10^{10},} \quad (836)$$

and the eigenvalues are

$$\boxed{\lambda_1 \approx -343.53287349, \quad \lambda_2 \approx 4771.21661139, \quad \lambda_3 \approx 18489.74428529.} \quad (837)$$

Hence

$$\boxed{M^{(4c)} \text{ is not positive definite.}} \quad (838)$$

So the exact four-class enlargement solves the quartic basis problem, but it does not by itself restore a simple stable quartic free-energy picture.

CCCXXXVII. QUARTIC RESIDUAL BASELINE AT THE CORRECTED SEED

Write

$$(a, e, o) = (0.0096, -0.0288, 0.0096), \quad r_A = a^2, \quad r_E = e^2, \quad r_O = o^2.$$

Then

$$\boxed{Q = M^{(4c)} \begin{pmatrix} r_A \\ r_E \\ r_O \end{pmatrix} = \begin{pmatrix} 8.494671573456258 \\ 8.241391656489691 \\ 2.580391735022780 \end{pmatrix}.} \quad (839)$$

Therefore the corrected-seed quartic residual baseline is

$$G^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} aQ_A \\ eQ_E \\ oQ_O \end{pmatrix} = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}, \quad (840)$$

with

$$\|G^{(4)}(\delta^{\text{final}})\| = 0.25219013527770096. \quad (841)$$

CCCXXXVIII. MISMATCH WITH THE ACTUAL CORRECTED RESIDUAL

The corrected local seed diagnostics give

$$\|G(\delta^{\text{final}})\| \approx 0.03791852545979427. \quad (842)$$

Hence the nonquartic mismatch is necessarily large:

$$\|G(\delta^{\text{final}}) - G^{(4)}(\delta^{\text{final}})\| \geq \|G^{(4)}(\delta^{\text{final}})\| - \|G(\delta^{\text{final}})\| = 0.2142716098179067. \quad (843)$$

So the corrected branch requires a substantial nonquartic completion beyond the exact four-class quartic baseline.

CCCXXXIX. NAIVE QUARTIC STATIONARY RECONSTRUCTION STILL FAILS

Suppose one nevertheless attempts a free-energy-like stationary system using the extracted orbit means:

$$\sigma + M^{(4c)}r = 0, \quad r = (r_A, r_E, r_O)^T, \quad \sigma = (\sigma_A, \sigma_E, \sigma_O)^T. \quad (844)$$

Solving (844) yields

$$r_A^{\text{crit}} \approx -1.031725 \times 10^{-2}, \quad r_E^{\text{crit}} \approx 1.872843 \times 10^{-2}, \quad r_O^{\text{crit}} \approx 1.618639 \times 10^{-2}. \quad (845)$$

This candidate is immediately inadmissible because

$$r_A^{\text{crit}} < 0. \quad (846)$$

Therefore the enlarged four-class basis does not rescue the free-energy reconstruction route.

CCCXL. MAINLINE CONCLUSION

The exact four-class enlargement has now done exactly what it was supposed to do: it removes the false (u, v, κ) compression and supplies the minimal exact quartic shell basis.

But after that enlargement, the correct reduced object is still not a free-energy stationary system. It remains the projected residual system

$$G_A(\delta) = 0, \quad G_E(\delta) = 0, \quad G_O(\delta) = 0. \quad (847)$$

Equivalently,

$$\text{the quartic basis problem is solved, but the residual-closure problem remains.} \quad (848)$$

So the next mainline step is not another branch search and not another quartic ansatz. It is to rerun the corrected projection directly on the enlarged four-class shell basis and solve the projected residual system itself.

CCCXLI. FINAL SUMMARY

$$\boxed{\text{Current stage: corrected residual quartic closure on the exact four-class shell kernel.}} \quad (849)$$

The corrected quartic data are

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813,$$

with orbit means

$$\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933.$$

The exact enlarged quartic matrix in the coarse (A, E, O) basis is

$$M^{(4c)} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}.$$

This matrix is not positive definite:

$$\det M^{(4c)} \approx -3.030598059046301 \times 10^{10},$$

with eigenvalues

$$-343.53287349, \quad 4771.21661139, \quad 18489.74428529.$$

At the frozen corrected seed

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096),$$

the quartic residual baseline is

$$\boxed{G^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}, \quad \|G^{(4)}(\delta^{\text{final}})\| = 0.25219013527770096.}$$

Since the actual corrected residual norm is

$$\|G(\delta^{\text{final}})\| \approx 0.03791852545979427,$$

the nonquartic mismatch is necessarily large:

$$\boxed{\|G(\delta^{\text{final}}) - G^{(4)}(\delta^{\text{final}})\| \geq 0.2142716098179067.}$$

Moreover the naive quartic stationary reconstruction

$$\sigma + M^{(4c)}r = 0$$

gives

$$\boxed{r_A^{\text{crit}} \approx -1.031725 \times 10^{-2}, \quad r_E^{\text{crit}} \approx 1.872843 \times 10^{-2}, \quad r_O^{\text{crit}} \approx 1.618639 \times 10^{-2},}$$

which is physically inadmissible because $r_A^{\text{crit}} < 0$.

Therefore the exact four-class enlargement solves the quartic basis problem but does not restore a viable free-energy stationary interpretation. The correct next reduced object is still the projected residual system

$$\boxed{G_A(\delta) = 0, \quad G_E(\delta) = 0, \quad G_O(\delta) = 0.}$$

$$\boxed{\text{Goal C: complete the corrected branch at the residual level, not by returning to a naive free-energy reconstruction.}} \quad (850)$$

$$\boxed{\text{Goal M: rerun the corrected projection directly on the enlarged four-class shell basis.}} \quad (851)$$

$$\boxed{\text{Goal S: solve the projected residual system itself.}} \quad (852)$$

CCCXLII. ACTUAL MAINLINE PROGRESS BEYOND THE FOUR-CLASS QUARTIC BASELINE

CCCXLIII. CURRENT STAGE

The current project stage is the corrected residual quartic closure stage. The carrier-side working branch is already effectively closed, while the actual remaining frontier is the enlarged quartic residual closure. The correct reduced object is the projected residual map

$$G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix}, \quad (853)$$

not a naive reduced free-energy critical point. The frozen operational backend and corrected seed are

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096). \quad (854)$$

The corrected projection extractor has already produced

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813, \quad (855)$$

and the corrected residual norm

$$\|G(\delta^{\text{final}})\| \approx 0.03791852545979427. \quad (856)$$

The notes state explicitly that the failed (u, v, κ) compression must be replaced by the four-class exact shell kernel and that the corrected closure must be handled directly at the residual level. :contentReference[oaicite:4]index=4 :contentReference[oaicite:5]index=5 :contentReference[oaicite:6]index=6 :contentReference[oaicite:7]index=7

CCCXLIV. 1. EXACT FOUR-CLASS QUARTIC REDUCED MATRIX

Using the four-class exact shell kernel, the coarse (A, E, O) quartic reduced matrix is

$$M^{(4c)} = \begin{pmatrix} \frac{U_d + U_{\text{opp}}}{2} & U_{\perp} & U_{\text{obl}} \\ U_{\perp} & \frac{U_d + U_{\text{opp}}}{2} & U_{\text{obl}} \\ U_{\text{obl}} & U_{\text{obl}} & \frac{U_d + U_{\text{opp}} + 2U_{\perp} + 4U_{\text{obl}}}{8} \end{pmatrix}. \quad (857)$$

Numerically,

$$M^{(4c)} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}. \quad (858)$$

Its determinant is

$$\det M^{(4c)} \approx -3.030598059046301 \times 10^{10}, \quad (859)$$

and its eigenvalues are

$$\lambda_1(M^{(4c)}) \approx -343.53287349, \quad \lambda_2(M^{(4c)}) \approx 4771.21661139, \quad \lambda_3(M^{(4c)}) \approx 18489.74428529. \quad (860)$$

Therefore

$$M^{(4c)} \text{ is indefinite.} \quad (861)$$

So the enlarged quartic basis solves the compression problem, but it does not restore a simple positive quartic free-energy picture.

CCCXLV. 2. QUARTIC RESIDUAL BASELINE AT THE CORRECTED SEED

Let

$$(a, e, o) = (0.0096, -0.0288, 0.0096), \quad r_A = a^2, \quad r_E = e^2, \quad r_O = o^2.$$

Then

$$Q = M^{(4c)} \begin{pmatrix} r_A \\ r_E \\ r_O \end{pmatrix} = \begin{pmatrix} 8.494671573456258 \\ 8.241391656489691 \\ 2.580391735022780 \end{pmatrix}. \quad (862)$$

Hence the quartic residual baseline is

$$G^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} aQ_A \\ eQ_E \\ oQ_O \end{pmatrix} = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}, \quad (863)$$

with

$$\|G^{(4)}(\delta^{\text{final}})\| = 0.25219013527770096. \quad (864)$$

Since the actual corrected residual norm is

$$\|G(\delta^{\text{final}})\| \approx 0.03791852545979427,$$

the nonquartic mismatch obeys

$$\|G(\delta^{\text{final}}) - G^{(4)}(\delta^{\text{final}})\| \geq 0.2142716098179067. \quad (865)$$

So the corrected branch requires a substantial nonquartic completion on top of the exact four-class quartic baseline.

CCCXLVI. 3. QUARTIC RESIDUAL JACOBIAN AT THE CORRECTED SEED

For the quartic residual model

$$G_X^{(4)}(a, e, o) = a_X \sum_Y M_{XY}^{(4c)} r_Y, \quad r_Y = a_Y^2,$$

the Jacobian is

$$(J_{\text{res}}^{(4c)})_{ij} = \delta_{ij} Q_i + 2a_i M_{ij}^{(4c)} a_j. \quad (866)$$

At the corrected seed this becomes

$$J_{\text{res}}^{(4c)}(\delta^{\text{final}}) = \begin{pmatrix} 10.095240102467715 & -4.991665531859216 & 0.4137780199567714 \\ -4.991665531859216 & 22.646508448870943 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 3.603395005929136 \end{pmatrix}. \quad (867)$$

Its eigenvalues are

$$\lambda_1(J_{\text{res}}^{(4c)}) \approx 3.52129589, \quad \lambda_2(J_{\text{res}}^{(4c)}) \approx 8.35219818, \quad \lambda_3(J_{\text{res}}^{(4c)}) \approx 24.47164948. \quad (868)$$

Therefore

$$J_{\text{res}}^{(4c)}(\delta^{\text{final}}) \text{ is positive definite.} \quad (869)$$

This is the crucial actual result: although the quartic matrix $M^{(4c)}$ is indefinite as a free-energy object, the quartic residual map itself is locally well-conditioned at the corrected seed.

CCCXLVII. 4. MINIMAL DIAGONAL QUADRATIC COMPLETION TARGET

If one introduces the minimal diagonal quadratic completion

$$G_X^{\text{model}} = a_X \left(q_X + \sum_Y M_{XY}^{(4c)} r_Y \right), \quad (870)$$

then exact stationarity at the corrected seed requires

$$q_X = -Q_X. \quad (871)$$

Therefore

$$\begin{array}{ccc} q_A = -8.494671573456258, & q_E = -8.241391656489691, & q_O = -2.580391735022780. \end{array} \quad (872)$$

CCCXLVIII. 5. DIAGONAL-ONLY COMPLETION DOES NOT CLOSE THE LOCAL BRANCH

For the diagonal-completed model, the Jacobian at the corrected seed is

$$J_{\text{res}}^{\text{diag-comp}} = \text{diag}(q^{+Q} + 2a_i M_{ij}^{(4c)} a_j = 2a_i M_{ij}^{(4c)} a_j). \quad (873)$$

Numerically,

$$J_{\text{res}}^{\text{diag-comp}}(\delta^{\text{final}}) = \begin{pmatrix} 1.600568529011483 & -4.991665531859216 & 0.4137780199567714 \\ -4.991665531859216 & 14.405116792381252 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 1.023003270906356 \end{pmatrix}. \quad (874)$$

Its eigenvalues are

$$\lambda_1 \approx -0.11554365, \quad \lambda_2 \approx 0.91059735, \quad \lambda_3 \approx 16.23363488. \quad (875)$$

Therefore

$$\text{the minimal diagonal quadratic completion makes the corrected seed stationary,} \quad (876)$$

but

$$\text{it does not make the corrected branch locally stable.} \quad (877)$$

So diagonal-only completion is not enough.

CCCXLIX. MAINLINE CONCLUSION

The actual four-class enlarged quartic basis is now numerically fixed. The free-energy route remains unreliable because $M^{(4c)}$ is indefinite and the naive stationary reconstruction is inadmissible.

But the quartic residual Jacobian itself is positive definite at the corrected seed, which confirms that the projected residual map is the correct mainline reduced object.

Moreover, the minimal diagonal quadratic completion

$$q = (-8.494671573456258, -8.241391656489691, -2.580391735022780)$$

is not sufficient to close the local corrected branch, because its Jacobian still has a negative mode.

Therefore the next microscopic completion cannot be diagonal-only. It must supply genuine off-diagonal and/or higher residual structure.

Equivalently, the actual next mainline task is:

$$\text{solve the projected residual system itself,} \quad (878)$$

with a nontrivial completion beyond the exact four-class quartic baseline and beyond a purely diagonal quadratic correction.

CCCL. FINAL SUMMARY

Current stage: corrected residual quartic closure on the exact four-class shell kernel.

 (879)

The corrected projected quartic data are

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813,$$

and the frozen corrected seed is

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096), \quad \|G(\delta^{\text{final}})\| \approx 0.03791852545979427.$$

The exact enlarged coarse (A, E, O) quartic matrix is

$$M^{(4c)} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}.$$

This matrix is indefinite:

$$\det M^{(4c)} \approx -3.030598059046301 \times 10^{10},$$

with eigenvalues

$$-343.53287349, \quad 4771.21661139, \quad 18489.74428529.$$

At the corrected seed,

$$Q = \begin{pmatrix} 8.494671573456258 \\ 8.241391656489691 \\ 2.580391735022780 \end{pmatrix}, \quad G^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix},$$

so the nonquartic mismatch is necessarily large:

$\|G(\delta^{\text{final}}) - G^{(4)}(\delta^{\text{final}})\| \geq 0.2142716098179067.$

However, the quartic residual Jacobian at the corrected seed is

$$J_{\text{res}}^{(4c)}(\delta^{\text{final}}) = \begin{pmatrix} 10.095240102467715 & -4.991665531859216 & 0.4137780199567714 \\ -4.991665531859216 & 22.646508448870943 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 3.603395005929136 \end{pmatrix},$$

with eigenvalues

3.52129589, 8.35219818, 24.47164948.

So the residual map itself is locally well-conditioned even though the free-energy quartic matrix is indefinite. The minimal diagonal quadratic completion required to make the corrected seed stationary is

$$\begin{array}{ccc} \overset{= -8.494671573456258,}{q_A} & \overset{= -8.241391656489691,}{q_E} & \overset{= -2.580391735022780.}{q_O} \end{array}$$

But the corresponding diagonal-completed Jacobian still has a negative eigenvalue:

-0.11554365, 0.91059735, 16.23363488.

Therefore diagonal-only completion is not enough.

Mainline result: the quartic basis problem is solved, the free-energy route still fails, and the next completion must be genuinely

(880)

Goal C: complete the corrected branch at the projected residual level.

(881)

Goal M: identify the first non-diagonal microscopic completion beyond the exact four-class quartic baseline.

(882)

Goal S: solve $G_A(\delta) = G_E(\delta) = G_O(\delta) = 0$ directly, not through a naive free-energy reconstruction.

(883)

CCCLI. ACTUAL LOCAL RESIDUAL-GEOMETRY COMPUTATION AT THE CORRECTED BRANCH

CCCLII. CURRENT STAGE

The current stage is the corrected residual quartic closure stage. The correct reduced object is the projected residual map

$$G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix}, \quad (884)$$

not a naive reduced free-energy critical point. The carrier-side working branch is already effectively closed, and the real remaining frontier is the enlarged quartic residual closure. The frozen operational backend and corrected seed are

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096). \quad (885)$$

CCCLIII. ACTUAL LOCAL RESIDUAL GEOMETRY ALREADY AVAILABLE IN THE NOTES

The current files already provide the reduced residual at the background:

$$G(0) = (0.0164535, 0.0150360, 0.0306818), \quad (886)$$

with norm

$$\|G(0)\| \approx 3.7923 \times 10^{-2}. \quad (887)$$

They also provide the numerical Jacobian of the reduced residual map:

$$J \approx \begin{pmatrix} 0.0055310 & 0.0055542 & 0.0032242 \\ 0.0055542 & 0.0052164 & 0.0028916 \\ 0.0032242 & 0.0028916 & 0.0124317 \end{pmatrix}. \quad (888)$$

Moreover the unconstrained Newton correction is

$$\delta_{\text{Newton}} \approx (0.373633, -2.133560, -2.068666), \quad (889)$$

which is effective at reducing the reduced residual norm but destroys the full BCC branch geometry. Therefore Newton-type descent is not the correct geometry-preserving solver. This is already established in the notes.

CCCLIV. LINEARIZED ACTION OF THE JACOBIAN ON THE CORRECTED DIRECTION

Now evaluate the current frozen corrected local seed in the local linear geometry:

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096)^T.$$

Using (888), one gets

$$J \delta^{\text{final}} = \begin{pmatrix} -7.591104 \times 10^{-5} \\ -6.915264 \times 10^{-5} \\ 6.701856 \times 10^{-5} \end{pmatrix}. \quad (890)$$

Its norm is

$$\|J \delta^{\text{final}}\| \approx 1.2267 \times 10^{-4}. \quad (891)$$

Thus the linearized residual prediction at the corrected seed is

$$G_{\text{lin}}(\delta^{\text{final}}) = G(0) + J \delta^{\text{final}} = \begin{pmatrix} 0.01637759 \\ 0.01496685 \\ 0.03074882 \end{pmatrix}, \quad (892)$$

with norm

$$\|G_{\text{lin}}(\delta^{\text{final}})\| \approx 0.03791730. \quad (893)$$

Compared with

$$\|G(0)\| \approx 0.037923,$$

this means that the frozen corrected direction produces only a tiny first-order residual change inside the local reduced geometry.

CCCLV. SOFT-MODE STRUCTURE OF THE REDUCED RESIDUAL JACOBIAN

The eigenvalues of J are approximately

$$\lambda_1 \approx -1.8549 \times 10^{-4}, \quad \lambda_2 \approx 7.2905 \times 10^{-3}, \quad \lambda_3 \approx 1.6074 \times 10^{-2}. \quad (894)$$

So the reduced residual Jacobian is nearly singular, with one very small soft (slightly unstable) mode.

Let v_{soft} be the normalized eigenvector associated with λ_1 . Numerically,

$$v_{\text{soft}} \approx (0.7014, -0.7126, -0.0159).$$

Normalize the corrected direction:

$$\hat{\delta} = \frac{\delta^{\text{final}}}{\|\delta^{\text{final}}\|}. \quad (895)$$

Then the overlap with the soft mode is

$$|\hat{\delta} \cdot v_{\text{soft}}| \approx 0.8513. \quad (896)$$

Therefore the corrected geometry-preserving direction is strongly aligned with the near-null soft mode of the current reduced residual geometry.

CCCLVI. MAINLINE INTERPRETATION

This computation explains the present stall in a precise way.

The issue is not that the corrected residual program lacks a reduced object. It already has one:

$$G_A(\delta) = 0, \quad G_E(\delta) = 0, \quad G_O(\delta) = 0.$$

The issue is also not that the current admissible correction direction is random. Rather, the frozen corrected direction lies almost along the soft mode of the local Jacobian.

Hence:

$$\boxed{\text{the current admissible correction direction is almost first-order invisible to the reduced residual map.}} \quad (897)$$

So a first-order local Jacobian correction is not enough to close the branch. This means that the next mainline step must genuinely go beyond the present local linear geometry.

CCCLVII. CORRECT NEXT STEP

Since the notes already state that the quartic basis must be enlarged from the failed (u, v, κ) truncation to the four-class exact shell kernel, and that the corrected branch projection must be rerun on that enlarged basis, the present computation shows why this is unavoidable:

$$\boxed{\text{the local first-order reduced residual geometry is nearly flat along the corrected direction,}} \quad (898)$$

so

$$\boxed{\text{one must move to the enlarged four-class residual rerun and/or second-order non-diagonal completion.}} \quad (899)$$

This is the actual mainline consequence of the currently available branch data.

CCCLVIII. FINAL SUMMARY

Using the already available numerical reduced residual Jacobian,

$$G(0) = (0.0164535, 0.0150360, 0.0306818), \quad J \approx \begin{pmatrix} 0.0055310 & 0.0055542 & 0.0032242 \\ 0.0055542 & 0.0052164 & 0.0028916 \\ 0.0032242 & 0.0028916 & 0.0124317 \end{pmatrix},$$

the frozen corrected local seed

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096)$$

satisfies

$$J\delta^{\text{final}} = (-7.591104 \times 10^{-5}, -6.915264 \times 10^{-5}, 6.701856 \times 10^{-5}),$$

so the linearized residual norm changes only at the 10^{-4} level.

The Jacobian has a soft eigenvalue

$$\lambda_1 \approx -1.8549 \times 10^{-4},$$

and the corrected direction has overlap

$$|\hat{\delta} \cdot v_{\text{soft}}| \approx 0.8513$$

with its soft mode.

Therefore the present geometry-preserving corrected direction is almost a near-null direction of the local reduced residual Jacobian.

This explains why progress stalls at first order and why the next genuine mainline step must be the enlarged four-class residual rerun (or an equivalent second-order non-diagonal completion), rather than further first-order polishing.

CCCLIX. FINAL SUMMARY

Current stage: corrected residual quartic closure stage, with the projected residual map $G(\delta)$ as the primary reduced object. (900)

The files already provide the local reduced residual geometry at the background:

$$G(0) = (0.0164535, 0.0150360, 0.0306818),$$

$$J \approx \begin{pmatrix} 0.0055310 & 0.0055542 & 0.0032242 \\ 0.0055542 & 0.0052164 & 0.0028916 \\ 0.0032242 & 0.0028916 & 0.0124317 \end{pmatrix},$$

together with the frozen corrected seed

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096).$$

Applying the local Jacobian to the corrected direction gives

$$J\delta^{\text{final}} = (-7.591104 \times 10^{-5}, -6.915264 \times 10^{-5}, 6.701856 \times 10^{-5}),$$

so the linearized residual changes only at the 10^{-4} level.

The Jacobian eigenvalues are

$$-1.8549 \times 10^{-4}, \quad 7.2905 \times 10^{-3}, \quad 1.6074 \times 10^{-2},$$

and the normalized corrected direction has overlap

$$|\hat{\delta} \cdot v_{\text{soft}}| \approx 0.8513$$

with the soft eigenmode.

Therefore the geometry-preserving corrected direction is strongly aligned with the near-null soft mode of the current reduced residual geometry.

Mainline consequence: first-order local correction is nearly exhausted, so the next real step must be the enlarged four-class residual geometry. (901)

Goal C: complete the corrected branch at the residual level. (902)

Goal M: rerun the corrected projection on the enlarged four-class shell basis. (903)

Goal S: go beyond the present local first-order Jacobian geometry, which is nearly flat along the corrected direction. (904)

CCCLX. CORRECTED FOUR-CLASS ORBIT EQUATIONS AND THE FAILURE OF THE FREE-ENERGY STATIONARY ROUTE

CCCLXI. CURRENT STAGE

The current stage is the corrected residual quartic closure stage. The carrier-side working branch is already effectively closed, and the remaining obstruction is the quartic reduced-kernel closure on the corrected branch. The frozen corrected seed and corrected projected coefficients are

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096), \quad (905)$$

$$\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933, \quad (906)$$

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813. \quad (907)$$

The exact orbit free-energy equations in the four-class basis are already given in closed form. Under the (A, E, O) orbit ansatz, the exact stationary system is

$$\sigma_A = (U_d + U_{\text{opp}})r_A + 2U_{\perp}r_E + 8U_{\text{obl}}r_O, \quad (908)$$

$$\sigma_E = 2U_{\perp}r_A + (U_d + U_{\text{opp}})r_E + 8U_{\text{obl}}r_O, \quad (909)$$

$$\sigma_O = 2U_{\text{obl}}(r_A + r_E) + (U_d + U_{\text{opp}} + 2U_{\perp} + 4U_{\text{obl}})r_O. \quad (910)$$

So the corrected four-class free-energy reconstruction problem reduces exactly to the linear system

$$M_{\text{stat}}^{(4c)} \begin{pmatrix} r_A \\ r_E \\ r_O \end{pmatrix} = \begin{pmatrix} \sigma_A \\ \sigma_E \\ \sigma_O \end{pmatrix}, \quad (911)$$

with

$$M_{\text{stat}}^{(4c)} = \begin{pmatrix} U_d + U_{\text{opp}} & 2U_{\perp} & 8U_{\text{obl}} \\ 2U_{\perp} & U_d + U_{\text{opp}} & 8U_{\text{obl}} \\ 2U_{\text{obl}} & 2U_{\text{obl}} & U_d + U_{\text{opp}} + 2U_{\perp} + 4U_{\text{obl}} \end{pmatrix}. \quad (912)$$

CCCLXII. NUMERICAL INSERTION OF THE CORRECTED DATA

Substituting the corrected projected quartic class means gives

$$M_{\text{stat}}^{(4c)} = \begin{pmatrix} 17367.28007137692 & 18054.34581835654 & 17959.11544951265 \\ 18054.34581835654 & 17367.28007137692 & 17959.11544951265 \\ 4489.778862378163 & 4489.778862378163 & 47600.68135400748 \end{pmatrix}. \quad (913)$$

Its determinant is

$$\det M_{\text{stat}}^{(4c)} \approx -9.697913788948185 \times 10^{11}, \quad (914)$$

and its eigenvalues are

$$\lambda_1 \approx -687.06574698, \quad \lambda_2 \approx 32134.87964772, \quad \lambda_3 \approx 47687.92985650. \quad (915)$$

Hence the corrected four-class stationary matrix is itself indefinite.

CCCLXIII. CLOSED-FORM REDUCTION BY SUM AND DIFFERENCE

Define

$$S := r_A + r_E, \quad \Delta_r := r_A - r_E.$$

Subtracting (909) from (908) gives

$$\Delta_r = \frac{\sigma_A - \sigma_E}{U_d + U_{\text{opp}} - 2U_{\perp}}. \quad (916)$$

Using the corrected data,

$$\Delta_r = \frac{-115.81015796064617 + 105.83201242243113}{2493.1087739398513 + 14874.171297437068 - 2(9027.17290917827)} \approx 0.01452283945470934. \quad (917)$$

Adding (908) and (909), one gets

$$(U_d + U_{\text{opp}} + 2U_{\perp})S + 16U_{\text{obl}}r_O = \sigma_A + \sigma_E. \quad (918)$$

Together with (910), this yields the 2×2 system

$$\begin{pmatrix} U_d + U_{\text{opp}} + 2U_{\perp} & 16U_{\text{obl}} \\ 2U_{\text{obl}} & U_d + U_{\text{opp}} + 2U_{\perp} + 4U_{\text{obl}} \end{pmatrix} \begin{pmatrix} S \\ r_O \end{pmatrix} = \begin{pmatrix} \sigma_A + \sigma_E \\ \sigma_O \end{pmatrix}. \quad (919)$$

Numerically this gives

$$S \approx -0.004205590866673833, \quad r_O \approx -0.0020232985378032353. \quad (920)$$

Finally

$$r_A = \frac{S + \Delta_r}{2} \approx 0.005158624294017749, \quad r_E = \frac{S - \Delta_r}{2} \approx -0.009364215160691582. \quad (921)$$

So the corrected four-class algebraic critical point is

$$r_A^{\text{crit}} \approx 5.158624294017749 \times 10^{-3}, \quad r_E^{\text{crit}} \approx -9.364215160691582 \times 10^{-3}, \quad r_O^{\text{crit}} \approx -2.0232985378032353 \times 10^{-3}. \quad (922)$$

CCCLXIV. PHYSICAL VERDICT

The corrected four-class stationary point is still physically inadmissible because

$$r_E^{\text{crit}} < 0, \quad r_O^{\text{crit}} < 0. \quad (923)$$

Therefore:

$$\text{the four-class exact shell kernel solves the basis problem,} \quad (924)$$

but

$$\text{it does not rescue the reduced free-energy stationary picture on the corrected branch.} \quad (925)$$

CCCLXV. MAINLINE CONCLUSION

This is now the corrected analogue of the earlier unphysical free-energy critical point: even after replacing the failed (u, v, κ) truncation by the exact four-class shell kernel, the free-energy stationary route remains inadmissible.

Hence the next mathematically correct reduced object is not

$$\partial_{r_I} \mathcal{F}_{\text{orb}} = 0,$$

but the projected residual system

$$G_A(r_A, r_E, r_O) = 0, \quad G_E(r_A, r_E, r_O) = 0, \quad G_O(r_A, r_E, r_O) = 0. \quad (926)$$

So the actual frontier is now sharply fixed:

$$\text{not branch search, not quartic-basis truncation, but the direct residual-level orbit reduction.} \quad (927)$$

CCCLXVI. FINAL SUMMARY

$$\boxed{\text{Current stage: corrected four-class quartic closure test on the frozen corrected branch.}} \quad (928)$$

Using the corrected projected data

$$\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933,$$

$U_d = 2493.1087739398513$, $U_{\text{opp}} = 14874.171297437068$, $U_{\perp} = 9027.17290917827$, $U_{\text{obl}} = 2244.8894311890813$,
the exact four-class orbit stationary system is

$$\sigma_A = (U_d + U_{\text{opp}})r_A + 2U_{\perp}r_E + 8U_{\text{obl}}r_O,$$

$$\sigma_E = 2U_{\perp}r_A + (U_d + U_{\text{opp}})r_E + 8U_{\text{obl}}r_O,$$

$$\sigma_O = 2U_{\text{obl}}(r_A + r_E) + (U_d + U_{\text{opp}} + 2U_{\perp} + 4U_{\text{obl}})r_O.$$

Its corrected algebraic critical point is

$$\boxed{r_A^{\text{crit}} \approx 5.158624294017749 \times 10^{-3}, \quad r_E^{\text{crit}} \approx -9.364215160691582 \times 10^{-3}, \quad r_O^{\text{crit}} \approx -2.0232985378032353 \times 10^{-3}.}$$

Therefore the corrected four-class free-energy stationary point is still physically inadmissible:

$$\boxed{r_E^{\text{crit}} < 0, \quad r_O^{\text{crit}} < 0.}$$

So the exact four-class shell kernel solves the quartic basis problem, but it does not rescue the free-energy stationary picture on the corrected branch.

$$\boxed{\text{Mainline result: the next reduced object must be the projected residual system } G_A = G_E = G_O = 0.} \quad (929)$$

CCCLXVII. FIRST GENUINE NON-DIAGONAL RESIDUAL COMPLETION AT THE CORRECTED BRANCH

CCCLXVIII. CURRENT STAGE

The current project stage is the corrected residual quartic closure stage. The carrier-side working branch is already effectively closed, while the actual remaining frontier is the enlarged quartic residual closure. The correct reduced object is the projected residual map

$$\boxed{G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix}}, \quad (930)$$

with frozen corrected seed

$$\boxed{\delta^{\text{final}} = (0.0096, -0.0288, 0.0096).} \quad (931)$$

The corrected projected quartic class data are

$$\boxed{U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813.} \quad (932)$$

CCCLXIX. EXACT FOUR-CLASS QUARTIC BASELINE

The coarse (A, E, O) four-class quartic reduced matrix is

$$M^{(4c)} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}. \quad (933)$$

At the corrected seed

$$(a, e, o) = (0.0096, -0.0288, 0.0096),$$

one gets

$$Q = \begin{pmatrix} 8.494671573456258 \\ 8.241391656489691 \\ 2.580391735022780 \end{pmatrix}, \quad (934)$$

hence

$$G^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}. \quad (935)$$

The quartic residual Jacobian at the corrected seed is

$$J_{\text{res}}^{(4c)}(\delta^{\text{final}}) = \begin{pmatrix} 10.095240102467715 & -4.991665531859216 & 0.4137780199567714 \\ -4.991665531859216 & 22.646508448870943 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 3.603395005929136 \end{pmatrix}. \quad (936)$$

CCCLXX. RESIDUAL-SIDE LOCAL COMPLETION PROBLEM

We now seek a local residual model of the form

$$G^{\text{model}}(\delta) = S\delta + G^{(4)}(\delta), \quad (937)$$

where S is a constant 3×3 completion matrix.

To make the corrected seed an exact stationary point of this local model, one must impose

$$G^{\text{model}}(\delta^{\text{final}}) = 0 \iff S \delta^{\text{final}} = -G^{(4)}(\delta^{\text{final}}). \quad (938)$$

This is the first residual-side completion problem beyond the failed free-energy reconstruction route.

CCCLXXI. MINIMAL SYMMETRIC COMPLETION

We restrict to symmetric completions

$$S = S^T,$$

and ask for the minimum-Frobenius-norm solution of (938).

Let

$$d := \delta^{\text{final}}, \quad g := G^{(4)}(\delta^{\text{final}}), \quad u := \frac{d}{\|d\|}, \quad w := \frac{-g}{\|d\|}.$$

Then the unique minimum-Frobenius symmetric solution is

$$S_{\min} = uw^T + wu^T - (u^T w) uu^T. \quad (939)$$

A direct substitution shows that $S_{\min} = S_{\min}^T$ and

$$S_{\min} d = -g.$$

Numerically,

$$S_{\min} = \begin{pmatrix} -0.839960218174805 & 2.450804313533534 & -0.302298414680853 \\ 2.450804313533534 & -7.145183917627953 & 0.837818903051675 \\ -0.302298414680853 & 0.837818903051675 & 0.235363388813100 \end{pmatrix}. \quad (940)$$

Its Frobenius norm is

$$\|S_{\min}\|_F \approx 8.087899329725037. \quad (941)$$

CCCLXXII. COMPARISON WITH THE DIAGONAL-ONLY COMPLETION

The diagonal-only completion that enforces stationarity is

$$D = \text{diag}(-Q_A, -Q_E, -Q_O) = \text{diag}(-8.494671573456258, -8.241391656489691, -2.580391735022780). \quad (942)$$

Its Frobenius norm is

$$\|D\|_F \approx 12.113562774127091. \quad (943)$$

So even before checking stability, the smallest symmetric completion is already genuinely non-diagonal and smaller in norm than the diagonal-only completion.

CCCLXXIII. OFF-DIAGONAL CONTENT IS UNAVOIDABLE

Decompose

$$S_{\min} = S_{\min}^{\text{diag}} + S_{\min}^{\text{off}}.$$

Numerically,

$$\|S_{\min}^{\text{diag}}\|_F \approx 7.198234666196914, \quad \|S_{\min}^{\text{off}}\|_F \approx 3.687754500794174. \quad (944)$$

Hence the off-diagonal sector contributes the non-negligible fraction

$$\frac{\|S_{\min}^{\text{off}}\|_F}{\|S_{\min}\|_F} \approx 0.4559594958409981. \quad (945)$$

So the first corrected residual completion is not merely a tiny perturbative off-diagonal tweak. Its off-diagonal part is order one relative to the whole completion.

CCCLXXIV. LOCAL JACOBIAN AFTER THE SYMMETRIC COMPLETION

The completed residual Jacobian at the corrected seed is

$$J_{\text{tot}} = J_{\text{res}}^{(4c)}(\delta^{\text{final}}) + S_{\min}. \quad (946)$$

Numerically,

$$J_{\text{tot}} = \begin{pmatrix} 9.255279884292909 & -2.540861218325682 & 0.111479605275918 \\ -2.540861218325682 & 15.501324531242990 & -0.403515156818639 \\ 0.111479605275918 & -0.403515156818639 & 3.838758394742236 \end{pmatrix}. \quad (947)$$

Its eigenvalues are

$$\lambda_1(J_{\text{tot}}) \approx 3.824699013629953, \quad \lambda_2(J_{\text{tot}}) \approx 8.352431280003644, \quad \lambda_3(J_{\text{tot}}) \approx 16.418232516644533. \quad (948)$$

Therefore

$$J_{\text{tot}} \text{ is positive definite.} \quad (949)$$

CCCLXXV. CONTRAST WITH THE DIAGONAL-ONLY COMPLETION

If one instead uses the diagonal completion D , the completed Jacobian is

$$J_{\text{diag}} = J_{\text{res}}^{(4c)}(\delta^{\text{final}}) + D. \quad (950)$$

Its eigenvalues are

$$\lambda_1(J_{\text{diag}}) \approx -0.11554365, \quad \lambda_2(J_{\text{diag}}) \approx 0.91059735, \quad \lambda_3(J_{\text{diag}}) \approx 16.23363488. \quad (951)$$

So the diagonal-only completion makes the corrected seed stationary, but still leaves a negative mode. By contrast, the minimum-norm symmetric non-diagonal completion both enforces stationarity and restores positive local residual geometry.

CCCLXXVI. MAINLINE CONSEQUENCE

This is the first genuine non-repetitive mainline result beyond the four-class quartic baseline:

1. the corrected branch cannot be closed by a diagonal-only completion;
2. the first viable local completion is genuinely non-diagonal;
3. the smallest symmetric completion already has a substantial off-diagonal sector;
4. once that non-diagonal completion is included, the corrected seed becomes both stationary and locally stable in the residual model.

Therefore the next microscopic completion source must reproduce a matrix of the same qualitative type as S_{min} , not merely the diagonal target q .

Equivalently,

$$\text{the next microscopic task is to identify which source generates the non-diagonal completion pattern } S_{\text{min}}. \quad (952)$$

CCCLXXVII. FINAL SUMMARY

$$\text{Current stage: first genuine non-diagonal residual completion at the corrected branch.} \quad (953)$$

At the corrected seed

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096),$$

the exact four-class quartic residual baseline is

$$G^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix},$$

and the quartic residual Jacobian is

$$J_{\text{res}}^{(4c)}(\delta^{\text{final}}) = \begin{pmatrix} 10.095240102467715 & -4.991665531859216 & 0.4137780199567714 \\ -4.991665531859216 & 22.646508448870943 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 3.603395005929136 \end{pmatrix}.$$

The unique minimum-Frobenius symmetric completion satisfying

$$S_{\delta^{\text{final}}} = -G^{(4)}(\delta^{\text{final}})$$

is

$$S_{\text{min}} = \begin{pmatrix} -0.839960218174805 & 2.450804313533534 & -0.302298414680853 \\ 2.450804313533534 & -7.145183917627953 & 0.837818903051675 \\ -0.302298414680853 & 0.837818903051675 & 0.235363388813100 \end{pmatrix}.$$

Its off-diagonal sector is not negligible:

$$\frac{\|S_{\text{min}}^{\text{off}}\|_F}{\|S_{\text{min}}\|_F} \approx 0.4559594958409981.$$

The completed residual Jacobian

$$J_{\text{tot}} = J_{\text{res}}^{(4c)} + S_{\text{min}}$$

has eigenvalues

$$3.824699013629953, \quad 8.352431280003644, \quad 16.418232516644533,$$

so it is positive definite.

By contrast, the diagonal-only completion

$$D = \text{diag}(-Q_A, -Q_E, -Q_O)$$

still gives one negative eigenvalue:

$$-0.11554365, \quad 0.91059735, \quad 16.23363488.$$

Therefore the corrected branch cannot be closed by a diagonal-only completion. The first viable local completion is genuinely non-diagonal.

Mainline result: the next microscopic source must reproduce the non-diagonal completion pattern S_{min} , not merely the diagonal. (954)

CCCLXXVIII. PROJECT: TECT — MASTER CONTENTS FOR CONTINUATION (CORRECTED RESIDUAL QUARTIC CLOSURE MAINLINE)

CCCLXXIX. CURRENT STAGE

The project has now reached the following precise stage:

$$\boxed{\text{carrier-side working-branch admissibility is effectively closed,}} \quad (955)$$

$$\boxed{\text{while the actual remaining frontier is the corrected residual quartic kernel closure.}} \quad (956)$$

Equivalently,

$$\boxed{\text{working-branch carrier witness: closed enough to move on,} \quad \text{corrected residual quartic closure: open.}} \quad (957)$$

Therefore the next conversation must *not* restart sign-polishing, broad branch search, or repeated carrier-side detours. It must begin directly from the enlarged quartic reduced shell kernel problem and the residual-level closure problem.

CCCLXXX. FINAL GOAL C

The final large-scale goal remains:

$$\boxed{\text{extract the actual branch-dependent coefficients } (\lambda_{\parallel}, \alpha, \beta) \text{ from the full microscopic TECT operator.}} \quad (958)$$

Equivalently, the target chain is

$$\boxed{L(k) \longrightarrow K_i := \partial_{k_i} L|_{G_*} \longrightarrow P_* K_i P_* \longrightarrow (\lambda_{\parallel}, \alpha, \beta).} \quad (959)$$

So the ultimate target is the actual microscopic PDE coefficient point, not merely a reduced ansatz.

CCCLXXXI. FROZEN OPERATIONAL BACKEND

The operational backend is frozen as

$$\boxed{(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62).} \quad (960)$$

This point should be treated as fixed unless a theorem-level reason appears to change it.

CCCLXXXII. CORRECT REDUCED OBJECT

The correct reduced object is the projected residual map, not a naive reduced free-energy critical point. Let

$$\Psi(\delta) = \Psi_{\text{BCC}} + \delta_A \Phi_A + \delta_E \Phi_E + \delta_O \Phi_O, \quad (961)$$

and define

$$\boxed{G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix} := \begin{pmatrix} \Re \langle \Phi_A, R[\Psi(\delta)] \rangle_{\Omega} \\ \Re \langle \Phi_E, R[\Psi(\delta)] \rangle_{\Omega} \\ \Re \langle \Phi_O, R[\Psi(\delta)] \rangle_{\Omega} \end{pmatrix}.} \quad (962)$$

This residual-based reduced object is primary.

CCCLXXXIII. FINAL CORRECTED LOCAL SEED

The final geometry-preserving corrected seed is frozen as

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096). \quad (963)$$

So the corrected local branch is operationally defined by

$$\Psi_{\text{corr}} = \Psi_{\text{BCC}} + 0.0096 \Phi_A - 0.0288 \Phi_E + 0.0096 \Phi_O. \quad (964)$$

The corresponding corrected diagnostics are

$$\|G(\delta^{\text{final}})\| \approx 0.03791852545979427, \quad (965)$$

$$\text{rel_residual}(\delta^{\text{final}}) \approx 0.036459908397927156, \quad (966)$$

$$\text{shell_fraction}(\delta^{\text{final}}) \approx 0.9156153162713265, \quad \text{orbit_balance}(\delta^{\text{final}}) \approx 0.8860846137625883. \quad (967)$$

CCCLXXXIV. PROJECTION EXTRACTOR STATUS

A residual-based projection extractor has already been implemented and executed.
Its extraction chain is

$$\Psi \implies h_\alpha \implies (\sigma_0, \eta), \quad (968)$$

$$\Psi \implies U_{\alpha\beta} \implies (U_d, U_{\text{opp}}, U_\perp, U_{\text{obl}}), \quad (969)$$

and optionally

$$(U_d, U_{\text{opp}}, U_\perp, U_{\text{obl}}) \implies (u, v, \kappa, \Delta_{\text{ker}}). \quad (970)$$

These must be interpreted honestly as residual-based effective reduced coefficients of the current solver family, not yet theorem-level coefficients of a fully closed microscopic energy functional.

CCCLXXXV. CORRECTED PROJECTION RESULTS

After inserting δ^{final} , the corrected projection gives

$$\sigma_0 = -109.41971030446824, \quad \eta = -6.502583100163014. \quad (971)$$

Orbit means:

$$\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933. \quad (972)$$

Quartic class averages:

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_\perp = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813. \quad (973)$$

If these are compressed to the simple quartic kernel

$$U_{\alpha\beta} = u\delta_{\alpha\beta} + v + \kappa(\hat{k}_\alpha \cdot \hat{k}_\beta)^2,$$

then one gets

$$u = -12381.062523497216, \quad v = 9027.17290917827, \quad \kappa = -27129.133911956756, \quad (974)$$

but the exact consistency defect becomes

$$\Delta_{\text{ker}} = 32976.132300215555. \quad (975)$$

Therefore the simple quartic (u, v, κ) truncation fails strongly even on the corrected local branch.

CCCLXXXVI. EXACT MEANING OF THE CURRENT OBSTRUCTION

The old (u, v, κ) kernel imposes the algebraic constraint

$$U_{\text{opp}} = 4U_{\text{obl}} - 3U_{\perp}. \quad (976)$$

Thus the exact defect is

$$\Delta_{\text{ker}} = U_{\text{opp}} - 4U_{\text{obl}} + 3U_{\perp}. \quad (977)$$

Since $\Delta_{\text{ker}} \neq 0$, the quartic reduced shell kernel basis must be enlarged.

CCCLXXXVII. EXACT FOUR-CLASS ENLARGED QUARTIC KERNEL

The minimal exact enlarged shell kernel is

$$U_{\alpha\beta}^{(4c)} = u_d P_{\alpha\beta}^{(d)} + u_{\text{opp}} P_{\alpha\beta}^{(\text{opp})} + u_{\perp} P_{\alpha\beta}^{(\perp)} + u_{\text{obl}} P_{\alpha\beta}^{(\text{obl})}, \quad (978)$$

with direct identification

$$u_d = U_d, \quad u_{\text{opp}} = U_{\text{opp}}, \quad u_{\perp} = U_{\perp}, \quad u_{\text{obl}} = U_{\text{obl}}. \quad (979)$$

After coarse orbit reduction to (A, E, O) , the exact four-class quartic matrix is

$$M^{(4c)} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}. \quad (980)$$

CCCLXXXVIII. CORRECTED-SEED QUARTIC BASELINE

At the corrected seed

$$(a, e, o) = (0.0096, -0.0288, 0.0096),$$

one has

$$Q = M^{(4c)} \begin{pmatrix} a^2 \\ e^2 \\ o^2 \end{pmatrix} = \begin{pmatrix} 8.494671573456258 \\ 8.241391656489691 \\ 2.580391735022780 \end{pmatrix}, \quad (981)$$

and hence

$$G^{(4)}(\delta^{\text{final}}) = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}, \quad \|G^{(4)}(\delta^{\text{final}})\| = 0.25219013527770096. \quad (982)$$

Since the actual corrected residual norm is only

$$\|G(\delta^{\text{final}})\| \approx 0.03791852545979427,$$

the nonquartic mismatch is necessarily large:

$$\|G(\delta^{\text{final}}) - G^{(4)}(\delta^{\text{final}})\| \geq 0.2142716098179067. \quad (983)$$

So the corrected branch requires a substantial nonquartic completion beyond the exact four-class quartic baseline.

CCCLXXXIX. WHY THE FREE-ENERGY RECONSTRUCTION ROUTE IS STILL DEAD

Even after the four-class enlargement, the free-energy stationary route still fails. The exact four-class stationary equations are

$$\sigma_A = (U_d + U_{\text{opp}})r_A + 2U_{\perp}r_E + 8U_{\text{obl}}r_O, \quad (984)$$

$$\sigma_E = 2U_{\perp}r_A + (U_d + U_{\text{opp}})r_E + 8U_{\text{obl}}r_O, \quad (985)$$

$$\sigma_O = 2U_{\text{obl}}(r_A + r_E) + (U_d + U_{\text{opp}} + 2U_{\perp} + 4U_{\text{obl}})r_O. \quad (986)$$

Solving this corrected four-class algebraic stationary system gives

$$r_A^{\text{crit}} \approx 5.158624294017749 \times 10^{-3}, \quad r_E^{\text{crit}} \approx -9.364215160691582 \times 10^{-3}, \quad r_O^{\text{crit}} \approx -2.0232985378032353 \times 10^{-3}. \quad (987)$$

This is physically inadmissible because

$$r_E^{\text{crit}} < 0, \quad r_O^{\text{crit}} < 0.$$

Therefore the exact four-class enlargement solves the quartic basis problem but still does not rescue the free-energy stationary picture on the corrected branch.

CCCXC. RESIDUAL-SIDE LOCAL GEOMETRY ALREADY KNOWN

The current files already provide the local reduced residual geometry at the background:

$$G(0) = (0.0164535, 0.0150360, 0.0306818), \quad (988)$$

$$J \approx \begin{pmatrix} 0.00555310 & 0.0055542 & 0.0032242 \\ 0.0055542 & 0.0052164 & 0.0028916 \\ 0.0032242 & 0.0028916 & 0.0124317 \end{pmatrix}. \quad (989)$$

The unconstrained Newton step

$$\delta_{\text{Newton}} \approx (0.373633, -2.133560, -2.068666)$$

reduces the reduced residual strongly, but destroys the full BCC branch geometry. So unconstrained Newton descent is not the correct solver.

CCCXCI. ACTUAL RESIDUAL-SIDE MAINLINE RESULT BEYOND THE QUARTIC BASELINE

The four-class quartic residual Jacobian at the corrected seed is

$$J_{\text{res}}^{(4c)}(\delta^{\text{final}}) = \begin{pmatrix} 10.095240102467715 & -4.991665531859216 & 0.4137780199567714 \\ -4.991665531859216 & 22.646508448870943 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 3.603395005929136 \end{pmatrix}, \quad (990)$$

which is positive definite.

The minimal diagonal completion target that forces stationarity is

$$\begin{matrix} q_A & q_E & q_O \end{matrix} \begin{matrix} = -8.494671573456258, \\ = -8.241391656489691, \\ = -2.580391735022780. \end{matrix} \quad (991)$$

But diagonal-only completion still leaves a negative eigenvalue in the completed Jacobian. By contrast, the minimum-Frobenius symmetric completion

$$S_{\min} = \begin{pmatrix} -0.839960218174805 & 2.450804313533534 & -0.302298414680853 \\ 2.450804313533534 & -7.145183917627953 & 0.837818903051675 \\ -0.302298414680853 & 0.837818903051675 & 0.235363388813100 \end{pmatrix} \quad (992)$$

satisfies

$$S_{\min} \delta^{\text{final}} = -G^{(4)}(\delta^{\text{final}})$$

and yields a positive-definite completed residual Jacobian.

Therefore the first viable corrected residual completion is already genuinely non-diagonal.

CCCXCII. WHAT IS CLOSED

The following points are already effectively closed:

1. broad backend search is finished;
2. a best confirmed actual BCC branch exists numerically;
3. a final geometry-preserving corrected seed has been extracted;
4. the reduced free-energy closure route is ruled out;
5. the reduced residual map is identified as the correct reduced object;
6. the simple quartic (u, v, κ) truncation is decisively falsified;
7. the exact four-class quartic shell basis is known;
8. diagonal-only residual completion is insufficient;
9. the first viable local corrected completion is genuinely non-diagonal.

CCCXCIII. WHAT REMAINS OPEN

Only the following mainline items remain open:

1. the corrected projection must be rerun directly on the enlarged four-class shell basis;
2. the residual system must be handled directly at the residual level, not through free-energy reconstruction;
3. the microscopic source of the non-diagonal completion pattern must be identified.

CCCXCIV. STRICT WORKING RULES FOR THE NEXT CONVERSATION

1. Do not reopen broad parameter scans over (μ^2, λ, γ) .
2. Keep the backend point fixed at $(0.26, -0.43, 1.62)$.
3. Keep the corrected seed fixed at $(0.0096, -0.0288, 0.0096)$.
4. Treat the projected residual map $G(\delta)$ as primary.
5. Do not return to naive free-energy stationary reconstruction.
6. Do not go back to carrier sign-polishing unless a direct contradiction appears.
7. Work only on enlarged quartic residual closure and non-diagonal residual completion.

CCCXCV. IMMEDIATE NEXT TASK

The next conversation must begin from the following precise objective:

$$\boxed{\text{rerun the corrected projection directly on the four-class exact shell kernel basis}} \quad (993)$$

and then

$$\boxed{\text{write the enlarged residual system directly in } (U_d, U_{\text{opp}}, U_{\perp}, U_{\text{obl}}) \text{ and identify the microscopic origin of the non-diagonal comp}} \quad (994)$$

CCCXCVI. FINAL SUMMARY

$$\boxed{\text{Current status: carrier-side working-branch admissibility is effectively closed, while the actual remaining frontier is corrected re}} \quad (995)$$

The backend and corrected seed are frozen as

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096).$$

The correct reduced object is the projected residual map

$$G(\delta) = (G_A(\delta), G_E(\delta), G_O(\delta))^T.$$

The corrected projected quartic data are

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813,$$

and the simple (u, v, κ) truncation fails with

$$\Delta_{\text{ker}} = 32976.132300215555.$$

The exact four-class quartic basis is therefore mandatory. However, even after this enlargement, the free-energy stationary reconstruction remains physically inadmissible.

At the residual level, the first viable corrected local completion is already known to be genuinely non-diagonal:

$$S_{\text{min}} \delta^{\text{final}} = -G^{(4)}(\delta^{\text{final}}),$$

with

$$S_{\text{min}} = \begin{pmatrix} -0.839960218174805 & 2.450804313533534 & -0.302298414680853 \\ 2.450804313533534 & -7.145183917627953 & 0.837818903051675 \\ -0.302298414680853 & 0.837818903051675 & 0.235363388813100 \end{pmatrix}.$$

Therefore the true next mainline task is no longer basis search or sign polishing.

$$\boxed{\text{Next step: rerun the corrected projection directly on the four-class exact shell kernel basis,}} \quad (996)$$

$$\boxed{\text{then write and solve the enlarged projected residual system itself.}} \quad (997)$$